

Resource limitation rewires chromosomal instability and ploidy evolution across in vitro and in vivo cancer models

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Abstract

Chromosomal instability (CIN) generates karyotypic diversity, but resource context may determine whether altered progeny persist. We combined lineage-matched near-diploid (2N) and near-tetraploid (4N) SUM159 culture and tumor datasets with a chromosome-number-structured model linking oxygen-dependent resource stress to proliferation, death, whole-genome doubling, chromosome missegregation, and post-missegregation survival. In culture, oxygen deprivation reduced growth and remodeled both low- and high-chromosome components, while the model attributed only a minor fraction of the resulting burden to death associated with resource stress. Tumor fits captured cohort-scale growth and ploidy remodeling but admitted multiple near-equivalent parameterizations. Joint fitting nevertheless resolved consistent function-level differences between contexts: tumors had a lower fitted proliferative ceiling, stronger effective chromosome missegregation under hypoxia, and higher absolute survival of altered daughters, whereas culture showed a steeper ploidy dependence of post-missegregation survival. Fixed-O₂ analyses further separated a switch-like low-versus high-ploidy response in culture from broader, family-dependent intermediate-ploidy regimes in tumors. These results support a model in which resource context changes both the production of chromosome-number variation and the ploidy-dependent filtering of viable descendants. The conclusions remain conditional on the model, data streams, parameter bounds, regularization, and sampled optimizer landscapes.

Introduction

Whole-genome doubling (WGD) and aneuploidy are common features of cancer evolution, yet their prevalence varies substantially across biological contexts.^{1–3} A central unresolved question is why high-ploidy states are tolerated in some environments but selected against in others.^{4,5} One clue comes from the striking difference between cultured cancer cell lines and primary tumors: across organ systems, cell lines exhibit markedly higher ploidy than primary tumors. This observation suggests that standard in vitro culture conditions, which provide abundant nutrients and simplified environmental constraints, may relax selection against high-ploidy states, whereas in vivo tumor environments impose stronger constraints on cells carrying excess genomic material.^{4,5}

A plausible basis for this context dependence is the energetic cost of increased chromosome content.⁴⁻⁶ WGD increases the biosynthetic burden associated with DNA replication, RNA production, and especially protein synthesis.⁴⁻⁶ Aneuploidy further compounds this cost by disrupting gene dosage balance, producing proteotoxic, osmotic, oxidative, replication, and endoplasmic-reticulum stress.⁷⁻⁹ These stresses require active homeostatic responses, including protein quality control, degradation of excess proteins, DNA repair, and metabolic adaptation.^{5,7-9} Thus, high-ploidy cells are expected to carry a resource-dependent fitness cost, particularly when oxygen, glucose, or other metabolic substrates become limiting.^{4,5}

Resource scarcity can, however, also increase genomic instability.^{4,5} Environmental stresses such as serum starvation, hypoxia, folate deficiency, and energy restriction have been linked to karyotypic instability, chromosome missegregation, polyploidization, and increased cellular heterogeneity. The evolutionary consequences of these events depend strongly on ploidy. Chromosome missegregation can trigger p53-mediated arrest, senescence, or death, particularly when chromosome loss creates severe dosage imbalance.^{8,10-13} WGD and high-ploidy states can buffer these effects by reducing the relative dosage impact of individual chromosome gains or losses, allowing some missegregation events to be converted into heritable karyotype variation rather than lethal outcomes.^{7,14,15} Consistent with this view, ALFA-K, a computational method that infers chromosome-level karyotype fitness from longitudinal single-cell karyotype data, showed that missegregation events impose stronger fitness costs near diploidy than near tetraploidy.¹⁶ Thus, ploidy is a quantitative determinant of whether CIN generates nonviable progeny or heritable karyotypic diversity.

Oxygen provides a tractable first-order proxy for this resource landscape.⁴⁻⁶ Across tissue sites, WGD prevalence correlates with peripheral oxygen partial pressure, consistent with the idea that well-oxygenated tissues are more permissive for WGD-positive cancers, whereas poorly oxygenated tissues are less permissive.^{5,6} However, oxygen is an imperfect surrogate for total energy availability. Tissue resource states also depend on amino acid and glucose availability, local metabolic consumption, and metabolic specialization. For example, tissues that rely heavily on glycolysis or have unusually high glucose availability may not experience low oxygen as equivalent to global energy scarcity.

Together, these observations identify a central tension in cancer ploidy evolution. Resource scarcity should select against WGD and other high-ploidy states because additional chromosome content increases biosynthetic, proteostatic, and homeostatic demands.⁴⁻⁶ Yet adverse environments can also promote CIN, chromosome missegregation, and polyploidization, generating the karyotypic variation on which selection acts.^{4,5} Whether this variation is eliminated or retained depends in part on ploidy: near-diploid cells are more vulnerable to lethal dosage imbalance after chromosome loss, whereas WGD or high-ploidy cells can better buffer individual segregation errors and convert them into heritable karyotype variation.^{7,14-16} Thus, resource limitation may simultaneously oppose and promote WGD, depending on how strongly it suppresses proliferation, induces CIN, and filters the survival of altered progeny. A key question is therefore which environments select for WGD, which select against it, and through which mechanisms. Answering this requires a quantitative framework that can jointly represent resource-dependent proliferation, stress-induced chromosomal instability, and ploidy-dependent survival of resulting karyotypic states.

Here, we used a chromosome-number-structured mechanistic model to connect resource stress with the generation and retention of chromosome-number variation in lineage-matched SUM159 culture and tumor data. Rather than interpret individual fitted parameters in isolation, we compared complete response functions, identified observations that the model did not fully reproduce, and defined the limits of mechanistic interpretation.

Methods

Study design and data streams. We analyzed near-diploid (2N) and near-tetraploid (4N) SUM159 lineages across two experimental contexts. *In vivo*, orthotopic tumors initiated from the 2N and 4N lineages were followed with longitudinal tumor-burden measurements and harvested for terminal single-cell chromosome-number profiling; where matched terminal histology was available, necrosis was quantified at harvest. The *in vivo* calibration used the untreated subset of eight tumors, four initiated from each lineage. *In vitro*, the corresponding lineage pair was analyzed in passage-structured culture experiments with measured passage growth, karyotype/chromosome-number observations, and G0/G1 flow-density profiles under controlled oxygen conditions. Each starting-ploidy cohort contained a control branch and two parallel oxygen-deprivation lineages, designated O1 and O2. O1 and O2 identify separately propagated lineages within the 2N- and 4N-starting cohorts and do not, by themselves, denote biological replication; their experimental design is defined in Supplementary Methods under *In vitro experiments* and summarized in Supplementary Table 1. The shared lineage origins placed the two datasets on a common modeling axis, but the experiments were not biological replicates of one another and did not constitute a randomized intervention comparing culture with tumors.

Model state and oxygen context. To compare these measurements within one framework, we represented viable cells as a distribution over total autosomal chromosome-number states. The model also tracks non-viable biomass in separate dead compartments, so simulated bulk tumor burden includes both live cells and uncleared dead material. Terminal single-cell chromosome-number profiles sample only the viable population at harvest, whereas tumor burden and necrotic fraction depend on live plus uncleared dead compartments. This state representation lets one simulation be compared simultaneously with longitudinal tumor volume, terminal necrosis, and single-cell chromosome-number distributions, while preserving the model’s focus on karyotype evolution.

Resource context was represented differently in the two experimental settings. *In vivo*, oxygen is modeled as an effective mean-field level that declines with chromosome-weighted tumor demand and relaxes toward a supply-demand target. *In vitro*, each passage segment is simulated at its assigned fixed oxygen level.

Mechanistic modules. Within this state representation, we modeled three routes through which resource context could select for or against WGD. First, oxygen-dependent resource stress suppresses proliferation through a chromosome-number-dependent growth penalty, so high-chromosome states pay a larger growth cost when resources are limiting. Stress also induces a chromosome-number-dependent live-to-dead hazard, and the model assigns the resulting burden to a dedicated compartment representing death associated with resource stress. This partition is an internal model attribution rather than a direct observation of the biological cause of death. Second, stress

can increase chromosome missegregation, while WGD is modeled as a constant-probability division-associated conversion branch that provides a route into high-ploidy states, generating the karyotypic variation on which selection can act. Third, daughters produced after missegregation pass through a ploidy-dependent survival filter: daughters of lower-ploidy mothers are more vulnerable to lethal dosage imbalance, whereas higher-ploidy mothers can better buffer chromosome gains or losses and convert some CIN events into heritable karyotype variation. These modules map directly onto the Introduction’s mechanistic alternatives by asking whether resource-poor environments suppress high-ploidy proliferation, increase CIN, alter the selective consequences of a constant-probability WGD route, or determine whether altered progeny die or persist.

Observation models. The observation model linked each measured data stream to a specific biological component of this mechanism, allowing growth dynamics and chromosome-number evolution to constrain the fitted trajectory jointly:

Data stream	Biological component constrained
Longitudinal tumor burden	Net growth, death, clearance, and resource-dependent proliferation
Terminal single-cell chromosome-number distributions	Karyotype evolution, WGD, missegregation, and survival of altered progeny
Terminal necrotic fraction	Uncleared dead biomass relative to total tumor burden
<i>In vitro</i> passage growth rates	Proliferation under controlled culture and oxygen conditions
<i>In vitro</i> karyotype/chromosome-number and flow-density profiles	Chromosome-number evolution in culture

Joint fitting. The separate and joint *in vivo*/*in vitro* fits shared the same cellular response model: chromosome-number transitions, oxygen-dependent resource-stress effects on proliferation and death, stress-induced missegregation, WGD generation, and post-missegregation survival were represented by one mechanistic structure. The contexts differed in how oxygen trajectories were supplied and in their observation models. In the reported joint analysis, all 14 cross-context mechanistic parameters were soft-coupled. Each received separate *in vivo* and *in vitro* values reconstructed around a transformed-scale center, and a bounded Welsch penalty regularized their separation (Supplementary Tables 8 and 7). The joint fit therefore tested which parameter contrasts were repeatedly recovered across optimizer starts and landscape-informed C-family ensembles. These contrasts are conditional on the model, data streams, regularization, and shared *in vitro* anchor; they identify model-supported hypotheses rather than direct measurements of environmental causes.

Interpretive hierarchy. Because these analyses combine observations, fitted quantities, and post-fit diagnostics, we distinguished three evidentiary levels throughout the study. Experimental burden, chromosome-number, flow-density, and necrosis records are direct observations. Fitted trajectories, latent *in vivo* O_2 , and context-specific response functions are conditional model outputs constrained by those observations. Fixed- O_2 dominant-eigenvector curves are post-fit asymptotic

diagnostics under standardized conditions and are therefore more extrapolative than fitted finite-time trajectories. Numerical optimizer seeds and C-family ensembles assess search dependence; they do not create additional biological replication or sampling-based confidence intervals.

This evidentiary hierarchy framed the central comparison: whether fitted differences between tumors and culture were associated with WGD-favoring or WGD-opposing regimes through resource-dependent proliferation, chromosome missegregation under stress, WGD generation, or ploidy-dependent survival of altered progeny.

Results

Lineage-matched culture and tumor trajectories motivate a resource-stress model of ploidy evolution

Longitudinal chromosome-number measurements revealed complementary ploidy remodeling in culture under oxygen deprivation: the 2N-starting cohort acquired a high-chromosome component, whereas the 4N-starting cohort shifted downward (Figure 1B). The tumor measurements complemented these trajectories by pairing longitudinal burden with chromosome-number and histology/necrosis measurements at harvest; the chromosome-number data therefore compare the starting cell-line references with distinct terminal tumor samples rather than tracking individual tumors longitudinally (Figure 1D). Shared near-diploid and near-tetraploid SUM159 lineages nonetheless placed the two settings on a common axis for fitted context comparisons (Figure 1A; Supplementary Table 1). This lineage match supports comparison across settings but does not make the culture and tumor experiments direct biological replicates.

These complementary trajectories motivated a model in which resource limitation can both oppose and promote high-ploidy states. Resource stress imposes growth and death costs on chromosome-rich states while increasing stress-linked chromosome missegregation; a constant per-division WGD branch provides a separate route into high-ploidy states, and high ploidy can then buffer some missegregation events that would be lethal or strongly deleterious near diploidy (Figure 2). Under this formulation, the resulting ploidy distribution reflects the balance among growth suppression, stress-associated death, CIN generation, the constant-probability WGD route, and post-missegregation survival filtering.

In vitro ploidy evolution under hypoxia occurs with little death attributed by the model to resource stress

Model behavior under oxygen deprivation was first assessed against the culture data. Separate *in vitro* fits indicated that oxygen stress could reshape ploidy without a large fitted contribution from the model-attributed resource-stress death compartment. Passage-growth observations shifted to a lower regime under oxygen deprivation in both starting-ploidy cohorts, and the fitted trajectories reproduced this cohort-level shift while not matching every passage-level observation (Figure 3A; Supplementary Table 1). Observed flow-density profiles contained low- and high-DNA-content components under both control and deprivation conditions, with pronounced changes in their relative weights during oxygen deprivation; the fitted model qualitatively reproduced this distributional remodeling across sampled passages (Figure 3B). The corresponding inferred chromosome-state distribution developed a broad high-chromosome mode while retaining a lower-chromosome mode

(Figure 3C). Together, these fits support a shared process that can generate and expand high-chromosome states under oxygen deprivation.

Partitioning the fitted nonviable burden distinguished death assigned by the model to resource stress from nonviable products of chromosome-number transitions. Post-misseggregation survival increased with chromosome number, allowing some chromosome-loss daughters from high-ploidy mothers to remain viable (Figure 3D–E). This mechanism reconciled the complementary severe-deprivation trajectories: the 2N lineage generated higher-chromosome descendants, whereas the 4N lineage shifted downward. Across the best-fitting solutions, the 4N mean fell by about four chromosomes, yet death assigned to resource stress never exceeded 1.9% of total burden (Figure 3F; Supplementary Table 6). The numerical searches converged repeatedly near parameter boundaries, so they establish neither posterior uncertainty nor a global optimum (Figure 3G). Within those limits, the culture fits favor redistribution through viable descendants over preferential loss of 4N cells through the model’s resource-stress death pathway.

In vivo fits capture cohort-scale behavior while retaining multiple compatible parameterizations

Tumor growth and terminal composition provided a complementary *in vivo* test of the same mechanistic structure. The near-equivalent best fits reproduced the broad burden trajectories and terminal mean chromosome numbers more closely than the full shapes of the terminal chromosome distributions (Supplementary Table 2). They also recovered the cohort-level scale of terminal necrosis, but not uniformly accurate trajectories for individual tumors. The dynamic parameters were shared within lineage cohorts rather than fitted as mouse-specific latent effects, so the supported result is cohort-scale behavior rather than mouse-level reconstruction.

Among these solutions, the lowest-objective separate fit (seed 25) illustrates the tumor dynamics reproduced by the model. It captured the cohort-scale increase in tumor burden while resolving markedly different ploidy trajectories for the two starting states (Figure 4A). The 2N-initiated cohort remained close to diploid throughout tumor growth, whereas the 4N-initiated cohort underwent a pronounced downward shift from near-tetraploid toward a state only slightly above diploid. In both cohorts, this remodeling occurred as fitted effective O_2 declined with increasing burden and modeled dead material accumulated alongside the viable compartment. CIN-associated dead burden constituted the largest modeled dead component by the end of both cohort trajectories, while the fitted contribution assigned to death associated with resource stress was greater in the 4N-initiated cohort. At harvest, the predicted distributions preserved the distinct central ploidy of the observed 2N and 4N cohorts, but were broader than the observed distributions. The representative fit therefore captured the direction and cohort-level endpoint of ploidy remodeling together with tumor growth, but its broader predicted terminal distributions and model-derived intermediate states preclude interpretation as precise mouse-level trajectory recovery (Supplementary Table 2).

Near-equivalent objectives nevertheless arose from different parameter combinations, motivating an ensemble-level analysis of the fitted ploidy response. Across fixed oxygen conditions, baseline chromosome-misseggregation probability was the only fitted parameter that remained positively associated with dominant ploidy throughout the evaluated range. Maximum division rate was associated with higher ploidy chiefly at very low oxygen, whereas the strongest negative associations involved stress-associated death and post-misseggregation viability loss (Figure 4B; Supplementary Table 3). Peak absolute correlations approached 0.69, but both their magnitude and, for several

parameters, their sign changed with oxygen. The heatmap therefore identifies oxygen-specific directions of fitted covariation rather than a fixed hierarchy of causal parameter effects.

Parameter variation was structured across the fitted landscape rather than concentrated around a single solution. Three exploratory groups separated mainly along axes governing stress-associated death, oxygen supply, proliferation, and chromosome missegregation (Figure 4C–D; Supplementary Figure 1; Supplementary Table 4). Because these groups were defined from the same fitted parameters being compared, their separation describes the internal organization of the optimizer landscape rather than biological subtypes or independent statistical evidence. Wide parameter ranges and recurrent boundary solutions reinforce the same conclusion: the data support several mechanistic parameterizations, not one uniquely resolved vector.

Joint fitting yields concordant function-level context contrasts across warm-start-conditioned search ensembles

Joint soft-coupled fitting enabled direct comparison of the culture and tumor parameterizations (Figure 5A). We focused on complete effective missegregation and post-missegregation survival functions because their scalar components act jointly. The pooled warm-start landscape defined three numerical-search regions, C01, C02, and C03. One *in vivo* representative from each region was paired with the common lowest-objective *in vitro* fit, and each configuration initiated an independent joint-search ensemble (Figure 5C; Supplementary Table 7). These labels identify search coverage rather than experimental groups or biological subtypes. All cross-context mechanistic parameters remained context specific but were regularized around a shared transformed-scale center by a bounded Welsch penalty. We therefore summarized repeated optimizer convergence within each C family and compared only directions reproduced across C01–C03.

The coverage audit supported the three C regions as a practical initialization design, but not as a unique partition of the fitted landscape. A four-region solution was nearly equivalent and introduced a singleton, while clustering in the original parameter space agreed poorly with the embedding-derived regions (Supplementary Figure 6A; Supplementary Table 7). C01–C03 therefore diversify the numerical search without defining biological subtypes.

Within this defined search scope, the selected joint fits retained the broad tumor-burden and terminal chromosome-number scales together with the low- and high-growth regimes in culture. Residual scatter remained, and several high-chromosome culture endpoints were underpredicted (Figure 5B). The fits therefore supported comparison of complete response functions without implying uniformly accurate prediction of every observation (Supplementary Tables 1 and 7).

Context contrasts were summarized at the C-family level rather than treating repeated optimizer starts as independent evidence (Supplementary Table 7). This retained variation across the three warm-start regions without converting numerical search multiplicity into biological replication.

The endpoint contrasts revealed a broadly conserved directional pattern, but not a common effect size, across the three search families (Figure 5D; Supplementary Figure 3; Supplementary Table 8). In all three families, tumor fits assigned lower values to the maximal division rate, the critical oxygen threshold, and the viability-loss coefficient governing post-missegregation survival, β_{buf} . Conversely, parameters governing resource-stress growth suppression, stress-associated death, stress-induced missegregation, and the survival-response exponent n_{exp} were higher in tumors. Maximum post-missegregation survival remained within the declared approximate-equality band. Three parameters did not follow a uniform context ordering. Oxygen-response steepness

and baseline missegregation were close to equality in some families but higher in tumors in others, whereas WGD probability was lower in tumors in C01 and C02 but higher in C03. The aligned log-ratio distributions further showed that a repeated direction did not imply a shared contrast magnitude. Several endpoint distributions were tightly concentrated within a family yet separated substantially among families, most prominently for baseline missegregation. Because the three families shared one culture anchor, their agreement evaluates sensitivity to the tumor warm start rather than replication across independent experiments (Supplementary Table 7).

Comparison with the reconstructed differential-evolution initial populations showed that the optimizer endpoint distributions were generally much narrower, indicating repeated numerical convergence under the tested design (Figure 5D; Supplementary Table 8). Recurrent active bounds, however, show that this concentration does not establish structural identifiability.

Scalar shifts alone do not determine model behavior because several parameters combine non-linearly within each response function. We therefore used the scalar results to identify candidate mechanisms and evaluated their complete functions before drawing context-level conclusions.

The complete effective per-chromosome missegregation function separated the contexts most clearly under hypoxia (Figure 5E). At severe and intermediate hypoxia, all three tumor families had higher fitted probabilities than culture at both reference ploidies, with the largest contrast approaching 21-fold. Under well-oxygenated conditions, the difference became ploidy and family dependent (Supplementary Table 16). The tumor-associated increase is therefore a hypoxia-linked functional contrast rather than a uniform offset in chromosome-error probability.

Post-missegregation survival showed a related but distinct context dependence (Figure 5F). Altered daughters had higher absolute fitted survival in tumors at both reference ploidies. Culture, however, showed the steeper ploidy-dependent increase: survival rose by about 0.63 between the diploid and tetraploid reference states, compared with no more than 0.19 in tumors (Supplementary Table 16). Resource context therefore separated the overall retention of altered daughters from the degree to which retention depended on ploidy.

The optimization design constrains the precision of these contrasts. The Welsch penalty saturates for large context separations, and every retained ensemble contained active parameter bounds (Supplementary Table 7). The repeated directions therefore define model-supported hypotheses across C01–C03, not unique context effect sizes or proof of a global optimum.

Oxygen and chromosome-error input distinguish in vivo and in vitro ploidy responses

The separate fits first asked whether parameterizations compatible with the observed endpoints implied a common asymptotic oxygen response. They did not. Tumor fits were dominated by complex nonmonotone curves, whereas culture contained more approximately flat and simpler non-monotone responses (Supplementary Figure 5A–B; Supplementary Table 9). This class difference alone was not decisive because spectral separation was weak for many curves, particularly in culture. We therefore treated the classifications as evidence of response multiplicity and tested the joint ensembles directly.

To avoid relying on a single numerical winner, we evaluated the lowest-objective joint endpoints from C01, C02, and C03 as matched context ensembles (Figure 6A). All retained endpoints passed the prespecified feasibility and operator checks, and the same conclusions were recovered across

broader objective cutoffs and after collapsing exact duplicate parameter vectors (Supplementary Figure 6B–C; Supplementary Tables 7 and 10).

The resulting surfaces exposed different oxygen–CIN–ploidy landscapes rather than a shared response shifted by one parameter offset (Figure 6A). The three culture surfaces were almost super-imposable, with 99.99% agreement in low, intermediate, or high ploidy class across C01–C03. They separated directly into low- and high-ploidy domains, with no intermediate median state at the evaluated resolution. Tumor surfaces instead varied among families and were dominated by broad intermediate-ploidy regions (Supplementary Table 11). Thus, changing oxygen at fixed $p_{\text{miss,eff}}$ produced a switch-like boundary in culture but a graded set of intermediate states in tumors. The fitted culture trajectories remained on the low- $p_{\text{miss,eff}}$ side of that boundary, whereas tumor trajectories traversed several ploidy domains. Because both contexts contained weak-gap regions, we next tested whether this topological contrast survived endpoint-level uncertainty in the dominant mode.

The weak-gap analysis preserved the qualitative regimes in both contexts (Supplementary Figure 7; Supplementary Table 13). More than 99.9% of weak-gap cells in every context-family panel retained strong endpoint agreement on whether dominant ploidy was low, intermediate, or high. Most cells also showed little spread across endpoints, although isolated boundary locations remained locally sensitive. A small spectral gap therefore signaled slow asymptotic mode separation more often than uncertainty about the ploidy regime itself. This robustness supports the culture switch-like pattern and the broader tumor intermediate states as stable features of the sampled ensembles.

Holding effective per-chromosome missegregation fixed clarified how the two landscapes generated these patterns. Dominant ploidy increased with oxygen across all three tumor families at each diagnostic input, with the largest response approaching five ploidy units. Culture was nearly insensitive at low and intermediate inputs and declined at the high input (Figure 6A; Supplementary Table 10). Restoring the fitted missegregation functions produced a different trajectory: culture remained nearly flat within its low- $p_{\text{miss,eff}}$ band, whereas tumors crossed multiple ploidy domains in a family-dependent direction. Together with the low culture missegregation function in Figure 5E and its steeper survival gradient in Figure 5F, this contrast points to different fitted routes of ploidy change. Differential retention of altered chromosome states is the more prominent filter in culture, whereas oxygen-linked chromosome-error generation contributes more strongly in tumors (Supplementary Tables 10 and 16). This mechanistic division is model based, but its direction was stable across the sampled objective thresholds and endpoint-weighting schemes.

The dense fixed-input curves revealed a second interaction within the tumor response (Figure 6B). At low $p_{\text{miss,eff}}$, dominant ploidy increased monotonically with oxygen. At higher inputs, the response became biphasic: ploidy first rose and then declined, and the downturn shifted to higher oxygen as $p_{\text{miss,eff}}$ increased (Supplementary Table 12). Oxygen and chromosome-error input therefore acted as coordinated determinants of ploidy rather than independent additive effects. This pattern is directionally consistent with the resource-dependent GBM simulations of Nowicka et al., in which high-ploidy clones were depleted in a low-oxygen brain region but persisted in a better-oxygenated region, and with their positive cross-tissue association between tissue oxygenation and cancer ploidy.⁴

The inverse maps posed the reciprocal question: how must $p_{\text{miss,eff}}$ change with oxygen to maintain a given ploidy (Figure 6B)? Culture produced nearly identical inverse domains across C01–C03,

and the effective missegregation input associated with ploidy 4 decreased as oxygen rose. Tumor maps were less uniform: their supported domains varied among families, and where a unique ploidy-4 solution existed, the required input increased with oxygen (Supplementary Table 10). Large unsupported regions in tumors further showed that oxygen and a single missegregation input were often insufficient to specify one asymptotic ploidy state. The inverse analysis therefore reinforces the forward-map result: culture follows a consistent trade-off between oxygen and chromosome-error input, whereas the tumor response is more heterogeneous and cannot generally be reduced to a single oxygen-dependent $p_{\text{miss,eff}}$ trajectory.

Importantly, this context contrast was not created by one response class simply fitting one dataset better. Response classes overlapped in both the pooled parameter embedding and relative fit quality, and no class uniquely contained the best solutions in both contexts (Supplementary Figure 5C–D; Supplementary Tables 9 and 4). The distinct forward and inverse maps instead emerged from how multiple compatible parameterizations were organized within each context. Together, these analyses support different oxygen–ploidy response architectures in culture and tumors rather than a universal response curve selected by fit quality alone.

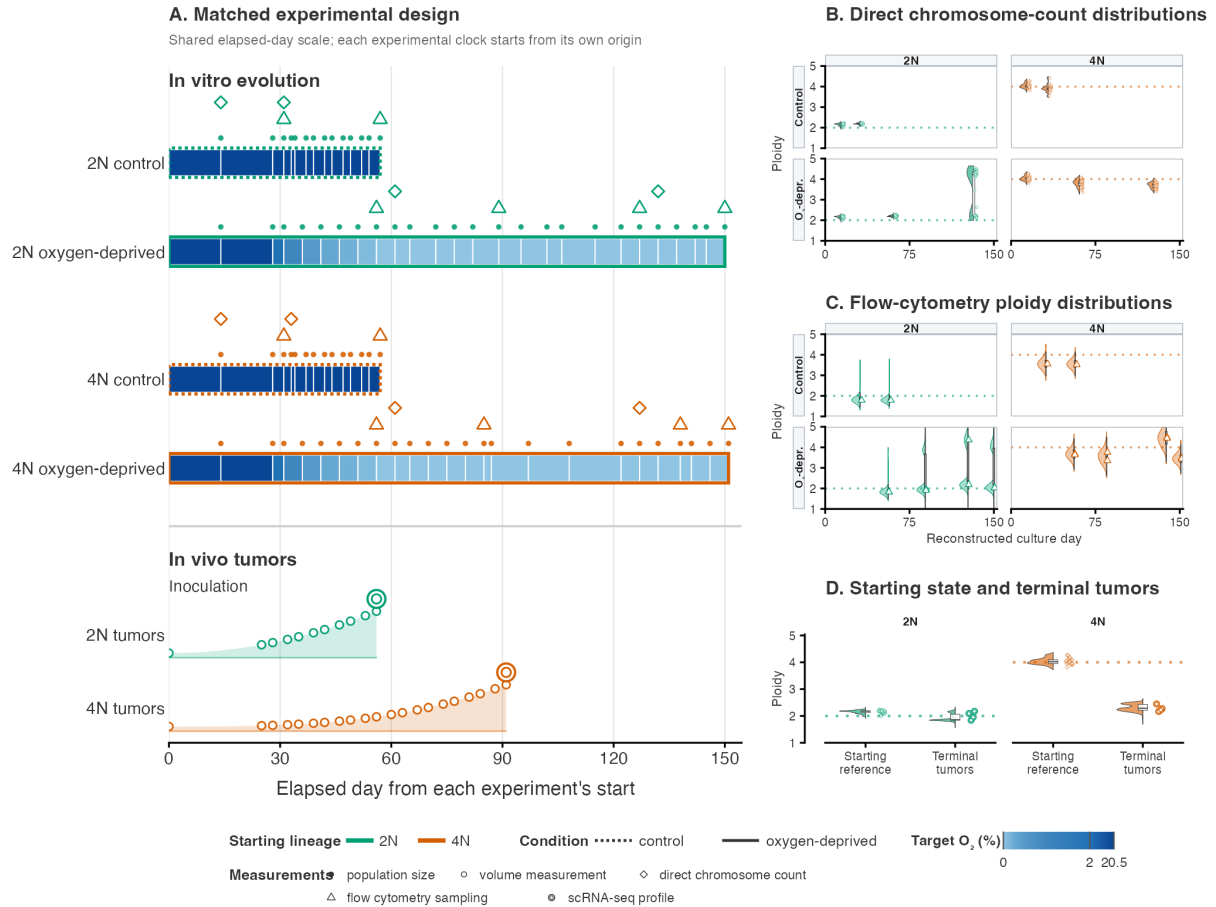


Figure 1: Matched experimental systems and ploidy readouts.

Green and orange identify the 2N- and 4N-initiated cohorts throughout; blue is reserved for target O_2 . (A) The culture and tumor designs are placed on the same 0–150-day elapsed-time axis,

although each experiment begins at its own origin and the two clocks do not denote synchronized biological states. Each O₂-colored culture block represents one reconstructed passage; the display scale expands 0–2% O₂ and compresses 2–20.5% O₂ so that the low-O₂ steps remain distinguishable, while the labels report the actual target concentrations. Dotted and solid cohort-colored outlines distinguish control and oxygen-deprived trajectories. Filled circles mark population-size measurements at every displayed passage endpoint, with shared pre-branch measurements repeated on the two aligned eventual branches only for visual comparison. Open triangles and diamonds mark flow-cytometry and direct chromosome-count sampling, respectively. In tumors, open circles mark volume-measurement days on schematic nonlinear growth wedges; wedge height is not a measured burden value. Double concentric circles mark single-cell RNA-sequencing-derived chromosome profiles; only the final 4N profile is displayed. Symbol height offsets in the culture tracks separate coincident assays and do not encode measurement magnitude. **(B)** Direct chromosome-count measurements are shown as an explicit condition-by-starting-ploidy grid, with control above oxygen-deprived cultures and 2N beside 4N. Counts were converted to ploidy by division by 22. At each sampled day, a left half-violin shows the distribution of directly counted cells, the central box shows the median and interquartile range, and small open diamonds to the right show the individual cell measurements. Horizontal offsets separate the three display layers and do not change sampling time. The shared directly measured starting distribution is repeated at the beginning of both condition rows for alignment and is not counted as an independent observation. **(C)** Observed flow-cytometry profiles use the same condition-by-starting-ploidy grid and reconstructed culture-day axis as panel B. At each sampled day, the cohort-, condition-, and day-specific G0/G1 density is shown as a left half-violin with a density-weighted box summary; parallel O1 and O2 profiles sampled on the same day are pooled with equal sample weight. Open triangles, matching the flow-cytometry sampling symbol in panel A, show the global density peak of each individual profile. Panels B and C use a common ploidy range of 1–5; the complete flow profiles were retained for all summary calculations, so upper tails beyond the displayed range are cropped rather than renormalized. The smallest density mass retained within the displayed range was 85.43%. **(D)** Direct starting-cell karyotypes and terminal tumor-cell chromosome estimates are displayed on the same ploidy range as panels B and C. Half-violins contain 40 direct starting-cell measurements (20 per starting cohort) and 6,502 terminal single-cell RNA-sequencing-derived chromosome estimates from eight tumors (four per cohort); boxes show medians and interquartile ranges. Small open diamonds identify individual starting cells, whereas double concentric circles identify one median per terminal tumor. Starting and terminal measurements are distinct samples rather than longitudinal measurements of the same tumor (Supplementary Table 1).

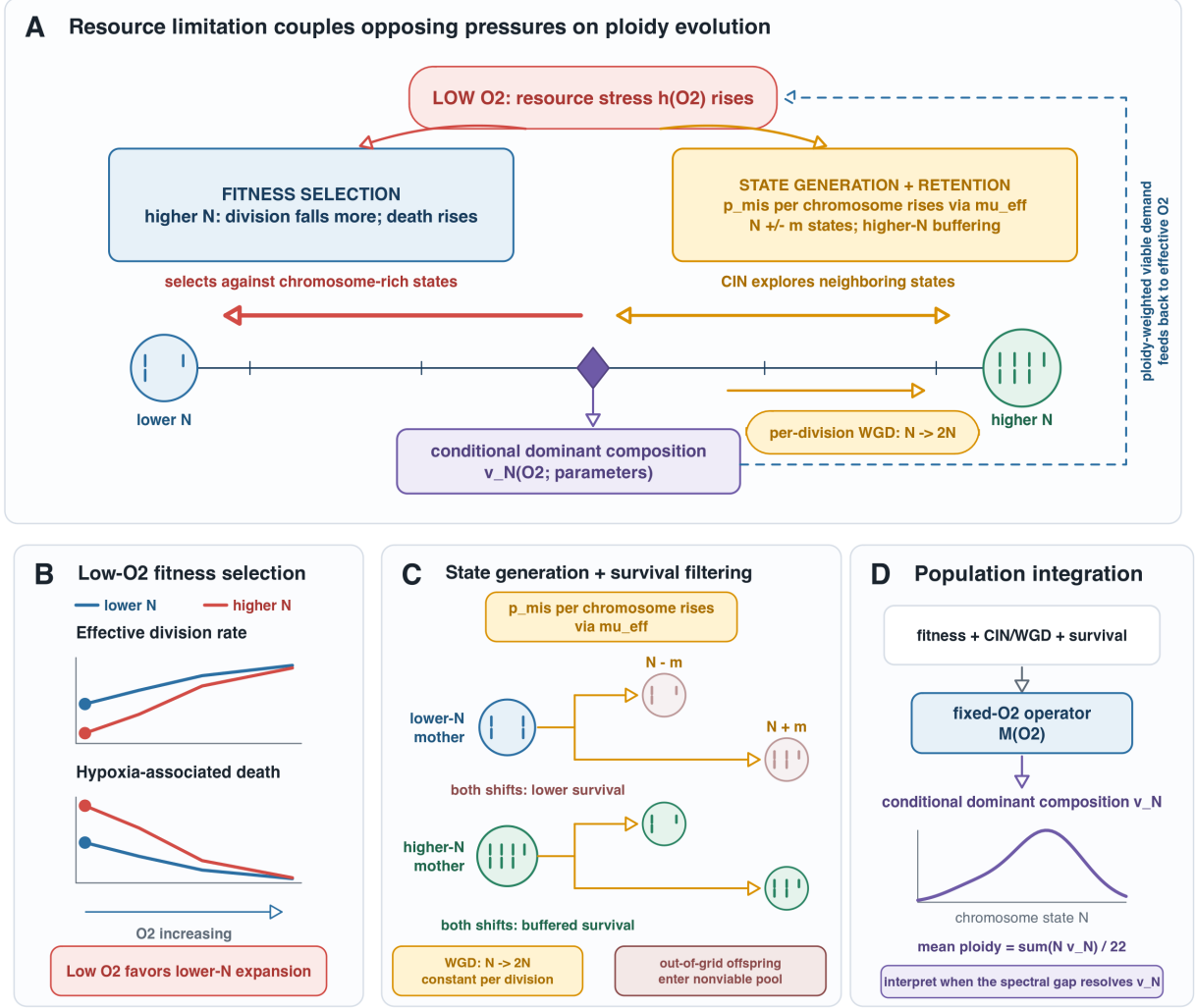


Figure 2: **Model overview: resource limitation can impose opposing pressures on ploidy evolution.**

(A) Low oxygen raises bounded resource stress and simultaneously strengthens selection against chromosome-rich states while increasing the per-chromosome missegregation probability that supplies chromosome-state variation. In the dynamic model, viable burden and ploidy-weighted oxygen demand close the feedback loop; the resulting composition depends jointly on oxygen and the fitted model parameters. (B) Under the ploidy-related stress mode, low oxygen suppresses effective division and increases the modeled death hazard associated with resource stress more strongly in higher-chromosome-count states, creating a relative expansion advantage for lower- N states. (C) The per-chromosome missegregation probability rises through the stress-linked death hazard and generates symmetric $N - m$ and $N + m$ daughter branches. For a given $|m|$, altered daughters from higher- N mothers have greater post-missegregation survival. Constant per-division WGD provides a separate tracked-lineage transition $N \rightarrow 2N$, and out-of-grid offspring enter the CIN-associated nonviable pool. (D) Growth, death, CIN/WGD transitions, and post-missegregation survival define the fixed- O_2 population operator. When its dominant mode is sufficiently separated, the

normalized dominant eigenvector gives the conditional asymptotic chromosome-state composition and mean ploidy.

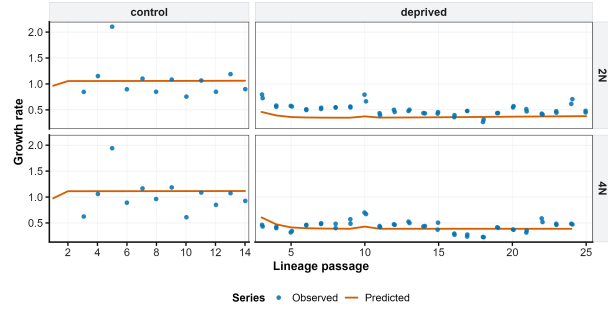
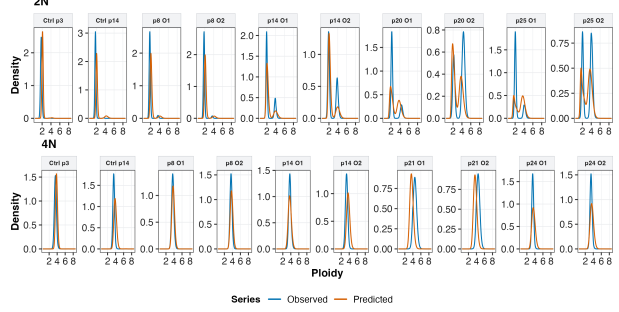
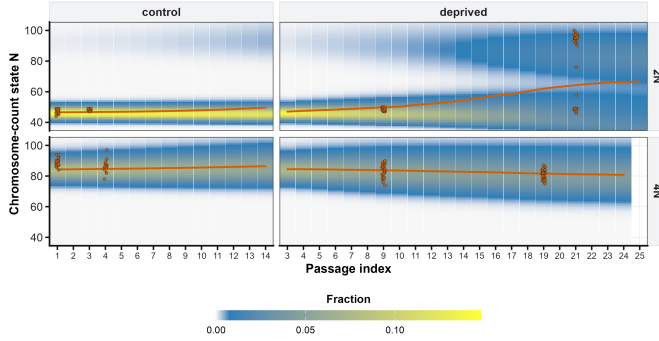
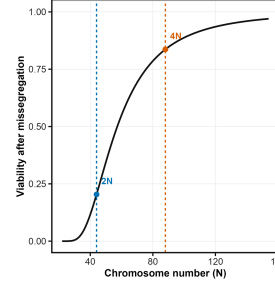
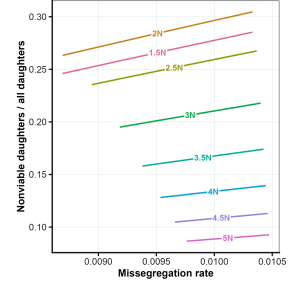
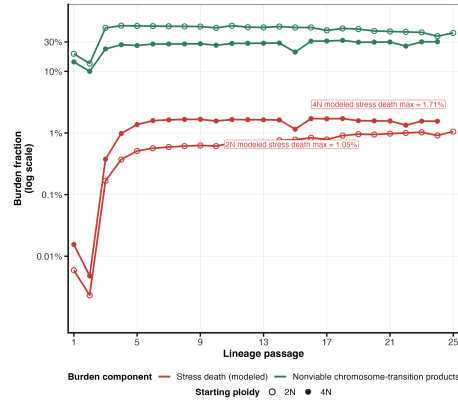
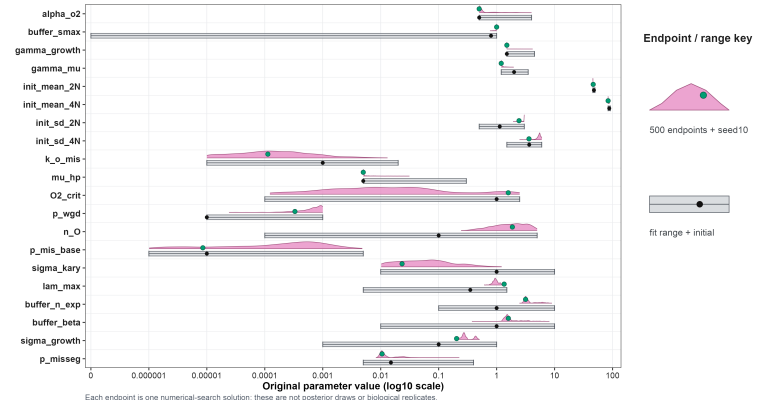
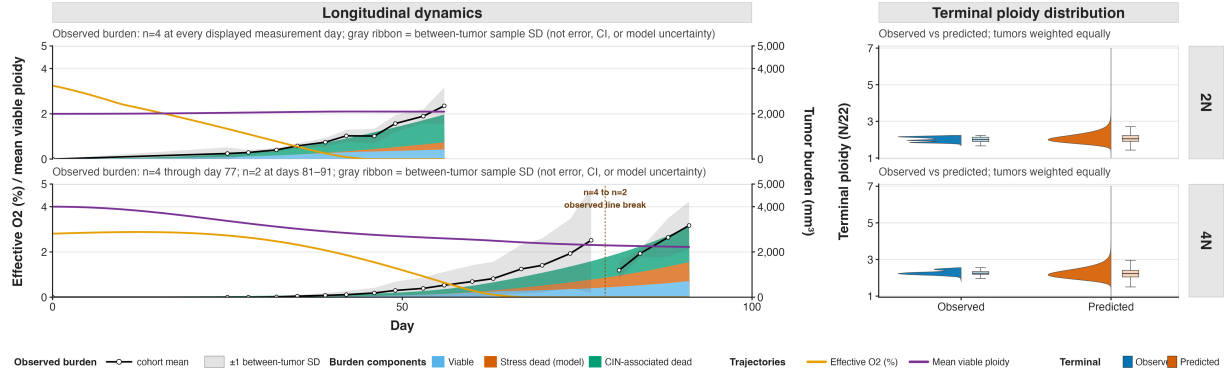
A. Observed versus predicted passage growth rates**B. Observed versus predicted G0/G1 ploidy-density curves****C. Predicted ploidy distribution across passages****D. Post-misseggregation survival****E. Nonviable daughter fraction****F. Severe-deprivation burden trajectories****G. Fitted parameter endpoints across 500 starts**

Figure 3: In vitro ploidy reshaping occurs with little death attributed by the model to resource stress.

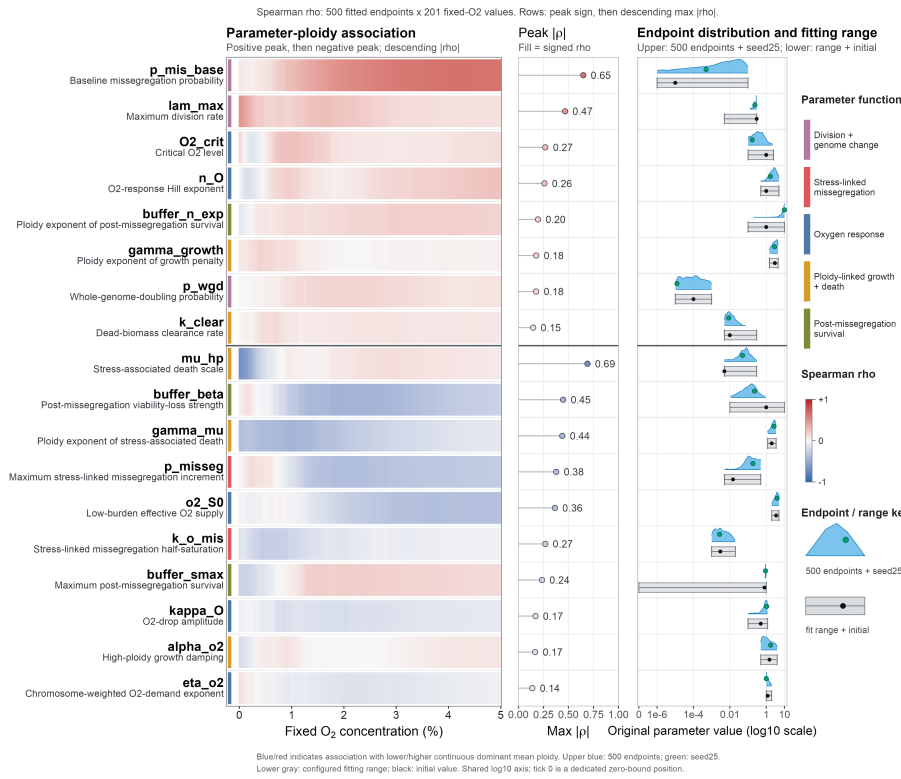
(A) Observed (blue points) and fitted (orange line) passage growth rates from the lowest-objective retained separate *in vitro* fit (seed 10), arranged as 2N/4N rows and control/deprived columns. The fitted trajectories reproduce the lower growth-rate regime under oxygen deprivation while individual passage measurements remain variable (Supplementary Table 1). (B) Observed (blue) and fitted (orange) G0/G1 DNA-ploidy density curves from the same fit. Parallel O1 and O2 measurements at the same deprived passage are displayed in separate sample-level subpanels rather than connected or overlaid as one observed curve; the corresponding shared prediction is repeated in each subpanel. Both sources are drawn as thin solid lines, with the fitted curve layered above the observed curve where they overlap. High-DNA-content components are present under both control and deprivation conditions, and their relative peak weights change prominently during deprivation.

(C) The predicted chromosome-state distribution across passages shows the oxygen-deprived 2N lineage developing a high-chromosome component while retaining a lower-chromosome mode. The orange line gives the fitted mean chromosome number and the orange points with dark outlines give direct single-cell karyotypes; these measurements occur only at sampled passages, whereas the intervening heatmap is model inferred. The displayed chromosome range spans the pooled weighted predicted and observed 1st–99th percentiles with a two-chromosome margin. (D) The fitted post-missegregation survival function increases with chromosome number. The black curve is evaluated at the diploid $N = 44$ and tetraploid $N = 88$ reference states, highlighted by the blue 2N and orange 4N dashed lines and their same-color curve intersections. (E) The model-implied nonviable daughter fraction plotted against missegregation rate shows how chromosome-loss opportunity and ploidy-dependent survival jointly determine which chromosome-segregation-error products remain viable across directly labeled reference ploidy states. (F) Passage-end burden predictions for the terminal severe-deprivation 2N and 4N lineages are overlaid in one coordinate system to permit direct comparison. Color distinguishes burden assigned by the model to death associated with resource stress (red) from nonviable chromosome-transition products (green), while open circles denote the 2N lineage and solid circles denote the 4N lineage. The maximum fitted contribution of this resource stress compartment is 1.05% and 1.71% of total burden for the 2N and 4N lineages, respectively (Supplementary Table 6). The larger dead-buffer component records nonviable chromosome-transition products, including out-of-grid boundary routing. Neither compartment is a direct observation of the biological cause of death. A $\log_{10}$ burden scale is used because the two components span approximately four orders of magnitude. (G) Fitted-parameter ranges and numerical-search endpoint distributions for the separate *in vitro* calibration. For each of the 20 fitted parameters, the upper pink half-violin shows the distribution of the best retained parameter value from each of the 500 numerical seeds, and the green point marks seed 10, the lowest-objective retained fit used in panels A–F. The lower gray box-shaped range glyph spans the configured lower and upper fitting bounds and the black point marks the configured initial value; this glyph describes the admissible fitting interval and is not an empirical quartile summary. The horizontal axis reports original parameter values on a $\log_{10}$ scale, with a dedicated plotting floor labeled zero for a non-positive configured bound. These optimizer-derived endpoints assess numerical solution spread; they are not posterior draws, confidence or credible intervals, biological replicates, or proof of a global optimum (Supplementary Table 6).

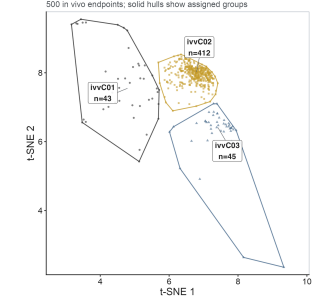
A. In vivo cohort dynamics and terminal ploidy



B. Oxygen-dependent parameter-ploidy landscape



C. Exploratory fitted-endpoint landscape



D. Top six cluster-separating fitted parameters

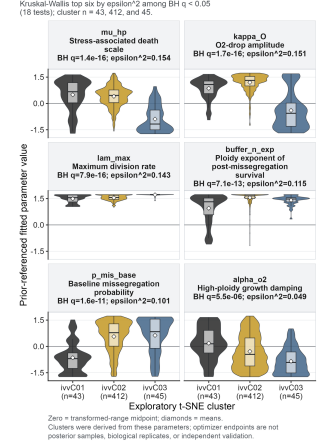


Figure 4: Separate in vivo fits capture burden, resource, and ploidy dynamics while retaining multiple resource-response parameterizations.

(A) Cohort-level summary of the lowest-objective retained separate fit (seed 25). The left column shows 2N-initiated and 4N-initiated longitudinal dynamics on a common 0–100-day axis. Stacked areas, averaged over the tumor trajectories represented at each day, are read from the right axis and show predicted viable burden, burden assigned by the model to death associated with resource stress, and CIN-associated dead burden. Black points and curves show observed cohort means at recorded measurement days. The gray ribbon is the cohort mean plus or minus one between-tumor sample SD; it is not a measurement-error bound, confidence interval, or model-uncertainty interval. Orange and purple curves show cohort-mean fitted effective O₂ and viable-cell mean ploidy, respectively, on the shared numerical 0–5 left axis. Mean ploidy is viable-cell weighted mean

chromosome number divided by $N_{\text{unit}} = 22$; sharing this numerical axis does not equate the physical units of O_2 and ploidy. The right column compares terminal observed and predicted chromosome-number distributions after conversion to ploidy, using blue for observed and orange for predicted distributions. In each half-violin/box summary, the left half shows the weighted distributional shape and the box-and-whisker at right shows the weighted median and quartiles. Each tumor receives equal total weight within its cohort and source. Effective O_2 , burden compartments, and intervening ploidy paths are model predictions; chromosome-number profiles directly constrain only the harvest endpoints (Supplementary Tables 1 and 2). **(B)** Continuous oxygen-dependent association and fitted-landscape structure. The left heatmap shows Spearman ρ between each of 18 fitted *in vivo* parameters and continuous dominant mean ploidy across 201 oxygen concentrations from 0% to 5%. Red denotes association of higher fitted parameter values with higher ploidy, blue denotes association with lower ploidy, and white denotes $\rho \approx 0$. Rows are divided by the sign of ρ at the maximum absolute association: positive-peak rows are shown first and negative-peak rows second, with decreasing maximum $|\rho|$ within each group. In the adjacent dot plot, horizontal position and the printed value give this absolute maximum, whereas dot fill retains the corresponding signed ρ . For each parameter, the next column uses a shared $\log_{10}$ axis of original parameter values. The upper pink half violin shows the distribution of the best retained endpoint from each of 500 numerical seeds, and the green point marks seed 25, the lowest-objective retained fit. The lower gray bar spans the configured fitting interval and its black point marks the configured initial value; these are optimizer settings, not empirical intervals. The zero tick is a dedicated plotting position for a non-positive configured bound and does not assign zero a positive logarithmic value. Row-side colors identify the five parameter functions, including the standardized post-missegregation-survival labels. The endpoint densities are not posterior distributions, confidence intervals, or biological replicates. **(C)** Two-dimensional coordinates of the same 500 best endpoints from an *in vivo*-only t-SNE fitted to 127,500 retained initial-population rows plus those 500 endpoints. All 18 displayed parameters were inputs; ρ_{2N} and σ_{burden} were excluded. Solid convex hulls contain every endpoint assigned to each exploratory cluster and are descriptive membership outlines, not confidence regions or validated subtype boundaries. **(D)** Prior-referenced distributions for the six parameters with the largest Kruskal–Wallis epsilon-squared among parameters passing Benjamini–Hochberg $q < 0.05$ across 18 three-group comparisons. Facets are ordered by decreasing effect size; violin shapes show the fitted-endpoint distributions, boxes show medians and interquartile ranges, and diamonds show means. Exact adjusted q values and effect sizes are printed in the facet strips (Supplementary Tables 3 and 4). Because all 18 parameters contributed to the t-SNE and cluster assignments, panel D is a descriptive within-landscape stratification, not an independent hypothesis test, validated biological subtype analysis, or evidence of biological replication. Panels B–D describe optimizer-endpoint structure within the fitted-solution ensemble and do not establish causal parameter effects or independently evaluated prediction accuracy.

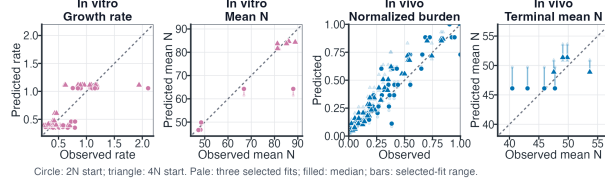
Joint-fit workflow, performance, mechanisms, and solution stability

Three primary-family pairs; Panel D compares pair-specific DE initial populations with 500 optimizer endpoints.

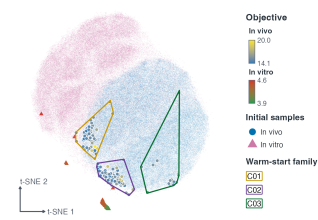
A. Joint-fitting workflow



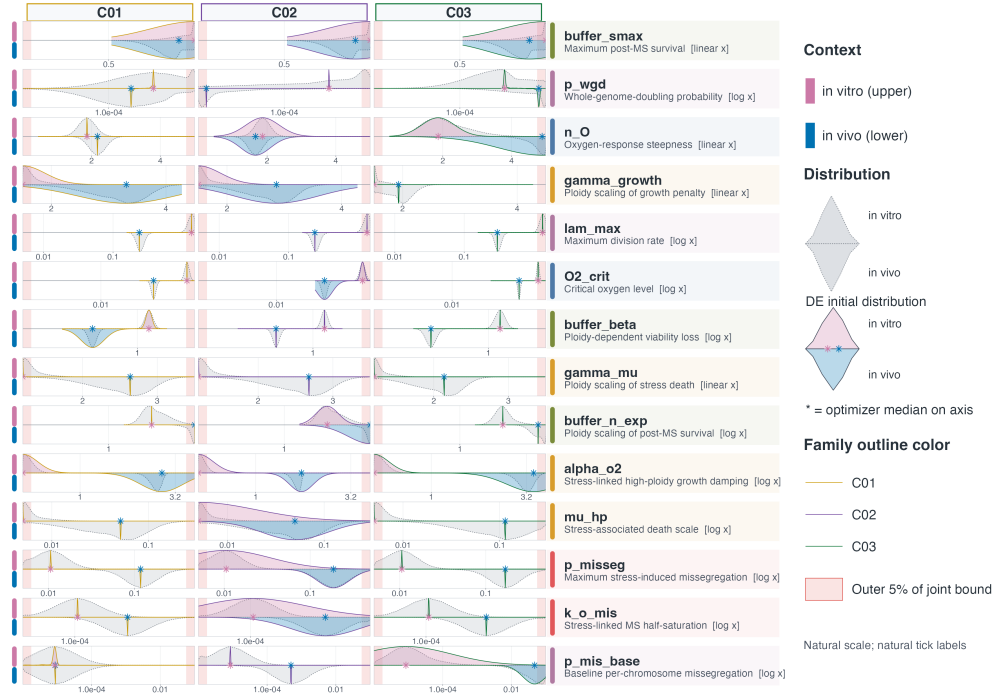
B. In-sample model performance



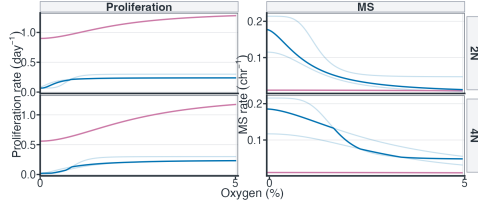
C. Warm-start map



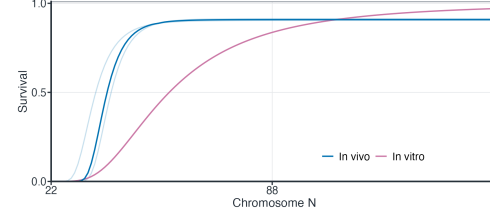
D. Context-specific DE initialization and optimizer endpoints



E. Oxygen-dependent fitted functions



F. Survival



B: circle = 2N start; triangle = 4N start. E-F: thin curves = three fits; thick curves = pointwise medians. D: upper = in vitro; lower = in vivo; gray fill/dashes = DE initialization; pink/blue fill with C-family solid outlines = optimizer endpoints. Red background = outer 5% of the complete joint bound. C01/C02/C03 retain the same family identities across panels; optimizer distributions are descriptive numerical-search summaries, not posterior or confidence distributions.

Figure 5: Joint-fit workflow, model performance, fitted mechanisms, and optimizer-solution stability.

(A) Joint-fitting workflow. Separate 500-seed *in vitro* and *in vivo* searches form two input lanes. One objective-minimum *in vivo* representative from each C01–C03 region is paired with the common

seed-10 *in vitro* anchor. Each C-family configuration enters the same chromosome-state model with 14 paired context-specific parameters linked by the bounded Welsch soft-coupling penalty. Five hundred numerical starts per family give 1,500 joint searches, and the lowest retained joint objective within each family supplies the three winners summarized in panels B, E, and F. Panel D uses all 500 feasible, unprojected endpoints from each family and its corresponding reconstructed DE initial populations (Supplementary Table 7). **(B)** Direct in-sample observed-versus-predicted summaries across the three C-family winners, arranged as one row of two *in vitro* followed by two *in vivo* comparisons. Circles identify 2N-initiated observations and triangles identify 4N-initiated observations; blue and pink retain the *in vivo*/*in vitro* context encoding. Pale points show individual winner predictions, filled points show their medians, and vertical ranges show the three-winner span where displayed (Supplementary Tables 1 and 7). These comparisons assess fitted-data agreement and are not held-out validation. **(C)** Pooled separate-fit t-SNE landscape used to distribute joint warm starts. Initial samples are blue circles for *in vivo* and pink triangles for *in vitro*; best-fit solutions retain their separate component-objective color scales. Solid outlines delimit C01, C02, and C03 using the same gold, purple, and green identities used in panel D. *In vitro* regions are not drawn because all joint searches use the same seed-10 culture anchor. The outlined regions are initialization inputs, not optimized joint endpoints, confidence regions, or biological subtypes. **(D)** Context-specific differential-evolution initial-population and optimizer-endpoint distributions for all 14 coupled parameters. Rows are ordered by the absolute distance from 1 of the arithmetic mean of the three best-winner *in vivo*/*in vitro* ratios, so parameters with the closest average agreement across C01–C03 appear first; columns denote C01, C02, and C03. Within every family subpanel, the upper half represents the *in vitro* parameter distribution and the lower half represents the *in vivo* distribution; narrow matching color strips outside the left edge of each plotting row identify these contexts. Parameter names and compact functional descriptions appear in aligned blocks outside the right edge, with adjacent equally narrow colors identifying functional groups. Gray-filled densities with gray dashed outlines show 200,000 initial values reconstructed from the selected family’s 500 DE runs (400 members per run). Pink upper and blue lower fills, bounded by solid dark-yellow, purple, or dark-green outlines, show the corresponding 500 optimizer endpoints for C01, C02, or C03. Context-colored asterisks on the shared central coordinate axis mark the context-specific endpoint medians (pink for *in vitro*, blue for *in vivo*); distribution-specific percentile spans are not drawn. The translucent red background at each horizontal extreme marks the lower 5% and upper 5% of the complete joint-union fitting interval for that parameter. Values are shown on the natural parameter scale; parameters optimized in $\log_{10}$ space use logarithmic horizontal axes labeled with natural values. Within a parameter, both contexts, both numerical roles, and all three families share the complete joint-union range. Densities are peak-normalized separately, so horizontal location, width, and shape can be compared but vertical peak heights cannot. Horizontal scales differ among parameters. The endpoint distributions are numerical-search summaries, not Bayesian posteriors or confidence intervals, and their concentration does not establish structural identifiability (Supplementary Table 8). **(E)** Joint comparison of fitted proliferation and complete stress-linked per-chromosome missegregation functions. Columns denote mechanisms and rows denote the 2N and 4N reference states. Curves are evaluated over the same 0–5% O₂ grid. **(F)** Complete fitted post-missegregation survival function across chromosome number. In panels E–F, thin curves show the three C-family winners and thick curves show pointwise medians; *in vivo* is blue and *in vitro* is pink (Supplementary Table 16).

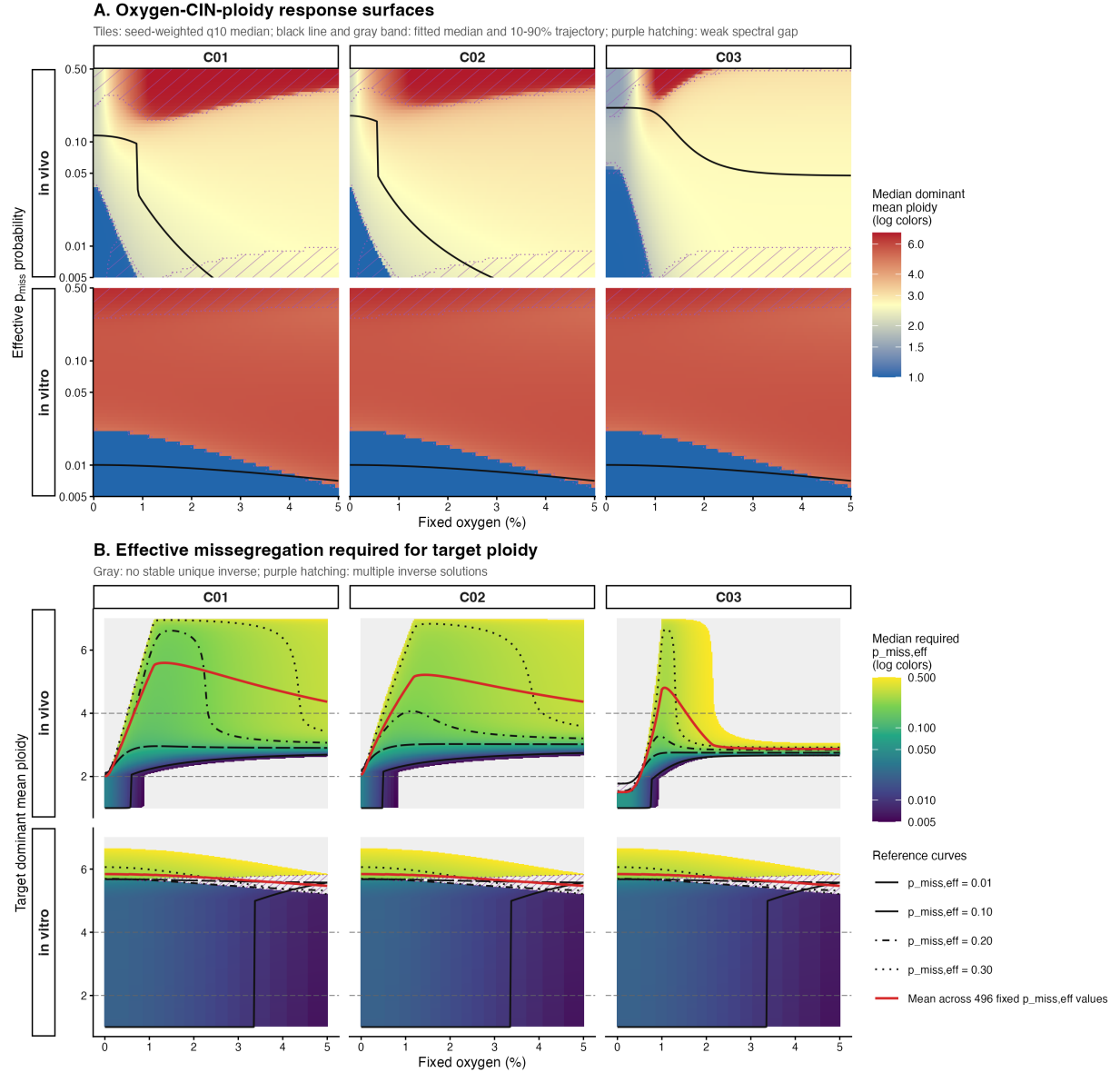


Figure 6: **Paired joint-fit context vectors yield context- and family-dependent oxygen–CIN–ploidy responses.**

(A) Joint-fit oxygen–CIN–ploidy response surfaces. Columns show C01, C02, and C03; rows show the context-specific *in vivo* and *in vitro* vectors from the same retained joint endpoints. Within each context–family panel, the complete grid of 201 oxygen concentrations and 60 forced effective per-chromosome missegregation probabilities was evaluated for the 50 joint endpoints in the family-specific lowest-objective decile. Fill reports seed-weighted median dominant mean ploidy in its original units; the color mapping is logarithmic and progresses from blue at low ploidy through yellow to red at high ploidy. Black curves show the pointwise median model-implied population-average missegregation probability along the unmodified context-specific fixed- O_2 trajectories, and gray ribbons show their 10th–90th percentile intervals. Purple diagonal hatching and dotted boundaries identify grid cells where at least half of eligible endpoint vectors have spectral gap < 0.005 ;

white crosses mark sampled positions where fewer than 80% agree on whether dominant mean ploidy is at least 2. Exact duplicate 14-parameter vectors were evaluated once within each context and restored to their original optimizer-seed multiplicities before aggregation. These surfaces summarize numerical fit multiplicity in fitted context-specific parameterizations; they are not independently fitted CIN measurements, causal oxygen-by-CIN interventions, or biological confidence regions (Supplementary Figures 6 and 7; Supplementary Tables 7 and 10). **(B) Context-specific inverse oxygen–ploidy maps with fixed-input reference trajectories.** Endpoint-level dense curves were evaluated separately in both contexts at 496 forced probabilities from 0.005 through 0.500 in increments of 0.001 and numerically inverted over target dominant mean ploidy 1–7 in increments of 0.025. Where at least 80% of the seed-weighted endpoint vectors had exactly one inverse solution and fewer than 20% had multiple solutions, color shows the seed-weighted median $p_{\text{miss,eff}}$ returned by the piecewise-linear inverse; color spacing is logarithmic and labels remain in original probability units. Purple hatching marks cells where at least 20% of endpoints had multiple solutions. Light gray marks cells without an ensemble-supported unique inverse in the evaluated probability interval, including targets that were not reached consistently. Black curves show the forward median-ploidy trajectories at fixed $p_{\text{miss,eff}} = 0.01, 0.10, 0.20,$ and 0.30 using solid, dash–dot–dot, dot–dash, and dotted lines, respectively. At each oxygen concentration, the red solid curve is the arithmetic mean of the 496 endpoint-median ploidy values. Dashed horizontal lines mark ploidy 2 and 4. The heatmap reverses the numerical question posed by the reference trajectories; it does not clamp ploidy in the operator, add fitted observations, or establish a biological CIN requirement. The red curve is a descriptive equal-weight average across the forced-input grid, not a fitted trajectory. All panels are post-fit asymptotic diagnostics rather than longitudinal measurements or biological uncertainty intervals (Supplementary Table 10).

Discussion

Across the lineage-matched culture and tumor datasets, the model resolved a small set of function-level context contrasts rather than a unique parameter vector. Tumors combined a lower fitted proliferative ceiling with stronger effective chromosome missegregation under hypoxia. Altered daughters also had higher absolute fitted survival in tumors, whereas culture imposed the steeper ploidy dependence on that survival (Supplementary Tables 8 and 16). Absolute retention and its dependence on ploidy are therefore separable axes of karyotype selection.

These function-level contrasts suggest that resource context may change not only how much chromosome-number variation is generated, but also how chromosome state determines whether that variation persists. The model therefore links CIN production to the retention of viable karyotypic diversity rather than treating CIN, WGD, and survival as interchangeable proxies. Comparing complete functions was essential: individual survival parameters showed mixed directions, whereas the evaluated nonlinear survival function yielded the same two-part context contrast in all three C-family winners.

The oxygen analysis extended this distinction from fitted functions to response topology. Culture separated sharply into low- and high-ploidy regimes and retained nearly the same surface across C01–C03. Tumors instead occupied broad intermediate-ploidy regions and showed greater family dependence. The fitted culture trajectory remained within the low- $p_{\text{miss,eff}}$ band, consistent with post-missegregation retention acting as the stronger filter on culture ploidy. Tumor trajectories crossed several ploidy domains, implicating oxygen-linked chromosome-error generation more strongly in the tumor response (Figures 5E–F and 6A; Supplementary Tables 10, 11, and 16). Fixed-input curves further showed that the tumor response changed direction as missegregation input increased, while inverse maps revealed opposite oxygen-dependent input requirements in culture and tumors (Figure 6B; Supplementary Table 12). These distinctions remained intact in weak-gap regions and across ensemble-sensitivity analyses (Supplementary Figure 7; Supplementary Table 13). Because the surfaces are post-fit asymptotic diagnostics rather than randomized perturbations or longitudinal CIN measurements, the inferred shift in the locus of selection remains model conditional.

These mechanistic interpretations remain conditional on the fitting framework. Soft coupling permits context-specific values while regularizing their separation on the transformed parameter scale, but the reported Welsch penalty is bounded, so sufficiently large differences contribute a saturated penalty rather than an indefinitely increasing cost. Repeated separation across C-family ensembles therefore identifies model-supported hypotheses conditional on the likelihoods, bounds, regularization, and common *in vitro* anchor; it is not an equality test and does not establish that environmental context caused the fitted difference. Likewise, fitted effective O_2 and stress-linked parameter contrasts are latent model quantities, not direct measurements of tissue oxygen, CIN, or death mechanisms.

A purely resource-dependent interpretation of ploidy evolution is limited by the absence of a separate oxygen-independent cost of abnormal chromosome content. Aneuploid states can incur proteotoxic, replication, mitotic, and gene-dosage stresses even outside the specific oxygen-dependent resource term represented here.^{7–9} Consequently, the fitted resource-dependent penalty should not be interpreted as an exhaustive account of high-ploidy or aneuploid fitness costs.

Oxygen also captures only one component of the broader resource environment experienced by

tumor cells. The analyzed datasets did not directly measure nutrient availability, perfusion, local consumption, or spatial oxygen heterogeneity. The fitted *in vivo* O₂ trajectory is therefore a latent model variable, and nominal *in vitro* oxygen and fitted *in vivo* oxygen should not be treated as experimentally equivalent energetic states. Oxygen is informative within the model but remains an incomplete proxy for resource stress.

Other omitted mechanisms could alter the generation or filtering of high-ploidy states, including context-dependent WGD generation, tissue-specific polyploidization programs, and immune or stromal selection. The present SUM159 analysis cannot distinguish these unmodeled possibilities from resource-mediated effects. Its conclusions should therefore remain restricted to the fitted mechanisms and matched culture/tumor datasets rather than generalized to all tissue-specific ploidy differences.

Practical identifiability and search coverage impose a separate constraint. Near-equivalent tumor fits arose from boundary-dependent parameter combinations, and all three joint ensembles contained active bounds. C01–C03 sampled distinct *in vivo* warm-start regions but shared one *in vitro* anchor, so they provide landscape coverage rather than independent replication. Qualitative oxygen-response conclusions were stable across objective thresholds and endpoint deduplication, yet trajectory direction and regime occupancy still varied among families (Supplementary Tables 7 and 10). Saturation of the Welsch penalty and weak spectral separation in many separate fits further limit estimation of a unique effect size or global optimum (Supplementary Table 9). The reported concordance should therefore be read as agreement across the three sampled C families, with behavior in unsampled regions unresolved.

XX Before submission, repeat the joint fitting with multiple independently selected *in vitro* anchors and a balanced set of *in vivo* and *in vitro* warm-start combinations. The sensitivity analysis should vary the coupling scale and saturation constant and should compare the bounded Welsch penalty with a nonsaturating quadratic coupling penalty. Report whether the directions and magnitudes of the complete function-level contrasts remain stable under these alternatives.

Finite-grid implementation adds another interpretive constraint. Transition mass outside the chromosome range is routed to the CIN-associated dead compartment, so that compartment combines biological nonviability with finite-grid boundary loss.

Even within these limits, the model suggests testable hypotheses for therapies that induce chromosome-segregation errors. Many anticancer agents elevate chromosome missegregation either directly, by disrupting mitosis, or indirectly, by producing replication and DNA-damage lesions that persist into mitosis.^{17–23} Within the fitted ploidy-dependent survival framework, such agents are predicted not to have uniform effects across chromosome-number states: near-diploid cells are predicted to be more vulnerable to missegregation-induced dosage imbalance, whereas WGD or high-ploidy cells may better buffer individual chromosome gains or losses and convert some errors into heritable karyotype variation. This offers a model-based hypothesis for prior drug-response analyses in which multiple mitotic spindle poisons, including paclitaxel, vinblastine, and vinorelbine, were among the strongest ploidy-associated agents.²⁴ It also suggests a possible explanation for why gemcitabine, despite being classically viewed as an S-phase antimetabolite, may show a strong association with ploidy: independent chromosome-loss assays have identified gemcitabine as a potent inducer of chromosomal instability.¹⁷ Thus, replication-stress-induced chromosome missegregation is a candidate ploidy-dependent therapeutic vulnerability generated by the model, not a response tested in the present SUM159 experiments.

1 Supplementary Information

1.1 *In vivo* experiments

The *in vivo* analysis used near-diploid and near-tetraploid variants of the SUM159 triple-negative breast cancer cell line, denoted SUM159_NLS_2N_A6M and SUM159_NLS_4N_A4M, respectively. For each tumor, 1×10^6 cells were injected orthotopically into mice. The subset analyzed here comprised eight tumors: four initiated from the near-diploid 2N line and four initiated from the near-tetraploid 4N line (Supplementary Table 1).

- **2N cohort:** four mice injected with SUM159_NLS_2N_A6M.
- **4N cohort:** four mice injected with SUM159_NLS_4N_A4M.

Tumors were harvested at predefined endpoints, and single-cell RNA sequencing was performed on all eight tumors. Chromosome-number distributions used for model calibration were derived from these endpoint single-cell profiles and analyzed together with the corresponding tumor-burden trajectories. Only this untreated subset was used for the modeling analyses presented here. Processed data and associated experimental metadata were organized and retrieved through CLONEID, a framework for structuring clone-specific genotypic and phenotypic measurements.²⁵

XX Before submission, complete the animal and ethics reporting by specifying the mouse strain and genetic background, sex, age at implantation, source, housing and husbandry conditions, allocation and randomization procedures, blinding procedures, injection matrix and volume, tumor measurement schedule, predefined experimental and humane endpoints, exclusion criteria, numbers excluded with reasons, euthanasia method, approving institutional committee, protocol number, and compliance with applicable animal welfare guidelines. Also report the tissue collection and processing procedures used for necrosis assessment and single cell profiling, together with the chromosome inference method, filtering criteria, quality control thresholds, uncertainty assessment, and number of cells retained for each tumor.

Supplementary Table 1: Experimental data inventory for the analyzed SUM159 culture and tumor datasets. Counts distinguish biological specimens, measurement records, and display summaries. Individual burden measurements, chromosome estimates, culture counts, passage records, and flow-density values are provided in the accompanying machine-readable supplementary tables.

Setting	Data stream or design quantity	Biological or measurement units	Count or value	Scope and interpretation
<i>In vivo</i>	Orthotopic inoculum	Cells per tumor	1×10^6	Experimental input for each analyzed tumor
<i>In vivo</i>	Untreated tumor cohort	Tumors	8	Four near-diploid initiated and four near-tetraploid initiated tumors
<i>In vivo</i>	Longitudinal burden	Tumor measurements	120	Measurements from the eight tumors used for joint burden and terminal chromosome-number calibration
<i>In vivo</i>	Terminal chromosome-number profiles	Single-cell estimates	6,502	3,728 estimates from four near-diploid initiated tumors and 2,774 from four near-tetraploid initiated tumors
<i>In vivo</i>	Matched terminal necrosis	Tumor-level fractions	6	Three tumors per starting-ploidy cohort had matched necrosis measurements used by the observation model
<i>In vitro</i>	Passage-structured fit records	Passage entries	131	All entries retained assigned oxygen, timing, and linked population measurements
<i>In vitro</i>	Passage growth	Finite estimates	114	Entries with a finite segment-level growth estimate
<i>In vitro</i>	Direct chromosome counts	Samples; cells	12; 220	Passage-linked chromosome-number samples and their total single-cell observations
<i>In vitro</i>	G0/G1 DNA-content profiles	Profiles; grid points per profile	20; 200	Normalized profiles used in the flow-density comparison
<i>In vitro</i>	Oxygen schedule	Target O ₂ values (%)	20.5, 2, 1, 0.5, 0.3, 0.2, 0.1, 0	Control and serial deprivation levels; a post-anoxia branch was evaluated at 1%
Figure 1 display	Culture population markers	Displayed endpoints	77	Representative cohort, condition, and reconstructed-day summaries
Figure 1 display	Flow summaries	Profiles; grouped time points	20; 12	Individual profile peaks and equal-weight grouped density summaries
Figure 1 display	Ploidy conversion	Autosomal chromosomes per ploidy unit	22	Used only to express chromosome counts on the plotted ploidy scale
Figure 1 display	Flow plotting viewport	Ploidy range; minimum retained mass	1–5; 85.43%	Full normalized profiles were used for calculation; only the display viewport was cropped
Figure 1 display	Starting chromosome counts	Directly counted cells	40	Twenty cells from each starting-ploidy cohort

Note: Counts summarize the analyzed data and do not imply equal replication across data streams. Display counts can exceed the number of unique biological samples when a shared starting distribution is repeated for alignment. The machine-readable companion tables preserve the individual observations and identifiers needed to reconstruct these totals.

Supplementary Table 2: Absolute fit-quality metrics for the ten lowest-objective separate *in vivo* fits. Burden errors are calculated on the log-burden scale. Terminal chromosome-number metrics and necrosis-fraction MAE are averaged equally across tumors or harvests.

Rank	Fit	Objective	Burden log-RMSE	Tumor-balanced log-RMSE	Terminal mean- <i>N</i> MAE	Terminal Wasserstein-1	Terminal total variation	Necrosis- fraction MAE
1	seed25	14.1193	0.6660	0.6317	2.41	4.51	0.697	0.091
2	seed366	14.1340	0.6680	0.6341	2.54	4.75	0.708	0.078
3	seed292	14.1372	0.6724	0.6384	2.79	4.59	0.698	0.064
4	seed392	14.1406	0.6784	0.6426	3.33	4.92	0.689	0.093
5	seed90	14.1524	0.6799	0.6439	3.65	4.86	0.684	0.092
6	seed391	14.1553	0.6797	0.6439	3.38	4.75	0.682	0.091
7	seed264	14.1558	0.6735	0.6400	2.56	4.18	0.682	0.079
8	seed109	14.1724	0.6629	0.6284	2.85	4.30	0.681	0.103
9	seed322	14.1724	0.6677	0.6331	2.52	4.86	0.714	0.078
10	seed26	14.1748	0.6784	0.6425	3.71	5.24	0.698	0.094

Note: Positive-observation burden RMSE weights every retained measurement equally; tumor-balanced RMSE first averages squared residuals within each tumor and then weights tumors equally. Wasserstein-1 and mean-*N* MAE are reported in chromosomes. The ten optimizer-selected fits are not biological replicates or confidence-interval samples.

Supplementary Table 3: Peak continuous fixed-O₂ fitted-parameter associations underlying Figure 4B. Parameters are grouped by the sign of the Spearman correlation at their maximum absolute association, with the positive group shown before the negative group, and ordered by decreasing maximum $|\rho|$ within each group. Signed peak ρ and peak O₂ report the first oxygen value attaining that maximum. Sign stability is the percentage of oxygen values sharing the predominant correlation sign and is not used for ordering.

Within-group rank	Parameter	Max $ \rho $	Signed peak ρ	Peak O ₂ (%)	Sign stability (%)
<i>Positive peak ρ</i>					
1	p_mis_base	0.649	+0.649	4.050	100.0
2	lam_max	0.467	+0.467	0.025	100.0
3	O2_crit	0.269	+0.269	1.100	93.0
4	n_0	0.261	+0.261	4.975	91.0
5	buffer_n_exp	0.196	+0.196	2.925	95.0
6	gamma_growth	0.178	+0.178	0.400	100.0
7	p_wgd	0.178	+0.178	1.900	97.5
8	k_clear	0.148	+0.148	0.700	100.0
<i>Negative peak ρ</i>					
1	mu_hp	0.691	-0.691	0.000	83.6
2	buffer_beta	0.447	-0.447	1.900	92.5
3	gamma_mu	0.439	-0.439	0.975	100.0
4	p_misseg	0.378	-0.378	2.075	85.6
5	o2_S0	0.365	-0.365	3.700	91.5
6	k_o_mis	0.269	-0.269	0.450	100.0
7	buffer_smax	0.236	-0.236	0.000	84.6
8	kappa_0	0.171	-0.171	2.050	100.0
9	alpha_o2	0.167	-0.167	0.025	78.1
10	eta_o2	0.139	-0.139	2.150	93.0

Interpretation: The display group uses the sign at maximum $|\rho|$, which can differ from the predominant grid sign when a strong association is localized to a narrow oxygen range. For example, **mu_hp** has a negative peak at 0% O₂ although 83.6% of its grid correlations are positive. The full signed heatmap, rather than the grouped peak order alone, should therefore be used to assess oxygen dependence. The gray fitting ranges, black initial values, endpoint densities, and seed25 markers in Figure 4B are optimizer-derived quantities, not posterior samples, credible intervals, measurement bounds, or biological confidence intervals.

Supplementary Table 4: Separate *in vivo* fitted-landscape and numerical-search summary. Cluster comparisons are descriptive because the groups were constructed from the same fitted parameters subsequently compared. Optimizer endpoints are numerical solutions, not biological replicates or posterior samples.

Analysis	Quantity	Value	Definition or interpretation
Fixed-O ₂ association audit	Evaluation dimensions	$500 \times 18 \times 201$	Fitted solutions, parameters, and oxygen concentrations from 0% to 5% in 0.025-percentage-point increments
Embedding input	Numerical rows	$127,500 + 500$	Retained initial-population rows plus fitted endpoints; 18 fitted parameters were embedded
Exploratory clustering	C01, C02, C03 sizes	43, 412, 45	Endpoint counts in the selected three-group solution
Exploratory clustering	Selected k ; average silhouette	3; 0.655	Descriptive separation on the saved <i>in vivo</i> -only embedding
Omnibus comparison	BH-significant parameters	14 of 18	Kruskal-Wallis tests with Benjamini-Hochberg $q < 0.05$
Top cluster contrast	Stress-associated death scale, μ_{hp}	0.154	Kruskal-Wallis epsilon-squared
Top cluster contrast	Oxygen-drop amplitude, κ_O	0.151	Kruskal-Wallis epsilon-squared
Top cluster contrast	Maximum division rate, λ_{max}	0.143	Kruskal-Wallis epsilon-squared
Top cluster contrast	Survival ploidy exponent, n_{exp}	0.115	Kruskal-Wallis epsilon-squared
Top cluster contrast	Baseline missegregation probability, $p_{mis,base}$	0.101	Kruskal-Wallis epsilon-squared
Top cluster contrast	Resource-dependent growth damping, α_{O_2}	0.049	Kruskal-Wallis epsilon-squared
Objective proximity	Starts within 1%, 5%, and 10%	29, 241, 405 of 500	Relative excess above the lowest retained objective
Optimizer record	DE termination flag	500 of 500	Implementation-specific differential-evolution stopping flag
Optimizer record	Local refinement attempted; accepted	500; 460	Recorded local-refinement outcomes
Optimizer record	Accepted local code 0	276	Accepted local refinements carrying convergence code 0
Selected solution	Active configured bounds	2 of 20	Exact active-bound count for the lowest-objective separate fit, seed 25

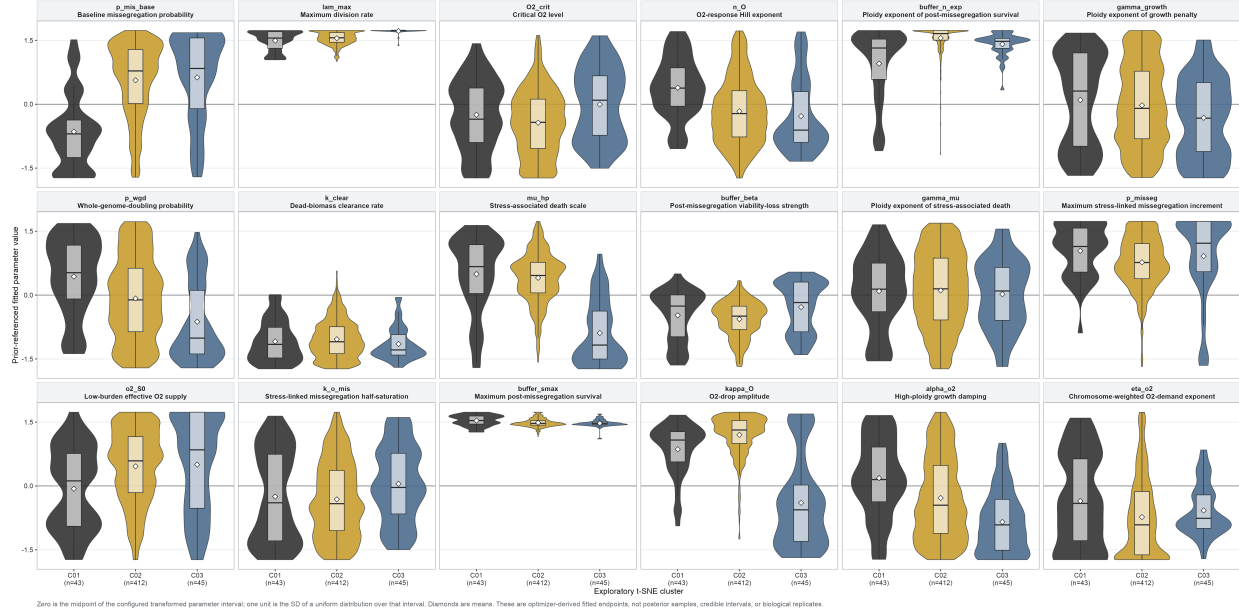
1.2 *In vitro* experiments

The *in vitro* analysis used matched near-diploid (2N) and near-tetraploid (4N) SUM159 lineages. The O1 and O2 nomenclature introduced in the main Methods is used consistently here and in Figures 1 and 3: O1 and O2 denote the two separately propagated oxygen-deprivation lineages within each starting-ploidy cohort. For each starting ploidy, a control branch was maintained at 20.5% O₂, and O1 and O2 were stepped over successive passages through target levels of 2%, 1%, 0.5%, 0.3%, 0.2%, 0.1%, and 0% O₂. The main deprivation branch was then maintained at target 0% O₂; an additional branch after the first 0% passage was evaluated at target 1% O₂. We describe O1 and O2 as parallel lineages because the available records do not establish whether they should be classified as biological replicates (Supplementary Table 1).

For calibration, each fit entry retained the assigned oxygen level, passage duration, initial and

All 18 fitted-parameter endpoint distributions across exploratory *in vivo* t-SNE clusters

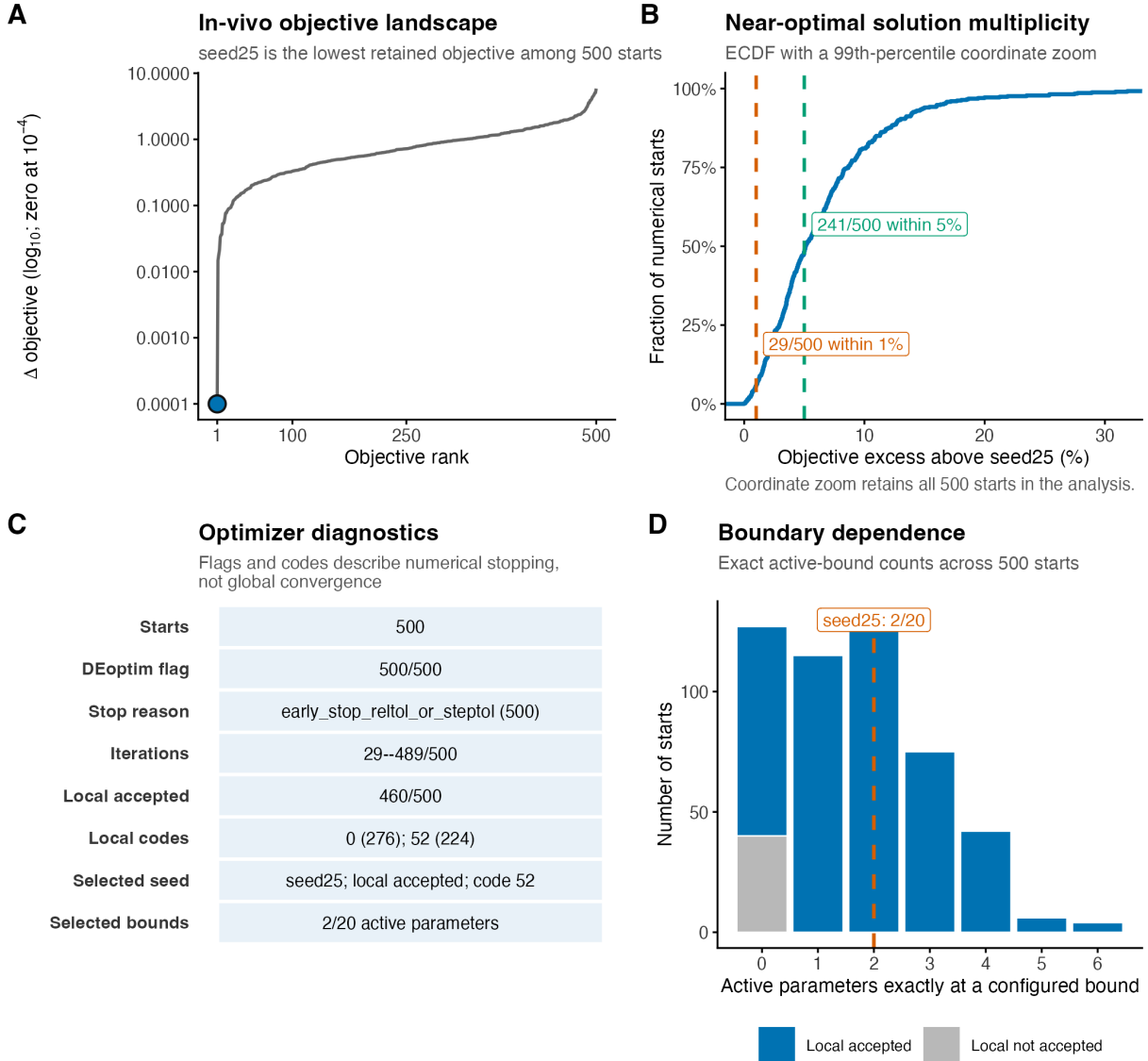
Parameters follow the Figure 4B positive-peak then negative-peak ordering, with descending $|\rho|$ within each group. All 18 parameters were inputs to the *in vivo*-only t-SNE.



Supplementary Figure 1: **All 18 fitted-parameter endpoint distributions across the exploratory separate-*in vivo* t-SNE clusters.** The 18 facets are arranged in three rows and six columns in the same positive-peak then negative-peak order as Figure 4B, with decreasing maximum $|\rho|$ within each sign group. Each facet shows the prior-referenced fitted-value distribution for clusters C01, C02, and C03; boxplots show medians and interquartile ranges and diamonds show means. Figure 4D displays the six-parameter subset selected from these same values by the pre-specified omnibus-test and effect-size rule. All displayed parameters contributed to the *in vivo*-only 18-parameter t-SNE and endpoint clustering. Zero is the midpoint of the configured transformed parameter interval and one unit is the standard deviation of a uniform distribution over that interval. Cluster sizes, tests, and effect sizes are reported in Supplementary Table 4. These are distributions of optimizer-derived fitted endpoints, not posterior samples, credible intervals, independent validation, or biological replication.

Separate in-vivo numerical-search performance

Optimizer starts are numerical endpoints, not posterior draws, confidence intervals, or biological replicates



Supplementary Figure 2: **Complete numerical-search diagnostics for the separate *in vivo* fit ensemble.** (A) Objective difference by rank across 500 starts, with the lowest retained fit, seed 25, highlighted. (B) Empirical cumulative distribution of relative objective excess above seed 25. Vertical markers show that 29 starts were within 1% and 241 were within 5%; the displayed x-range ends at the 99th percentile by coordinate zoom without removing starts from the analysis. (C) Recorded differential-evolution termination, iteration, local-refinement, selected-seed, and boundary diagnostics. Flags and codes describe implementation-specific numerical stopping and do not establish global convergence. (D) Distribution of exact active-parameter boundary counts across all 500 retained endpoints, separated by local-refinement acceptance status; the dashed line marks seed 25, for which two of 20 active parameters were exactly at configured bounds. The corresponding counts are reported in Supplementary Table 4. Optimizer starts are numerical endpoints, not posterior draws, biological replicates, or inferential uncertainty.

final corrected cell counts, and a segment-level growth-rate estimate. Chromosome-number measurements and G0/G1 DNA-content density profiles were linked to the corresponding passage when available. The current fit objects contain 131 entries, including 114 with finite growth estimates, 12 with chromosome-number measurements (220 single-cell values in total), and 20 with matched G0/G1 profiles. Each G0/G1 profile was represented on a 200-point ploidy grid and normalized before comparison with the simulated viable chromosome-number distribution (Supplementary Table 1).

For the Figure 1 timeline, population-size measurement availability was audited against the raw passage snapshot: all 131 passage IDs had a positive seeding count and at least one positive linked harvest count. The displayed representative tracks collapse matching records to cohort, condition, and reconstructed day; common pre-branch endpoints are repeated on both aligned eventual branches. For panel B, direct autosomal chromosome counts were converted to ploidy by division by $N_{\text{unit}} = 22$. For panel C, each observed flow profile was normalized to unit density mass. Profiles from parallel O1 and O2 lines at the same cohort, condition, and reconstructed day were averaged with equal sample weight for the half-violin and density-weighted box summary, while the global density maximum of each individual profile supplied its open-triangle marker. The complete normalized profiles were used to calculate peaks, quantiles, and whiskers before applying the ploidy 1–5 plotting viewport. This viewport retained at least 85.43% of normalized density mass in every sample; density above ploidy 5 was cropped without within-window renormalization (Supplementary Table 1). These display summaries did not alter the flow-density likelihood.

Within the passage-structured model, oxygen was held fixed at the assigned value for each segment. For segments represented by parallel records, passage duration and endpoint cell counts were summarized by the median, while all available growth, chromosome-number, and flow-density observations contributed to their corresponding likelihood terms.

XX Before submission, complete the cell culture and assay reporting by specifying cell line provenance and authentication, mycoplasma testing, passage range, culture medium and supplements, incubation conditions, oxygen control equipment, independent verification of oxygen exposure, equilibration procedures, seeding and passaging protocols, replicate structure, allocation procedures, cell counting method, sampling schedule, chromosome counting method, flow cytometry preparation and gating, assay specific quality control, and all prespecified exclusion criteria.

Supplementary Table 5: Final karyotype distributions of the parallel 2N oxygen-deprivation lines and their shared fitted state. O1 and O2 were propagated as separate lines from the same 2N precursor through the matched nominal oxygen schedule. The fitted row is the single model distribution evaluated against both observed lines at the final matched chromosome-count segment.

Distribution	n	Mean N	Median N	$N \geq 80$	Component structure
O1 observed	20	66.85	49	8/20 (40.0%)	12 cells at $N = 46$ –49; 8 cells at $N = 90$ –100
O2 observed	20	88.05	95	16/20 (80.0%)	4 cells at $N = 49$ –76; 16 cells at $N = 93$ –99
Shared fitted state	—	64.27	60	32.5% probability	dominant local modes at $N = 40$ and $N = 83$

Note: $N \geq 80$ is a descriptive bin chosen within the unoccupied interval between the low- and high-chromosome components of the final observed samples; it is not a fitted threshold or inferential cutoff. Observed entries are single-cell counts. The fitted entry is a probability distribution and therefore has no observed sample size. Values are descriptive and do not constitute an additional statistical test.

Supplementary Table 6: Modeled 4N chromosome-number decline and the fitted contribution of the compartment representing death associated with resource stress across the ten lowest-objective separate *in vitro* fits. Initial and terminal mean chromosome numbers refer to the first and last modeled states of the selected 4N lineage; its terminal portion was maintained at target 0% O₂. The last column is the maximum daily burden assigned by the model to death associated with resource stress, expressed as a percentage of total burden over the full modeled 4N trajectory.

Rank	Fit	Objective	Initial mean N	Terminal mean N	Decline in mean N	Maximum modeled stress death burden (%)
1	seed10	3.8525	84.34	80.76	3.58	1.76
2	seed132	3.8533	84.21	80.66	3.55	1.76
3	seed81	3.8541	84.22	80.67	3.54	1.77
4	seed294	3.8594	84.71	80.89	3.82	1.74
5	seed337	3.8598	84.66	80.76	3.91	1.74
6	seed106	3.8605	84.45	80.53	3.91	1.82
7	seed317	3.8610	84.30	80.13	4.17	1.76
8	seed140	3.8610	84.21	80.54	3.67	1.74
9	seed285	3.8618	84.24	80.20	4.04	1.77
10	seed464	3.8623	84.68	80.73	3.96	1.77

Seed-10 audit quantity	Value	Scope
Maximum passage-end stress-associated dead-burden fraction	2N: 1.05%; 4N: 1.71%	Severe-deprivation passage-end summaries shown in Figure 3F
Numerical starts and fitted parameters	500; 20	Separate <i>in vitro</i> calibration
Differential-evolution convergence flags	0 of 500	Recorded implementation-specific flag
Selected local refinement	accepted, code 1	Lowest-objective retained seed 10
Selected active configured bounds	5 of 20	Exact active-bound count

Note: Fits are ordered by the retained objective value. The decline is the initial minus terminal predicted mean chromosome number. The dead burden percentage is the maximum, over all daily 4N states, of the burden assigned by the model to death associated with resource stress as a percentage of total burden. It is therefore a trajectory maximum rather than an endpoint-only value. These optimizer-selected solutions are not biological replicates or a confidence interval.

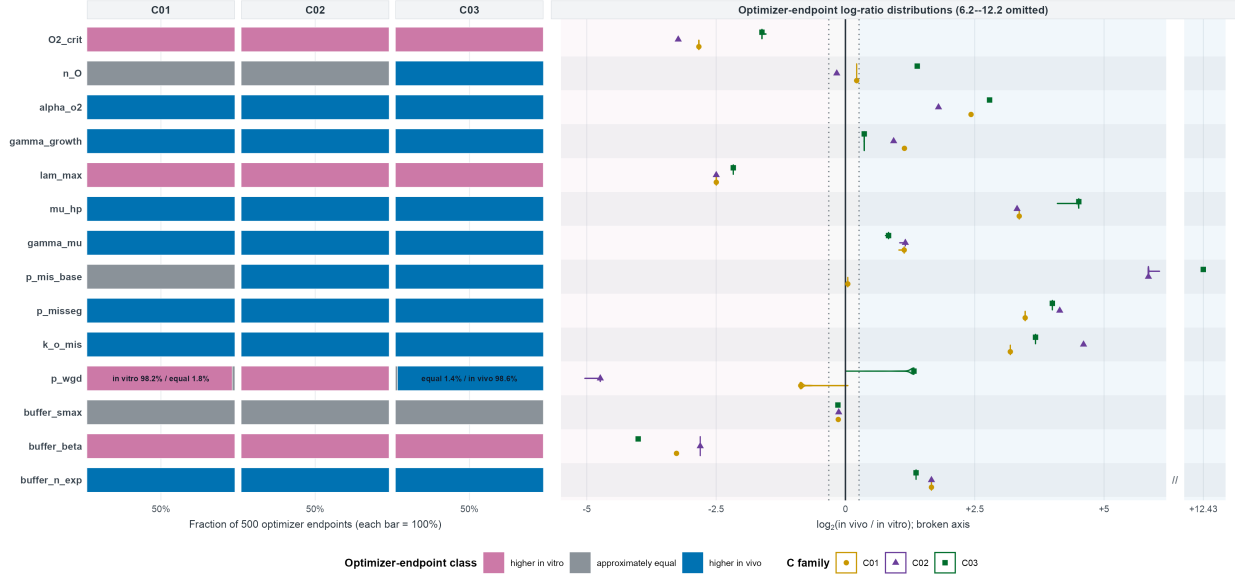
Supplementary Table 7: Warm-start selection, joint numerical-search diagnostics, and objective eligibility for the three retained C families. Every family used the same separate *in vitro* seed-10 anchor and 500 numerical starts. The lowest-objective decile retained 50 endpoints per family. Numerical endpoints and cluster assignments are search diagnostics rather than biological replicates, confidence sets, or biological groups.

Family	Separate <i>in vivo</i> start	Joint winner	Minimum objective	Iterations of 500	DE flags of 500	Local refinement	Winner bounds of 20	Decile cutoff above minimum
C01	366	472	18.8523	26–82	500	rejected, code 52	2	0.038700
C02	25	497	18.9705	26–46	500	rejected, code 52	2	0.003593
C03	311	18	19.4145	26–73	500	rejected, code 52	2	0.061229

Audit quantity	Value	Interpretation
Complete endpoints passing hard configuration and feasibility checks	1,500 of 1,500	Three families, 500 endpoints per family
Widest retained objective set passing post-fit operator checks	300 of 300	Lowest-objective 20%, 100 endpoints per family
Primary complete response surfaces	300	Two contexts, three families, and 50 endpoints per family
Parameter-specific context-ratio records passing feasibility checks	21,000 of 21,000	Three families, 500 endpoints, and 14 paired parameters
Pooled warm-start embedding	228,000 vectors	127,500 <i>in vivo</i> population points, 99,500 <i>in vitro</i> population points, and 500 final solutions per context
Pooled embedding configuration	seed 123; perplexity 30; $\theta = 0.5$; 1,000 iterations	Saved t-SNE coordinates used for warm-start selection
Saved-embedding silhouette, $k = 3$; $k = 4$; difference	0.815694; 0.816651; 0.000957	$k = 4$ was numerically near-tied and produced a singleton region
Selected k across 100 subsamples retaining 80% of endpoints	$k = 3$: 58; $k = 4$: 42	Region-count resampling on the saved embedding
Fixed- $k = 3$ label agreement under the same subsampling	Median ARI 1.00	Agreement with saved labels
Standardized 14-parameter-space $k = 3$ agreement	ARI 0.052	Poor agreement with the saved-embedding regions

Joint-fit parameter differences across three selected primary families

Left: endpoint-level directional composition under the 0.8/1.2 natural-scale thresholds. Right: horizontal distributions retain within-family spread and contrast magnitude.



Supplementary Figure 3: **Scalar joint-fit parameter contrasts across the three retained C-family ensembles.** For each parameter and C family, the left matrix partitions all 500 numerical optimizer endpoints into higher *in vitro* ($0 < \text{in vivo}/\text{in vitro} \leq 0.8$), approximately equal ($0.8 < \text{in vivo}/\text{in vitro} < 1.2$), and higher *in vivo* ($\text{in vivo}/\text{in vitro} \geq 1.2$) fractions. Pink, gray, and blue show these three classes, respectively; percentages are printed only for mixed cells. The aligned plot on the right shows the complete optimizer-endpoint $\log_2(\text{in vivo}/\text{in vitro})$ distributions separately for C01, C02, and C03. Horizontal violins show kernel-density shapes where endpoint ratios vary, and colored symbols mark the family medians. A median symbol without a violin denotes a degenerate distribution in which all 500 ratios are identical. Solid zero and dotted threshold lines mark equal fitted values and the 0.8/1.2 natural-scale boundaries. The displayed axis extends below the observed minimum and explicitly omits the data-empty interval from 6.2 to 12.2; the short continuation retains the C03 $p_{\text{mis},\text{base}}$ group at 12.43. The composition matrix therefore reports class stability, whereas the aligned distributions distinguish modest from large context differences and expose within-family concentration or spread. Each C family retains a distinct *in vivo* warm start and shares the same *in vitro* anchor. These optimizer-endpoint summaries assess numerical-search and warm-start sensitivity; they are not posterior draws, confidence intervals, biological replicates, or proof of structural identifiability (Supplementary Tables 7 and 8).

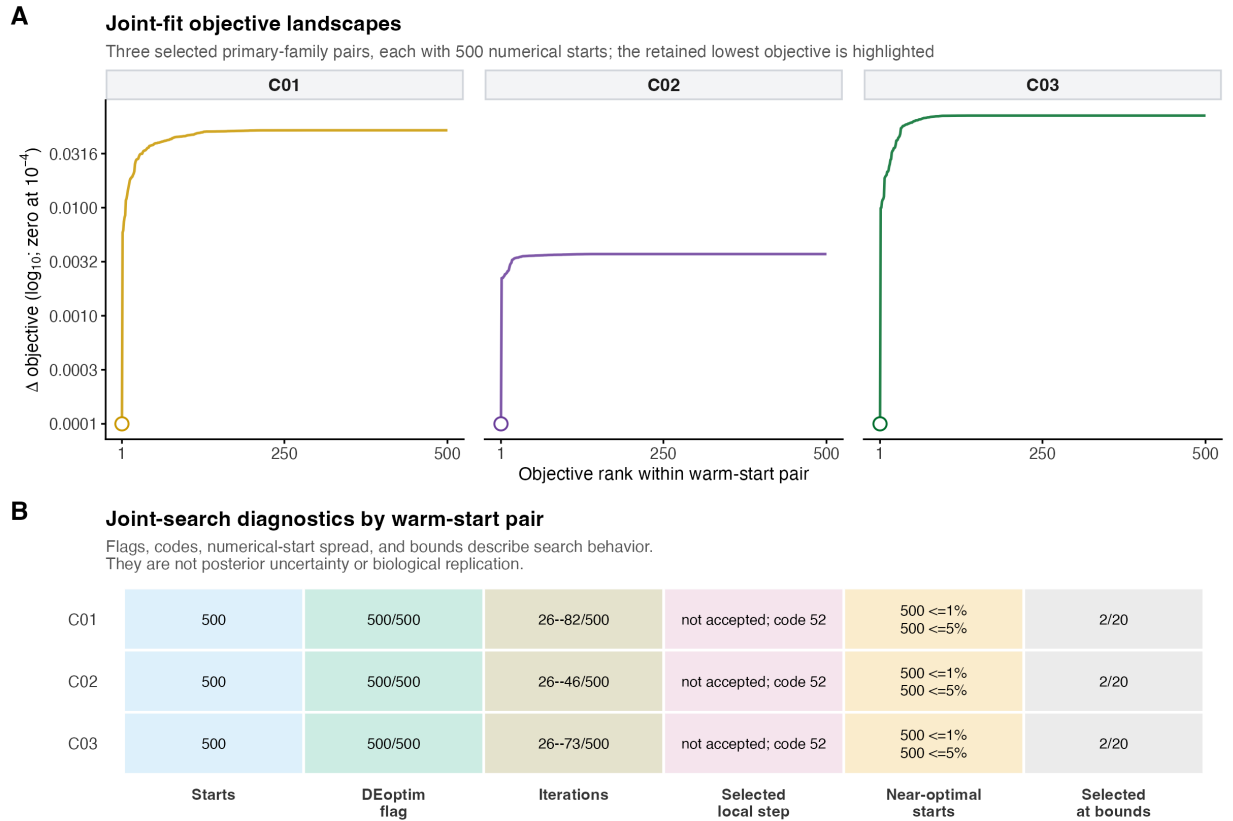
Supplementary Table 8: Context-specific differential-evolution initial-population and optimizer-endpoint distributions for the 14 paired parameters. Each C-family cell reports the in vivo and in vitro natural-scale parameter distributions separately. The DE-initial distribution comprises 200,000 members reconstructed from the 500 runs (400 members per run); the endpoint distribution comprises the corresponding 500 feasible final solutions. Entries report median [5th, 95th percentile], followed by context-specific active-bound occupancy. Cross-family direction is an auxiliary sign-based audit of the paired endpoint ratios and does not apply the 0.8/1.2 equivalence band used for the family-specific classes in Supplementary Figure 3. These numerical-search distributions are not posterior or confidence intervals.

Parameter	C01: in vivo; in vitro	C02: in vivo; in vitro	C03: in vivo; in vitro	Directional/search audit
lam_max	in vivo: DE init 0.247 [0.201, 0.306]; endpoint 0.239 [0.239, 0.239]; bound=0%; in vitro: DE init 1.32 [1.09, 1.48]; endpoint 1.35 [1.35, 1.35]; bound=0%	in vivo: DE init 0.247 [0.202, 0.306]; endpoint 0.239 [0.239, 0.239]; bound=0%; in vitro: DE init 1.32 [1.09, 1.48]; endpoint 1.35 [1.35, 1.35]; bound=0%	in vivo: DE init 0.308 [0.259, 0.366]; endpoint 0.300 [0.300, 0.300]; bound=0%; in vitro: DE init 1.33 [1.13, 1.48]; endpoint 1.35 [1.35, 1.35]; bound=0%	lower in vivo; max paired W.E/W.I=0; max bound=0%
p_mis_base	in vivo: DE init 0.00000988 [0.00000155, 0.0000954]; endpoint 0.00000882 [0.00000337, 0.00000882]; bound=0%; in vitro: DE init 0.00000958 [0.00000151, 0.0000925]; endpoint 0.00000854 [0.00000329, 0.00000856]; bound=0.2%	in vivo: DE init 0.000495 [0.0000634, 0.00356]; endpoint 0.000495 [0.000495, 0.000495]; bound=0%; in vitro: DE init 0.00000854 [0.00000133, 0.0000615]; endpoint 0.00000854 [0.00000854, 0.00000854]; bound=0%	in vivo: DE init 0.0428 [0.00795, 0.0918]; endpoint 0.0471 [0.0471, 0.0471]; bound=0%; in vitro: DE init 0.00000982 [0.00000172, 0.000123]; endpoint 0.00000854 [0.00000854, 0.00000854]; bound=0%	higher in vivo; max paired W.E/W.I=0.372; max bound=0.2%
p_wgd	in vivo: DE init 0.000164 [0.0000334, 0.000617]; endpoint 0.000182 [0.000182, 0.000202]; bound=0%; in vitro: DE init 0.000299 [0.0000607, 0.000964]; endpoint 0.000331 [0.000298, 0.000331]; bound=0%	in vivo: DE init 0.0000144 [0.0000103, 0.000197]; endpoint 0.0000124 [0.0000124, 0.0000124]; bound=0%; in vitro: DE init 0.000333 [0.0000181, 0.000947]; endpoint 0.000331 [0.000331, 0.000331]; bound=0%	in vivo: DE init 0.000626 [0.000163, 0.000988]; endpoint 0.000830 [0.000657, 0.000830]; bound=0%; in vitro: DE init 0.000250 [0.0000648, 0.000841]; endpoint 0.000331 [0.000331, 0.000405]; bound=0%	family-dependent or overlaps equal-ity; max paired W.E/W.I=0.690; max bound=0%
p_misseg	in vivo: DE init 0.117 [0.0542, 0.239]; endpoint 0.117 [0.117, 0.117]; bound=0%; in vitro: DE init 0.0106 [0.00562, 0.0215]; endpoint 0.0106 [0.0106, 0.0106]; bound=0%	in vivo: DE init 0.186 [0.0887, 0.368]; endpoint 0.187 [0.187, 0.187]; bound=0%; in vitro: DE init 0.0106 [0.00572, 0.0213]; endpoint 0.0106 [0.0106, 0.0106]; bound=0%	in vivo: DE init 0.169 [0.0799, 0.337]; endpoint 0.169 [0.169, 0.169]; bound=0%; in vitro: DE init 0.0105 [0.00570, 0.0211]; endpoint 0.0106 [0.0106, 0.0106]; bound=0%	higher in vivo; max paired W.E/W.I=0; max bound=0%
k_o_mis	in vivo: DE init 0.00103 [0.000205, 0.00513]; endpoint 0.00103 [0.00103, 0.00103]; bound=0%; in vitro: DE init 0.000113 [0.0000225, 0.000563]; endpoint 0.000113 [0.000113, 0.000113]; bound=0%	in vivo: DE init 0.00274 [0.000591, 0.0126]; endpoint 0.00274 [0.00274, 0.00274]; bound=0%; in vitro: DE init 0.000113 [0.0000243, 0.000525]; endpoint 0.000113 [0.000113, 0.000113]; bound=0%	in vivo: DE init 0.00144 [0.000296, 0.00698]; endpoint 0.00144 [0.00144, 0.00144]; bound=0%; in vitro: DE init 0.000113 [0.0000231, 0.000548]; endpoint 0.000113 [0.000113, 0.000113]; bound=0%	higher in vivo; max paired W.E/W.I=0; max bound=0%
02_crit	in vivo: DE init 0.223 [0.178, 0.278]; endpoint 0.223 [0.223, 0.223]; bound=0%; in vitro: DE init 1.59 [1.27, 1.98]; endpoint 1.59 [1.59, 1.59]; bound=0%	in vivo: DE init 0.169 [0.130, 0.219]; endpoint 0.169 [0.169, 0.169]; bound=0%; in vitro: DE init 1.59 [1.22, 2.05]; endpoint 1.59 [1.59, 1.59]; bound=0%	in vivo: DE init 0.518 [0.463, 0.580]; endpoint 0.518 [0.518, 0.518]; bound=0%; in vitro: DE init 1.59 [1.42, 1.78]; endpoint 1.59 [1.59, 1.59]; bound=0%	lower in vivo; max paired W.E/W.I=0; max bound=0%

Parameter	C01: in vivo; in vitro	C02: in vivo; in vitro	C03: in vivo; in vitro	Directional/search audit
n_0	in vivo: DE init 2.17 [1.77, 2.58]; endpoint 2.17 [2.17, 2.17]; bound=0%; in vitro: DE init 1.87 [1.46, 2.27]; DE init 1.87 [1.87, 1.87]; endpoint 1.87 [1.87, 1.87]; bound=0%	in vivo: DE init 1.66 [1.31, 1.66]; endpoint 1.66 [1.66, 1.66]; bound=0%; in vitro: DE init 1.87 [1.52, 2.22]; DE init 1.87 [1.87, 1.87]; endpoint 1.87 [1.87, 1.87]; bound=0%	in vivo: DE init 4.77 [4.15, 4.98]; endpoint 4.89 [4.89, 4.89]; bound=0%; in vitro: DE init 1.97 [1.13, 3.27]; DE init 1.97 [1.97, 1.97]; endpoint 1.87 [1.87, 1.87]; bound=0%	family-dependent; max paired; W_E/W_I=0; max bound=0%
alpha_o2	in vivo: DE init 2.50 [2.17, 2.74]; endpoint 2.69 [2.69, 2.69]; bound=0%; in vitro: DE init 0.537 [0.503, 0.614]; DE init 0.537 [0.500, 0.500]; endpoint 0.500 [0.500, 0.500]; bound=100%	in vivo: DE init 1.65 [1.49, 1.75]; endpoint 1.74 [1.74, 1.74]; bound=0%; in vitro: DE init 0.527 [0.502, 0.582]; DE init 0.527 [0.500, 0.500]; endpoint 0.500 [0.500, 0.500]; bound=100%	in vivo: DE init 3.15 [2.64, 3.60]; endpoint 3.45 [3.45, 3.45]; bound=0%; in vitro: DE init 0.544 [0.504, 0.636]; DE init 0.544 [0.500, 0.500]; endpoint 0.500 [0.500, 0.500]; bound=100%	higher in vivo; max paired; W_E/W_I=0; max bound=100%
gamma_growth	in vivo: DE init 3.21 [2.26, 3.82]; endpoint 3.31 [3.31, 3.31]; bound=0%; in vitro: DE init 1.62 [1.51, 2.01]; DE init 1.62 [1.50, 1.50]; endpoint 1.50 [1.50, 1.50]; bound=100%	in vivo: DE init 2.79 [1.95, 2.86]; endpoint 2.86 [2.86, 2.86]; bound=0%; in vitro: DE init 1.60 [1.51, 1.96]; DE init 1.60 [1.50, 1.50]; endpoint 1.50 [1.50, 1.50]; bound=100%	in vivo: DE init 1.96 [1.55, 2.29]; endpoint 1.93 [1.93, 1.93]; bound=0%; in vitro: DE init 1.55 [1.50, 1.86]; DE init 1.55 [1.50, 1.50]; endpoint 1.50 [1.50, 1.50]; bound=99.8%	higher in vivo; max paired; W_E/W_I=0; max bound=100%
mu_hp	in vivo: DE init 0.0459 [0.00957, 0.121]; endpoint 0.0513 [0.0513, 0.0513]; bound=0%; in vitro: DE init 0.00592 [0.00507, 0.0118]; endpoint 0.00500 [0.00500, 0.00500]; bound=99.8%	in vivo: DE init 0.0447 [0.00935, 0.118]; endpoint 0.0499 [0.0499, 0.0499]; bound=0%; in vitro: DE init 0.00591 [0.00507, 0.0118]; endpoint 0.00500 [0.00500, 0.00500]; bound=100%	in vivo: DE init 0.0964 [0.0219, 0.244]; endpoint 0.114 [0.114, 0.114]; bound=0%; in vitro: DE init 0.00612 [0.00509, 0.0117]; endpoint 0.00500 [0.00500, 0.00500]; bound=99.8%	higher in vivo; max paired; W_E/W_I=0; max bound=100%
gamma_mu	in vivo: DE init 2.56 [1.80, 3.04]; endpoint 2.64 [2.63, 2.64]; bound=0%; in vitro: DE init 1.30 [1.21, 1.61]; DE init 1.30 [1.20, 1.20]; endpoint 1.20 [1.20, 1.20]; bound=94%	in vivo: DE init 2.60 [1.83, 2.68]; endpoint 2.68 [2.68, 2.68]; bound=0%; in vitro: DE init 1.30 [1.21, 1.61]; DE init 1.30 [1.20, 1.20]; endpoint 1.20 [1.20, 1.20]; bound=98.6%	in vivo: DE init 2.09 [1.46, 2.49]; endpoint 2.14 [2.14, 2.14]; bound=0%; in vitro: DE init 1.27 [1.21, 1.55]; DE init 1.27 [1.20, 1.20]; endpoint 1.20 [1.20, 1.20]; bound=98.4%	higher in vivo; max paired; W_E/W_I=0.0191; max bound=98.6%
buffer_smain	in vivo: DE init 0.857 [0.695, 0.984]; endpoint 0.909 [0.909, 0.909]; bound=0%; in vitro: DE init 0.949 [0.786, 0.999]; DE init 0.949 [1.00, 1.00]; endpoint 1.00 [1.00, 1.00]; bound=100%	in vivo: DE init 0.861 [0.699, 0.984]; endpoint 0.914 [0.914, 0.914]; bound=0%; in vitro: DE init 0.947 [0.785, 0.999]; DE init 0.947 [1.00, 1.00]; endpoint 1.00 [1.00, 1.00]; bound=100%	in vivo: DE init 0.854 [0.691, 0.984]; endpoint 0.904 [0.904, 0.904]; bound=0%; in vitro: DE init 0.950 [0.787, 0.999]; DE init 0.950 [1.00, 1.00]; endpoint 1.00 [1.00, 1.00]; bound=100%	lower in vivo; max paired; W_E/W_I=0; max bound=100%
buffer_beta	in vivo: DE init 0.165 [0.127, 0.215]; endpoint 0.165 [0.165, 0.165]; bound=0%; in vitro: DE init 1.59 [1.22, 2.07]; DE init 1.59 [1.59, 1.59]; endpoint 1.59 [1.59, 1.59]; bound=0%	in vivo: DE init 0.227 [0.183, 0.283]; endpoint 0.227 [0.227, 0.227]; bound=0%; in vitro: DE init 1.59 [1.28, 1.98]; DE init 1.59 [1.59, 1.59]; endpoint 1.59 [1.59, 1.59]; bound=0%	in vivo: DE init 0.0990 [0.0710, 0.138]; endpoint 0.0990 [0.0990, 0.0990]; bound=0%; in vitro: DE init 1.59 [1.14, 2.22]; DE init 1.59 [1.59, 1.59]; endpoint 1.59 [1.59, 1.59]; bound=0%	lower in vivo; max paired; W_E/W_I=0; max bound=0%
buffer_n_exp	in vivo: DE init 9.24 [6.95, 9.93]; endpoint 10.0 [10.0, 10.0]; bound=99.8%; in vitro: DE init 3.35 [2.19, 6.61]; endpoint 3.16 [3.16, 3.16]; bound=0%	in vivo: DE init 9.24 [6.95, 9.93]; endpoint 10.0 [10.0, 10.0]; bound=100%; in vitro: DE init 3.35 [2.19, 6.60]; endpoint 3.16 [3.16, 3.16]; bound=0%	in vivo: DE init 8.13 [5.80, 9.84]; endpoint 8.13 [8.13, 8.13]; bound=0%; in vitro: DE init 3.16 [2.25, 5.15]; DE init 3.16 [3.16, 3.16]; endpoint 3.16 [3.16, 3.16]; bound=0%	higher in vivo; max paired; W_E/W_I=0; max bound=100%

All panels use the same 201 oxygen concentrations, 60 fixed effective per-chromosome missegregation probabilities, and family-specific lowest-objective 50-joint-endpoint ensembles as Figure 6A. Columns show C01, C02, and C03; the top row shows the *in vivo* parameter vectors and the bottom row shows the paired *in vitro* vectors from the same retained joint endpoints. The dotted purple boundary encloses grid cells where at least half of the context-specific endpoint vectors have dom-

Joint-fit numerical-search performance



Supplementary Figure 4: **Joint-fit numerical-search performance across the three retained C-family ensembles.** (A) Objective-difference landscapes for the C01, C02, and C03 500-start joint fits. Within each family, starts are ranked by objective, the lowest retained winner is highlighted, and exact zero is displayed at 10^{-4} on the $\log_{10}$ axis. (B) C-family-resolved run diagnostics, including numbers of starts, recorded differential-evolution convergence flags, completed-iteration ranges relative to the target, selected local-step status and code, near-optimal start counts, and exact active-bound counts for the selected winners. The corresponding C-family rows are reported in Supplementary Table 7. These diagnostics characterize the sampled numerical searches; they are not biological replication, posterior distributions, confidence intervals, bootstrap uncertainty, profile likelihoods, held-out validation, or proof of a global optimum.

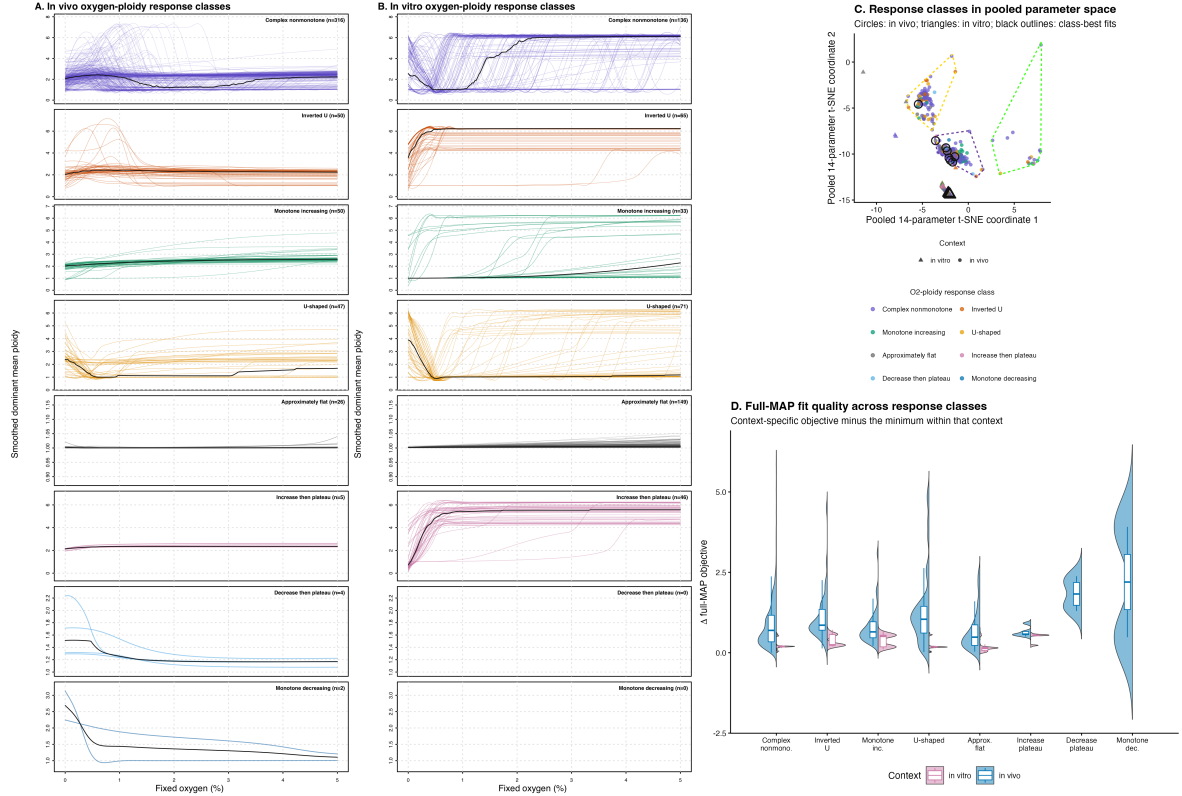
Supplementary Table 9: Fixed-O₂ response-shape and spectral-gap classifications for the 500 separate fitted solutions in each context. Reliability was classified from the fraction of the 201 oxygen-grid evaluations below the prespecified spectral-gap thresholds. These categories describe asymptotic numerical diagnostics and are not additional fitted-data hypothesis tests.

Evaluation quantity	Value			
Separate fitted solutions and oxygen grid	500 solutions per context; 201 values from 0% to 5% in 0.025-percentage-point increments			
Total seed-oxygen evaluations	100,500 per context			
Unreliable criterion	At least 25% of grid values with spectral gap below 0.005			
Caution criterion	Any grid value below 0.005, or at least 10% below 0.01			

Response-shape class	<i>In vivo</i>		<i>In vitro</i>	
	Solutions	Fraction (%)	Solutions	Fraction (%)
Complex nonmonotone	316	63.2	136	27.2
Inverted U-shaped	50	10.0	65	13.0
Monotone increasing	50	10.0	33	6.6
U-shaped	47	9.4	71	14.2
Approximately flat	26	5.2	149	29.8
Increase then plateau	5	1.0	46	9.2
Decrease then plateau	4	0.8	0	0.0
Monotone decreasing	2	0.4	0	0.0

Spectral-gap class	<i>In vivo</i>		<i>In vitro</i>	
	Solutions	Fraction (%)	Solutions	Fraction (%)
Reliable	141	28.2	32	6.4
Caution	168	33.6	86	17.2
Unreliable	191	38.2	382	76.4

Warm-start representative	Separate-fit seed	Spectral-gap class
C01	366	unreliable
C02	25	caution
C03	311	caution
Shared <i>in vitro</i> anchor	10	reliable



Supplementary Figure 5: **Separate-fit fixed- O_2 response classes and their fitted-landscape context.** (A–B) **Context-specific oxygen–ploidy response classes.** Analytical dominant-eigenvector ploidy curves from the 500 separate *in vivo* fits (A) and 500 separate *in vitro* fits (B) were grouped by the same regression-smoothed response-shape rule. Both columns use the same fixed eight-class order; an empty plotting region therefore records a class absent from that context rather than omitted data. Class names and seed counts are printed in the upper-right corner. Thin colored curves show individual optimizer-seed endpoints and the black curve is the pointwise class median. Panel-specific ploidy ranges preserve visibility of both broad and nearly flat responses. These class counts include reliable, caution, and unreliable spectral-gap strata. (C) **Response classes in pooled parameter space.** All 1,000 separate-fit endpoints are positioned in the pooled 14-parameter t-SNE coordinate system used for the joint warm-start landscape in Figure 5C and colored by fixed- O_2 response class. Circles denote *in vivo* and triangles denote *in vitro*. These unitless visualization coordinates have no direct mechanistic interpretation and are distinct from the *in vivo*-only 18-parameter embedding in Figure 4B. A black outline marks the lowest-objective endpoint within every nonempty context–class combination. Dashed outlines delimit the C01, C02, and C03 *in vivo* warm-start regions using the same gold, purple, and green colors as Figure 5; they are search-design regions rather than biological groups. (D) **Context-relative full-MAP fit quality across response classes.** For each context, the minimum full MAP objective across its 500 separate fits was subtracted before comparison. Blue left and pink right half-violins show the *in vivo* and *in vitro* distributions, respectively, with narrow context-colored boxplots showing medians and interquartile ranges. Empty halves identify classes absent from that context. Because the two fits use different observation models, this within-context objective difference is used only to compare relative fit quality among response classes; vertical values are not cross-context likelihood ratios. Class and reliability counts are reported in Supplementary Table 9. Together, panels C–D diagnose remaining fitted-landscape multiplicity rather than selecting one response shape a priori.

Supplementary Table 10: Context-specific joint-ensemble fixed-O₂ and inverse-response summaries for the lowest-objective decile of C01–C03. Response entries report the median change in dominant mean ploidy from 0% to 5% O₂ across 50 endpoints. Fixed low, intermediate, and high denote prespecified effective-missegregation inputs; the unmodified trajectory retains the fitted oxygen dependence. Inverse-response entries summarize the Figure 6B oxygen–target–ploidy grid.

Context	Family	Unmodified trajectory	Fixed low	Fixed intermediate	Fixed high	Trajectory direction
<i>In vivo</i>	C01	−1.112	+1.624	+0.814	+4.888	decrease
<i>In vivo</i>	C02	−1.172	+1.652	+0.809	+4.874	decrease
<i>In vivo</i>	C03	+1.175	+1.678	+1.693	+1.552	increase
<i>In vitro</i>	C01	+0.000194	+0.00113	+0.0807	−0.823	approximately flat
<i>In vitro</i>	C02	+0.000219	+0.00126	+0.0807	−0.823	approximately flat
<i>In vitro</i>	C03	+0.000273	+0.00157	+0.0807	−0.823	approximately flat

		Median population-average $p_{\text{miss,eff}}$	
Context	Family	at 0% O ₂	at 5% O ₂
<i>In vivo</i>	C01		0.115
<i>In vivo</i>	C02		0.178
<i>In vivo</i>	C03		0.212
<i>In vitro</i>	C01	0.01004	
<i>In vitro</i>	C02	0.01004	
<i>In vitro</i>	C03	0.01004	

Design or robustness quantity	Value	Scope
Complete fixed-input surface per endpoint	201 × 60	Oxygen by effective-missegregation grid
Complete-surface evaluations	3,618,000	300 context-specific endpoint vectors and 12,060 conditions per vector
Dense fixed-missegregation line grid	496 values	0.005–0.500 in increments of 0.001
Distinct dense-grid operator evaluations	22,930,080	11,465,040 per context after exact-endpoint deduplication
Family-diagnostic modal results unchanged across the 5%, 10%, and 20% sets	48 of 48	Two contexts, three families, and eight qualitative diagnostics
Seed-weighted versus unique-parameter comparisons with the same modal result	144 of 144	Two contexts, three families, eight diagnostics, and three cutoffs
Within-set modal support across all qualitative audits	100%	Minimum over all context, family, diagnostic, and cutoff combinations
Unique context-specific parameter endpoints in a 50-seed decile ensemble	15–50	Family-specific range in both contexts; C02 had 15

		Stable unique	Multiple	No stable unique
Context	Family	inverse (%)	solutions (%)	inverse (%)
<i>In vivo</i>	C01	69.84	0.03	30.13
<i>In vivo</i>	C02	69.30	0.05	30.65
<i>In vivo</i>	C03	27.67	0.63	71.70
<i>In vitro</i>	C01	84.28	4.27	11.45
<i>In vitro</i>	C02	84.28	4.27	11.45
<i>In vitro</i>	C03	84.28	4.28	11.45

Context	Family	Target ploidy 4 at 0% O ₂	At 1% O ₂	At 5% O ₂
<i>In vivo</i>	C01	–	0.156 (0.156–0.156)	0.323 (0.323–0.323)
<i>In vivo</i>	C02	–	0.198 (0.197–0.198)	0.327 (0.327–0.327)
<i>In vivo</i>	C03	–	0.246 (0.246–0.246)	–
<i>In vitro</i>	C01	0.02174 (0.02174–0.02174)	0.01973 (0.01973–0.01973)	0.006725 (0.006725–0.006725)
<i>In vitro</i>	C02	0.02174 (0.02174–0.02174)	0.01973 (0.01973–0.01973)	0.006725 (0.006725–0.006725)
<i>In vitro</i>	C03	0.02174 (0.02174–0.02174)	0.01973 (0.01973–0.01973)	0.006725 (0.006725–0.006725)

Supplementary Table 11: Descriptive topology of the joint-ensemble oxygen–CIN–ploidy response. Grid-domain percentages classify the seed-weighted median dominant ploidy at each of the 12,060 matched oxygen–effective-missegregation cells per context and C family using the same low, intermediate, and high thresholds as Supplementary Figure 7. Across-family agreement is the percentage of matched cells assigned to the same three-class regime in C01, C02, and C03.

Context	Family	Low (%)	Intermediate (%)	High (%)	Across-family agreement (%)
<i>In vivo</i>	C01	4.41	79.39	16.20	77.36
<i>In vivo</i>	C02	3.53	82.60	13.86	77.36
<i>In vivo</i>	C03	12.48	85.03	2.49	77.36
<i>In vitro</i>	C01	20.56	0.00	79.44	99.99
<i>In vitro</i>	C02	20.56	0.00	79.44	99.99
<i>In vitro</i>	C03	20.56	0.00	79.44	99.99

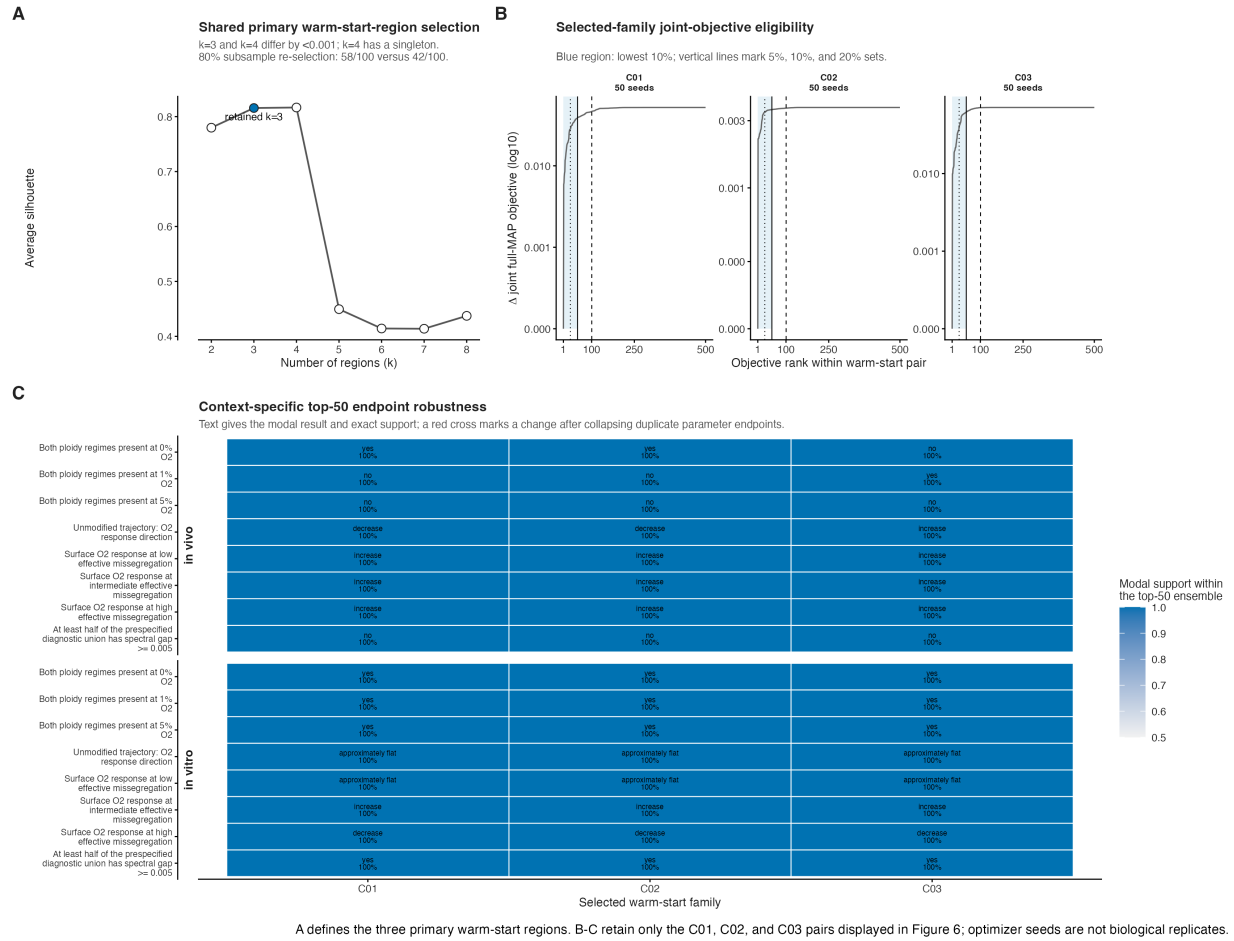
Supplementary Table 12: Descriptive features of the *in vivo* fixed-input reference curves in Figure 6B. The sharpest sampled downturn is the oxygen coordinate at the end of the largest negative dominant-ploidy change between adjacent 0.025-percentage-point oxygen values. It is a grid-descriptive feature rather than a fitted biological threshold.

<i>In vivo</i> family	$p_{\text{miss,eff}} = 0.01$ over 0–5% O ₂	Sharpest downturn O ₂ (%) at $p_{\text{miss,eff}} = 0.20$	Sharpest downturn O ₂ (%) at $p_{\text{miss,eff}} = 0.30$
C01	monotone increase	2.275	4.375
C02	monotone increase	1.550	3.850
C03	monotone increase	1.100	1.300

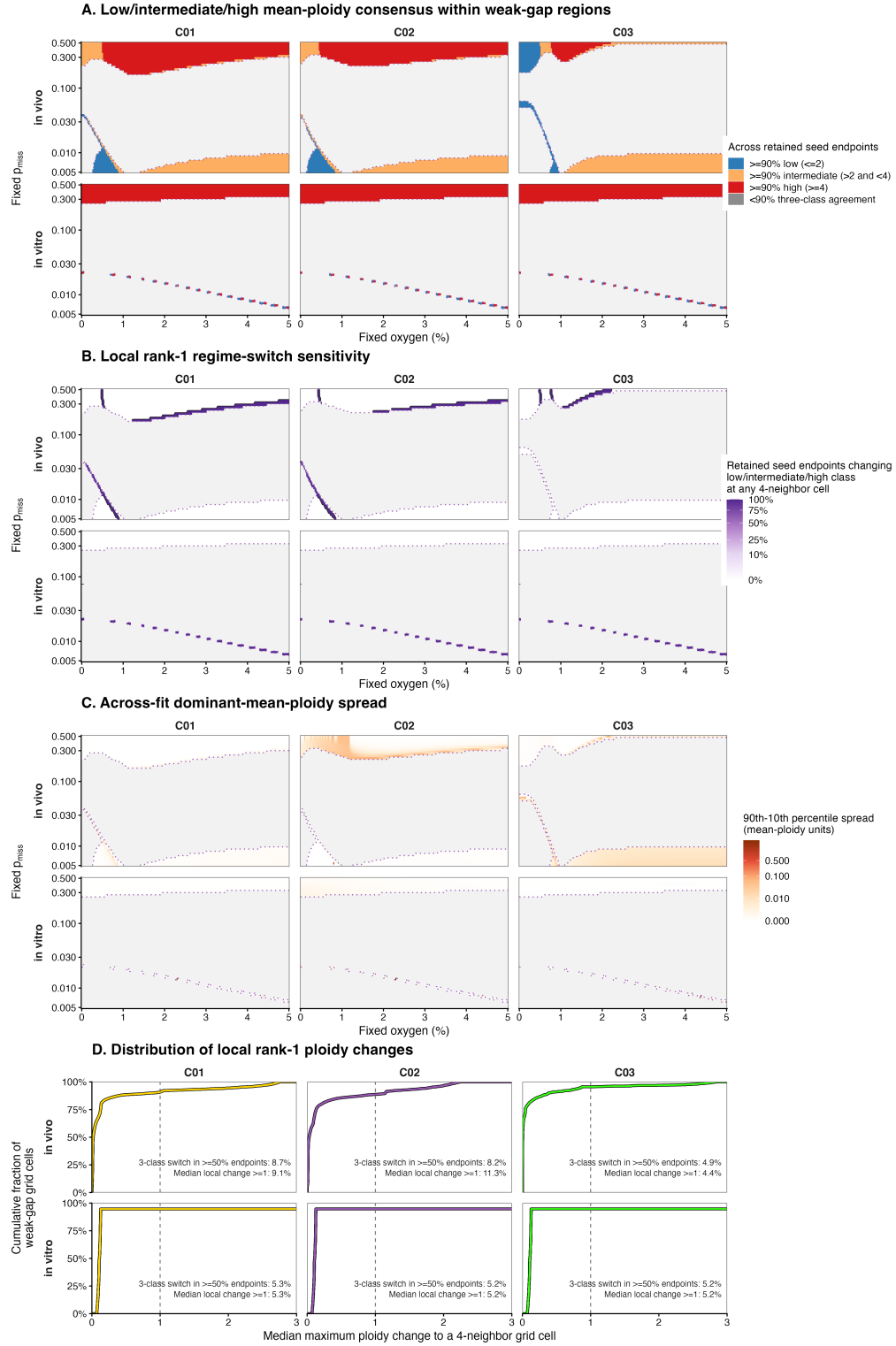
Supplementary Table 13: Context-specific dominant-ploidy robustness within weak-spectral-gap regions of the three displayed joint ensembles. Stable classes require at least 90% endpoint agreement. Spread is the across-endpoint 90th minus 10th percentile range of dominant mean ploidy. Local-switch and local-jump quantities compare each grid cell with its immediately adjacent oxygen and fixed-missegregation cells. These grid summaries are numerical diagnostics, not biological transition probabilities.

Context	Family	Weak-gap cells	Stable low (%)	Stable intermediate (%)	Stable high (%)	Mixed (%)	Majority local class switch (%)
<i>In vivo</i>	C01	3,287	5.9	36.3	57.8	0.00	8.67
<i>In vivo</i>	C02	2,838	7.0	37.2	55.8	0.00	8.17
<i>In vivo</i>	C03	2,118	17.4	69.0	13.6	0.00	4.86
<i>In vitro</i>	C01	1,527	2.55	0.00	97.38	0.07	5.30
<i>In vitro</i>	C02	1,526	2.56	0.00	97.38	0.07	5.24
<i>In vitro</i>	C03	1,526	2.49	0.00	97.44	0.07	5.24

Context	Family	Median spread	90th-percentile spread	Cells with median local jump at least 1 (%)
<i>In vivo</i>	C01	0.000035	0.000733	9.13
<i>In vivo</i>	C02	0.000317	0.0237	11.31
<i>In vivo</i>	C03	0.00672	0.0107	4.39
<i>In vitro</i>	C01	0.000043	0.000061	5.30
<i>In vitro</i>	C02	0.000395	0.00121	5.24
<i>In vitro</i>	C03	0.000083	0.000109	5.24



Supplementary Figure 6: **Cluster-selection, objective-eligibility, and multi-seed robustness diagnostics for Figure 6.** (A) Average silhouette over candidate primary-region counts in the frozen pooled 14-parameter t-SNE coordinates. The retained $k = 3$ solution is within 0.001 of the numerical $k = 4$ maximum, whereas $k = 4$ creates a singleton region. This profile and the associated resampling audit condition on the saved embedding rather than re-estimating t-SNE. (B) Full joint MAP objective differences ordered across all 500 numerical endpoints within C01, C02, and C03. The blue region is the primary lowest-objective decile; vertical lines mark the nested 5%, 10%, and 20% eligibility thresholds. Facet labels report the 50 retained optimizer seeds. (C) Qualitative oxygen–CIN–ploidy diagnostics for the primary 50-endpoint ensembles, evaluated separately from the *in vivo* and *in vitro* parameter vectors of the same three C families. Text reports the modal result and its exact support; fill gives the modal support within the primary ensemble. A red cross would mark a modal result that changed when repeated exact context-specific parameter endpoints were collapsed to one equal-weight analysis unit. The corresponding cluster, objective, response, and robustness values are reported in Supplementary Tables 7 and 10. This categorical audit does not assert pointwise identity of the response surfaces. Cluster labels, objective ranks, and support fractions describe numerical search behavior; they are not biological groups, likelihood-ratio confidence sets, posterior probabilities, or biological replication.



inant spectral gap below 0.005; areas outside that weak-gap region are gray. **(A)** Across-endpoint regime consensus within the weak-gap region. Stable low, intermediate, and high require at least 90% of endpoints to place dominant mean ploidy at or below 2, between 2 and 4, or at or above 4, respectively; gray within the colored region would denote less than 90% three-class agreement. **(B)** Proportion of endpoints for which the rank-1 mean-ploidy classification changes among low, intermediate, and high at any of the up to four immediately adjacent O_2 -missegregation grid cells, shown with square-root color spacing and percentage labels in the original scale. The black contour marks a proportion of 0.5. **(C)** Across-endpoint 90th–10th percentile spread of rank-1 dominant mean ploidy, shown in original ploidy units with pseudo-logarithmic color spacing. **(D)** Empirical cumulative distribution of the endpoint-median maximum absolute rank-1 ploidy change to an immediately adjacent grid cell, restricted to weak-gap cells. The vertical dashed line marks one ploidy unit; annotations report the fraction with a three-class change in at least half of endpoints and the fraction with median local ploidy change of at least one unit. The ECDF is calculated from all weak-gap cells, while the displayed horizontal range is limited to 0–3 ploidy units; a curve can therefore remain below 100% at the right boundary when larger changes occur. Exact duplicate context-specific parameter endpoints were evaluated once and restored to their original optimizer-seed multiplicities before aggregation. Family- and context-level counts and robustness summaries are reported in Supplementary Table 13. These are post-fit numerical robustness diagnostics, not biological state frequencies, transition probabilities, or uncertainty intervals.

1.3 A Mechanistic Model of Chromosome-Number Evolution

We model tumor population dynamics as a deterministic state-structured ODE system over chromosome-number classes. The formulation separates viable cells from non-viable biomass, couples local oxygen availability to resource-dependent growth and stress-associated death, and uses chromosome missegregation and whole-genome doubling to move viable mass across chromosome-number states.

State space and compartments. We consider tumor cells stratified by modeled total autosomal chromosome count; sex chromosomes are not included in this state variable. Let

$$\mathcal{N} = \{N_{\min}, \dots, N_{\max}\}. \quad (1)$$

This set denotes the allowed autosomal chromosome-count grid. For the simulations described here, $N_{\min} = 22$ and $N_{\max} = 154$. Let $N_{\text{dip}} = 44$ denote the diploid autosomal chromosome-count reference state, and let $N_{\text{unit}} = 22$ denote the one-ploidy autosomal chromosome-count unit, so that $2N_{\text{unit}} = N_{\text{dip}}$.

The viable abundance vector is

$$\mathbf{x}^L(t) = \{x_N^L(t)\}_{N \in \mathcal{N}}, \quad (2)$$

where $x_N^L(t)$ is the number of viable cells with modeled autosomal chromosome count N at time t . The ODE state is indexed only by autosomal chromosome count and by live/dead compartment. Unless explicitly marked as a dead compartment, chromosome-number distributions in the model are defined over the viable population. The total viable population size is

$$N_{\text{tot}}(t) = \sum_{N \in \mathcal{N}} x_N^L(t). \quad (3)$$

To represent non-viable biomass contributing to macroscopic tumor volume, we track two dead pools:

$$\mathbf{x}^{D,\text{hyp}}(t) = \{x_N^{D,\text{hyp}}(t)\}_{N \in \mathcal{N}}, \quad \mathbf{x}^{D,\text{CIN}}(t) = \{x_N^{D,\text{CIN}}(t)\}_{N \in \mathcal{N}}. \quad (4)$$

Here $\mathbf{x}^{D,\text{hyp}}$ receives cells assigned to the modeled pathway for death associated with resource stress, and $\mathbf{x}^{D,\text{CIN}}$ receives non-viable daughter mass generated by chromosome missegregation or WGD; the superscript CIN denotes missegregation-origin loss or out of grid WGD. The superscript hyp is an internal label for a model assigned compartment. It does not indicate that the biological cause of death was directly observed or uniquely identified. Their sum is

$$\mathbf{x}^D(t) = \mathbf{x}^{D,\text{hyp}}(t) + \mathbf{x}^{D,\text{CIN}}(t). \quad (5)$$

The model's bulk tumor burden is therefore

$$N_{\text{burden}}(t) = \sum_{N \in \mathcal{N}} \left(x_N^L(t) + x_N^{D,\text{hyp}}(t) + x_N^{D,\text{CIN}}(t) \right), \quad (6)$$

which includes viable cells and dead biomass. This distinction is used throughout: tumor-volume observations are compared with total burden, whereas single-cell chromosome-number observations sample the viable distribution.

Environmental stress functions. Let $O_2(t)$ denote the mean-field approximation of the effective oxygen level experienced by the tumor population. Oxygen-dependent resource stress is represented by the bounded Hill function

$$h(O_2) = \frac{O_{2,c}^{n_O}}{O_{2,c}^{n_O} + O_2^{n_O}}, \quad (7)$$

where $O_{2,c}$ is the critical oxygen level at which the stress term is half-maximal, and n_O controls the steepness of the transition from low stress at high oxygen to strong stress under oxygen limitation. A Hill-type stress function is biologically justified because the literature indicates that oxygen-dependent cell fate is graded but threshold-like rather than linear: “The severity of hypoxia determines whether cells become apoptotic or adapt to hypoxia and survive”,²⁶ and recent spheroid work identifies a “critical survival pO₂”;²⁷ accordingly, Eq. (7) provides a smooth, bounded approximation to an oxygen-linked resource-stress threshold.

Proliferation and stress-associated death rates. Given chromosome count N and local oxygen level O_2 , the effective division rate is

$$\lambda_{\text{eff}}(N, O_2) = \lambda_{\text{max}} \frac{1}{1 + \alpha_{O_2} h(O_2) \left(\frac{N}{N_{\text{dip}}} \right)^{\gamma_{\text{growth}}}}. \quad (8)$$

Here α_{O_2} controls the strength of resource-stress-dependent growth damping, and γ_{growth} controls how strongly this damping increases with chromosome content. When oxygen-linked stress is small, $h(O_2) \approx 0$ and the reciprocal factor approaches one; as oxygen decreases, the reciprocal factor suppresses proliferation more strongly, especially for high-chromosome-count states.

Stress-associated death is applied as an explicit live-to-dead flow. The per-capita death hazard is

$$\mu_{\text{eff}}(N, O_2) = \mu_{\text{hp}} h(O_2) \left(\frac{N}{N_{\text{dip}}} \right)^{\gamma_{\mu}}. \quad (9)$$

Thus μ_{hp} sets the stress-associated death scale, $h(O_2)$ makes death depend on the same oxygen-linked stress function (Eq. (7)), and γ_{μ} controls the chromosome-number penalty on stress-associated death.

Cell division, WGD, and chromosomal transitions. With the stress-modulated rates defined, divisions are represented by a live-state generator matrix

$$G_{\text{div}}(O_2) \in \mathbb{R}^{R \times R}, \quad R = N_{\text{max}} - N_{\text{min}} + 1. \quad (10)$$

Let

$$L(O_2) = \text{diag}(\lambda_{\text{eff}}(N, O_2)) \quad (11)$$

be the diagonal matrix of state-specific division rates. A division follows the ordinary chromosome-missegregation branch with probability $1 - p_{\text{wgd}}$ and follows a whole-genome-doubling branch with probability p_{wgd} . The corresponding division generator is

$$G_{\text{div}}(O_2) = (1 - p_{\text{wgd}}) B(O_2) L(O_2) + p_{\text{wgd}} W L(O_2) - L(O_2), \quad (12)$$

where $B(O_2)$ is the viable offspring-allocation matrix generated by chromosome missegregation and post-missegregation viability filtering, W maps a mother state N to the within-grid doubled state $2N$, and the final $-L(O_2)$ term removes the dividing mother cell. The ordinary branch B accounts for the two daughter branches of a division and therefore has total mass up to two before viability losses. By contrast, the WGD branch is implemented as an effective tracked-lineage conversion: W has unit mass in the doubled state, so a WGD event replaces the mother lineage by one viable doubled lineage when $2N$ lies in the modeled grid. Out-of-grid WGD mass is not retained in the viable state space and is routed to the CIN-associated dead compartment.

Chromosome missegregation and daughter viability. For a mother cell with chromosome count N , the per-chromosome missegregation probability increases with the stress-associated death hazard defined in Eq. (9):

$$p_{\text{mis}}(N, O_2) = p_{\text{mis}, \text{base}} + p_{\text{misseg}} \frac{\mu_{\text{eff}}(N, O_2)}{\mu_{\text{eff}}(N, O_2) + k_{o, \text{mis}}}, \quad (13)$$

with values clipped to $[0, 1]$. Here $p_{\text{mis}, \text{base}}$ is the baseline per-chromosome missegregation probability under low stress, p_{misseg} is the maximal stress-induced increment above baseline, and $k_{o, \text{mis}}$ is the death-hazard scale at which the stress-induced increase in missegregation is half-maximal.

Let m denote the number of missegregated chromosome copies in a division of a mother cell with chromosome count N . We model

$$m \sim \text{Binomial}\{N, p_{\text{mis}}(N, O_2)\}. \quad (14)$$

For each value of m , the two daughter branches are assigned coarse chromosome-count shifts $+m$ and $-m$. Before applying the daughter-viability modifier, the shift kernel is

$$d_N(\Delta; O_2) = \sum_{m=0}^N \binom{N}{m} \{p_{\text{mis}}(N, O_2)\}^m \{1 - p_{\text{mis}}(N, O_2)\}^{N-m} [\mathbf{1}\{\Delta = m\} + \mathbf{1}\{\Delta = -m\}]. \quad (15)$$

This kernel sums over two daughter branches, so its total mass is two before viability filtering.

For a daughter with chromosome-count shift Δ , post-missegregation survival is approximated by the symmetric ploidy-dependent modifier

$$S_N(\Delta) = s_N^{|\Delta|}, \quad s_N = s_{\text{max}} \exp \left[-\beta_{\text{buf}} \left(\frac{N_{\text{dip}}}{N} \right)^{n_{\text{exp}}} \right]. \quad (16)$$

The parameter s_{max} is the maximum per-copy survival factor, β_{buf} controls the severity of post-missegregation viability loss at lower ploidy, and n_{exp} controls the steepness of the ploidy dependence. This form lets higher-chromosome-count states tolerate a given chromosome-count shift more readily than lower-chromosome-count states. This survival modifier enters the retained-grid offspring allocation and dead-pool accounting described in the boundary-handling subsection below.

Boundary handling on the chromosome-count domain. The chromosome-count state space is finite. We define the viable chromosome-count domain as

$$\mathcal{N} = \{N_{\min}, N_{\min} + 1, \dots, N_{\max}\},$$

where $N_{\min} = 22$ and $N_{\max} = 154$ in the simulations described here. The viable state vector

$$\mathbf{x}^L(t) = \{x_N^L(t)\}_{N \in \mathcal{N}}$$

therefore tracks only cells whose modeled autosomal chromosome count lies within this retained grid.

This finite state space requires an explicit rule for division events that would produce daughter or doubled-lineage states outside $\mathcal{N}$. We use absorbing boundary handling on the viable chromosome-count grid. That is, daughter cells or WGD-derived lineages whose chromosome count would fall below $N_{\min}$ or exceed $N_{\max}$ are removed from the viable offspring allocation and routed to the CIN-associated dead compartment.

For ordinary missegregating divisions, the viable offspring-allocation matrix is defined only over retained daughter states $N' \in \mathcal{N}$:

$$B_{N',N}(O_2) = d_N(N' - N; O_2) S_N(N' - N), \quad N, N' \in \mathcal{N}.$$

Here $d_N(\Delta; O_2)$ specifies the probability of generating a daughter chromosome-count shift Δ , and $S_N(\Delta)$ specifies the probability that a daughter with that shift remains viable. Any probability mass corresponding to $N + \Delta \notin \mathcal{N}$ is excluded from $B_{\cdot,N}$. Similarly, for whole-genome-doubling events, the doubling operator W retains only within-grid transitions $N \mapsto 2N \in \mathcal{N}$; out-of-grid WGD offspring mass is excluded from $W_{\cdot,N}$ and routed to the CIN-associated dead compartment.

Because the viable offspring allocation can be subunitary, either because buffering reduces daughter survival or because daughter or doubled-lineage states fall outside the retained chromosome-count grid, we define the CIN-associated event-level nonviable inflow as

$$\phi_{\text{CIN},N}(\mathbf{x}^L, O_2) = \lambda_{\text{eff}}(N, O_2) x_N^L \left\{ (1 - p_{\text{wgd}}) \left[2 - \sum_{N' \in \mathcal{N}} B_{N',N}(O_2) \right] + p_{\text{wgd}} \left[1 - \sum_{N' \in \mathcal{N}} W_{N',N} \right] \right\}. \quad (17)$$

The first bracket is the ordinary-division deficit between two potential daughters and the viable daughter mass retained within $\mathcal{N}$. The second bracket accounts for WGD events for which the doubled state $2N$ lies outside the modeled grid. This term is distinct from the modeled continuous death hazard associated with resource stress, $\mu_{\text{eff}}(N, O_2)$; it is added to the CIN-associated dead compartment $x^{D,\text{CIN}}$.

Thus, the chromosome-count boundaries are absorbing for viable offspring mass, with out-of-domain and non-buffered daughter mass transferred to the dead compartment.

Oxygen demand and supply dynamics. To couple tumor size to oxygen demand, we define the chromosome-number-weighted oxygen-demand proxy

$$N_{\text{eff}}(t) = \sum_N x_N^L(t) \left(\frac{N}{N_{\text{dip}}} \right)^\eta, \quad (18)$$

where η controls how oxygen demand scales with chromosome content. The reference scale N_{ref} is a fixed cell-count scale used to nondimensionalize demand, so that $N_{\text{eff}}/N_{\text{ref}}$ is a relative effective tumor size.

The oxygen supply target is the phenomenological supply–demand relation

$$O_2^*(N_{\text{eff}}) = \max \left(O_{2,\text{min}}, S_0 - \kappa_O \log \left(1 + \frac{N_{\text{eff}}}{N_{\text{ref}}} \right) \right), \quad (19)$$

where S_0 is the low-burden oxygen supply level, $O_{2,\text{min}}$ is the oxygen floor, and κ_O controls the oxygen-drop amplitude per unit log-burden increase. The logarithmic term is dimensionless because of the normalization by N_{ref} . Oxygen values are interpreted as percentages and are restricted to the physical interval $[0, 100]$.

Effective oxygen is updated by first-order relaxation toward this target:

$$\alpha_\tau = 1 - \exp \left(-\frac{\Delta t}{\tau_{O_2}} \right), \quad O_{2,k+1} = O_{2,k} + \alpha_\tau (O_2^*(N_{\text{eff},k}) - O_{2,k}). \quad (20)$$

Thus oxygen changes gradually toward the demand-adjusted target rather than instantaneously matching it at each simulation step.

Discrete-time simulation scheme. The simulations use a time step of $\Delta t = 0.05$ days. At each step $k \rightarrow k+1$, the oxygen state is first relaxed toward the supply–demand target in Eq. (20), using the current viable population to calculate $N_{\text{eff},k}$. The resulting relaxed value, $O_{2,k+1}$, is then used to evaluate division-linked chromosome transitions and explicit stress-associated death for that population step:

$$\mathbf{x}_{k+1}^L = \mathbf{x}_k^L + \Delta t G_{\text{div}}(O_{2,k+1}) \mathbf{x}_k^L - \Delta t \mu_{\text{eff}}(N, O_{2,k+1}) \odot \mathbf{x}_k^L, \quad (21)$$

where $\odot$ denotes element-wise multiplication over chromosome-count states.

Mass that becomes nonviable through the modeled stress pathway or failed divisions is transferred into the dead compartments, which contribute to tumor burden. The dead pool assigned by the model to resource stress is updated as

$$\mathbf{x}_{k+1}^{D,\text{hyp}} = \mathbf{x}_k^{D,\text{hyp}} + \Delta t \mu_{\text{eff}}(N, O_{2,k+1}) \odot \mathbf{x}_k^L - \Delta t k_{\text{clear}} \mathbf{x}_k^{D,\text{hyp}}, \quad (22)$$

and the CIN-associated dead pool is updated as

$$\mathbf{x}_{k+1}^{D,\text{CIN}} = \mathbf{x}_k^{D,\text{CIN}} + \Delta t \phi_{\text{CIN}}(\mathbf{x}_k^L, O_{2,k+1}) - \Delta t k_{\text{clear}} \mathbf{x}_k^{D,\text{CIN}}. \quad (23)$$

Here k_{clear} is the first-order clearance rate for dead biomass, and ϕ_{CIN} is defined component-wise in Eq. (17). Together, the oxygen relaxation and Eqs. (21)–(23) define the coupled viable/dead

population dynamics used in the simulations.

1.3.1 Model calibration and observation models

Model calibration uses a common transformed-parameter optimizer for *in vivo*, *in vitro*, and joint fits. Natural-scale parameter specifications provide initialization values and admissible bounds; each component maps its fitted parameters to the optimizer scale before evaluating its objective. The mechanistic dynamics described above are shared, while the observation models differ between the *in vivo* and *in vitro* data.

***In vivo* observation model and likelihood.** The *in vivo* component uses three data modalities: longitudinal tumor burden, terminal necrosis where matched histology is available, and terminal single-cell chromosome-number profiles.

Tumor burden. For each mouse $i \in \mathcal{I}$, where $\mathcal{I}$ denotes the eight retained tumors used for calibration (section 1.1), tumor burden is measured at observation times $\{t_{ij}^{(V)}\}_{j=1}^{m_i}$, yielding observed tumor volumes

$$V_{ij}^{\text{obs}} \equiv V_i^{\text{obs}}(t_{ij}^{(V)}). \quad (24)$$

Let $x_{i,N}^L(t)$, $x_{i,N}^{D,\text{hyp}}(t)$, and $x_{i,N}^{D,\text{CIN}}(t)$ denote viable abundance, dead abundance assigned by the model to loss associated with resource stress, and CIN-associated dead abundance, respectively, for tumor i in chromosome-count state N at time t . The model predicts tumor volume by scaling total burden:

$$\widehat{V}_{ij} = \sum_N \left(x_{i,N}^L(t_{ij}^{(V)}) + x_{i,N}^{D,\text{hyp}}(t_{ij}^{(V)}) + x_{i,N}^{D,\text{CIN}}(t_{ij}^{(V)}) \right) v_{\text{cell}}, \quad (25)$$

where each cell is assigned the effective volume

$$v_{\text{cell}} = \frac{1}{\rho_{2N}}. \quad (26)$$

This observation model does not impose a ploidy-dependent cell-volume correction: low- and high-ploidy cells contribute the same effective per-cell volume to the bulk burden. Tumor-burden observations are modeled on the logarithmic scale,

$$\log V_{ij}^{\text{obs}} \sim \mathcal{N}\left(\log \widehat{V}_{ij}, \sigma_{\text{burden}}^2\right), \quad (27)$$

where σ_{burden} is the burden measurement-noise scale. For longitudinal tumor burden, the fitted loss is a Gaussian negative log-likelihood on the logarithmic scale, averaged first within tumor and then across tumors. Let $\mathcal{J}_i$ denote the retained observation times for tumor i after excluding day 0. Then

$$\mathcal{L}_B(\theta) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \frac{1}{|\mathcal{J}_i|} \sum_{j \in \mathcal{J}_i} \left[\log \sigma_{\text{burden}} + \frac{1}{2} \log(2\pi) + \frac{\left(\log V_{ij}^{\text{obs}} - \log \widehat{V}_{ij}\right)^2}{2\sigma_{\text{burden}}^2} \right]. \quad (28)$$

In implementation, each volume entering a logarithm is replaced by $\max(V, \varepsilon_{\log B})$ with $\varepsilon_{\log B} = 10^{-12} \text{ mm}^3$; this numerical floor is suppressed in Eqs. (27)–(28) for readability. Because σ_{burden} is

estimated jointly, the $\log \sigma_{\text{burden}}$ term is retained in the likelihood contribution. Omitting it would allow the optimizer to reduce the burden loss by inflating the assumed measurement noise.

Terminal necrosis. For tumors with matched terminal histology, necrosis is compared with the simulated terminal fraction of total dead biomass. This is an observation-model term, not an additional live-to-dead flux. The stress-associated death hazard $\mu_{\text{eff}}(N, O_2)$ contributes to $\mathbf{x}^{D, \text{hyp}}$, while mitotic nonviability and boundary-dropped offspring contribute to $\mathbf{x}^{D, \text{CIN}}$. The terminal necrosis comparison uses the sum of these two dead pools.

For tumor i , define the terminal total-dead volume and terminal total tumor volume as

$$\widehat{V}_i^{D, \text{tot}}(\theta) = v_{\text{cell}} \sum_N \left[x_{i,N}^{D, \text{hyp}}(T_i^{(H)}; \theta) + x_{i,N}^{D, \text{CIN}}(T_i^{(H)}; \theta) \right], \quad \widehat{V}_i^{\text{tot}}(\theta) = v_{\text{cell}} \sum_N \left[x_{i,N}^L(T_i^{(H)}; \theta) + x_{i,N}^{D, \text{hyp}}(T_i^{(H)}; \theta) + \right. \\ \left. \right] \quad (29)$$

The predicted terminal necrotic fraction is

$$\widehat{f}_i^{\text{nec}}(\theta) = \frac{\widehat{V}_i^{D, \text{tot}}(\theta)}{\widehat{V}_i^{\text{tot}}(\theta)}. \quad (30)$$

Because the same effective cell-volume factor appears in both numerator and denominator, it cancels in this ratio. Thus the fitted necrosis term constrains the terminal total-dead fraction of the simulated tumor burden, after both dead pools have accumulated their respective inflows and have been cleared by the shared first-order clearance process.

Let $f_i^{\text{nec, obs}}$ be the observed terminal necrotic fraction for tumors in the matched necrosis set $\mathcal{I}_{\text{nec}}$. Fractions are compared on a clipped logit scale. Define

$$\text{logit}_\epsilon(f) = \log \left(\frac{\widetilde{f}}{1 - \widetilde{f}} \right), \quad \widetilde{f} = \min\{1 - \epsilon_{\text{nec}}, \max(\epsilon_{\text{nec}}, f)\}. \quad (31)$$

The terminal necrosis loss is the mean standardized squared discrepancy on this scale,

$$\mathcal{L}_{\text{nec}}(\theta) = \frac{1}{|\mathcal{I}_{\text{nec}}|} \sum_{i \in \mathcal{I}_{\text{nec}}} \left[\frac{\text{logit}_\epsilon(\widehat{f}_i^{\text{nec}}(\theta)) - \text{logit}_\epsilon(f_i^{\text{nec, obs}})}{\sigma_{\text{nec}}} \right]^2. \quad (32)$$

Here σ_{nec} sets the tolerated discrepancy in terminal necrosis on the logit-fraction scale, and ϵ_{nec} prevents fractions at 0 or 1 from producing infinite logits. In the run configuration these quantities are named `sigma_necrosis_logit` and `necrosis_fraction_eps`, respectively.

Ploidy. Each mouse has an initial ploidy group $g_i \in \{2N, 4N\}$ and harvest time $T_i^{(H)}$. Retained burden observations are nonzero-time measurements up to harvest, so the number and timing of observations may differ across mice.

For each tumor $i \in \mathcal{I}$, endpoint single-cell copy-number profiling provides modeled total autosomal chromosome-number estimates. If n_i cells are profiled from tumor i , the observed chromosome-number values are denoted

$$z_{i\ell}, \quad \ell = 1, \dots, n_i. \quad (33)$$

These single-cell observations sample the viable population at harvest. The model-predicted viable

chromosome-number distribution at harvest is

$$\hat{\pi}_{i,N}(\theta) = \frac{x_{i,N}^L(T_i^{(H)}; \theta)}{\sum_{N'} x_{i,N'}^L(T_i^{(H)}; \theta)}. \quad (34)$$

This distribution represents the predicted probability that a viable cell sampled from tumor i at harvest has true chromosome count N .

Observed single-cell chromosome-number estimates are then modeled conditional on this predicted viable distribution. Specifically, each observed value $z_{i\ell}$ is treated as a noisy measurement of an unobserved true chromosome-count state N , with mixture weights given by $\hat{\pi}_{i,N}(\theta)$:

$$p(z_{i\ell} | \theta) = \sum_N \hat{\pi}_{i,N}(\theta) \mathcal{N}(z_{i\ell}; N, \sigma_{\text{ploidy}}^2). \quad (35)$$

where σ_{ploidy} represents measurement variability in single-cell chromosome-number estimates. For terminal single-cell chromosome-number data, the fitted loss first computes a tumor-level average negative log-likelihood. Define

$$\ell_i(\theta) = -\frac{1}{n_i} \sum_{\ell=1}^{n_i} \log p(z_{i\ell} | \theta). \quad (36)$$

Let $\mathcal{I}_{2N}$ and $\mathcal{I}_{4N}$ denote the retained tumors initiated from the near-diploid and near-tetraploid lineages, respectively. With both cohorts present, the endpoint contribution gives equal weight to the two initial-ploidy cohorts and equal weight to tumors within each cohort:

$$\mathcal{L}_C(\theta) = \frac{1}{2} \left[\frac{1}{|\mathcal{I}_{2N}|} \sum_{i \in \mathcal{I}_{2N}} \ell_i(\theta) + \frac{1}{|\mathcal{I}_{4N}|} \sum_{i \in \mathcal{I}_{4N}} \ell_i(\theta) \right]. \quad (37)$$

Maximum-a-posteriori objective. The *in vivo* component is formulated as a maximum-a-posteriori objective combining burden likelihood, chromosome-number likelihood, terminal necrosis loss, and soft-prior penalties:

$$\mathcal{L}_{\text{vivo}}(\theta) = \mathcal{L}_B(\theta) + \mathcal{L}_C(\theta) + \lambda_{\text{nec}} \mathcal{L}_{\text{nec}}(\theta) + \lambda_{\text{prior}} \mathcal{L}_{\text{prior}}(\theta), \quad (38)$$

where θ denotes the fitted parameter vector for this component, $\mathcal{L}_B$ is the burden contribution, $\mathcal{L}_C$ is the terminal chromosome-number contribution, $\mathcal{L}_{\text{nec}}$ is the terminal total-dead-fraction loss, λ_{nec} is its relative objective weight, and $\mathcal{L}_{\text{prior}}$ is the soft-prior penalty.

For each prior-constrained parameter, let $z_i = T_i(\theta_i)$ denote the calibration-scale value used by the optimizer. The soft-prior penalty is

$$\mathcal{L}_{\text{prior}}(\theta) = \frac{1}{2} \sum_{i \in \mathcal{P}} \left(\frac{z_i - \mu_i}{\sigma_i} \right)^2, \quad (39)$$

where $\mathcal{P}$ is the subset of parameters with configured soft priors. The transform T_i is the parameter-specific optimizer transform (including $\log_{10}$ transforms for several positive parameters and the identity transform where configured), and μ_i and σ_i are the prior center and scale on that same

calibration scale.

***In vitro* observation model and likelihood.** The *in vitro* component uses the same chromosome-number transition model but applies it to passage-structured cell-line data. Each passage segment s has an initial cohort label, a fixed oxygen level during the segment, and a duration Δt_s ; available growth observations are represented as segment-level growth-rate data rather than being restricted to a single initial–final cell-count pair. The model propagates the corresponding viable chromosome-number distribution through the segment and compares the simulated passage outcome with three measured data modalities: passage growth-rate observations, passage-level chromosome-number observations, and flow-density measurements. For each passage, the propagated chromosome-number distribution used for reseeding and endpoint comparison was taken from the simulated time point whose live-cell count was closest to the observed final or next-passage initial cell count.

For growth-rate data, let g_s^{obs} be the observed passage growth rate and $\hat{g}_s(\theta)$ the model-predicted growth rate for segment s . The averaged growth log-likelihood is

$$\bar{\ell}_{\text{growth}}(\theta) = \frac{1}{|\mathcal{S}_g|} \sum_{s \in \mathcal{S}_g} \log \mathcal{N}(g_s^{\text{obs}}; \hat{g}_s(\theta), \sigma_{\text{growth}}^2), \quad (40)$$

where $\mathcal{S}_g$ is the set of segments with observed growth-rate data and σ_{growth} is the growth measurement-noise scale.

For chromosome-number observations in segment s , let $\pi_{s,N}$ denote the predicted viable chromosome-number distribution at the comparison time and let $z_{s\ell}$ be the observed integer chromosome-number value for cell ℓ . The implementation first smooths the predicted distribution on the finite modeled chromosome grid $\mathcal{N}$. For an observed-grid state m and true modeled state N , define the column-normalized Gaussian kernel

$$K_{\sigma_{\text{kary}}}(m | N) = \frac{\phi(m; N, \sigma_{\text{kary}}^2)}{\sum_{r \in \mathcal{N}} \phi(r; N, \sigma_{\text{kary}}^2)}, \quad m, N \in \mathcal{N}, \quad (41)$$

where $\phi(\cdot; N, \sigma_{\text{kary}}^2)$ is a Gaussian density evaluated only on the discrete chromosome grid. The smoothed discrete probability mass is

$$\tilde{\pi}_{s,m} = \sum_{N \in \mathcal{N}} K_{\sigma_{\text{kary}}}(m | N) \pi_{s,N}. \quad (42)$$

The averaged chromosome-number log-likelihood is then

$$\bar{\ell}_{\text{ploidy}}(\theta) = \frac{1}{|\mathcal{S}_z|} \sum_{s \in \mathcal{S}_z} \frac{1}{n_s} \sum_{\ell=1}^{n_s} \log [\max \{\tilde{\pi}_{s,z_{s\ell}}, 10^{-12}\}], \quad (43)$$

where $\mathcal{S}_z$ is the set of passage comparisons with chromosome-number observations, n_s is the number of observations in segment s , and σ_{kary} is the chromosome-number smoothing scale for the *in vitro* karyotype term. Observations that do not match a modeled grid state receive the same 10^{-12} probability floor. Thus, this term is a discrete grid likelihood after finite-grid kernel smoothing, not a continuous Gaussian-mixture density.

For flow-density data, let ω_{sp}^{obs} be the normalized observed density mass at flow grid point p for segment s , and let $\hat{f}_s(p; \theta)$ be the model-predicted density after Gaussian smoothing of the simulated viable chromosome-state distribution onto the observed flow grid and renormalization over that grid. The Gaussian width was set separately from the fitted karyotype smoothing parameter to the larger of twice the median flow-grid spacing and 1/22 ploidy units; for the seed-10 grid this was 0.0804020100502587. The averaged flow log-likelihood is

$$\bar{\ell}_{\text{flow}}(\theta) = \frac{1}{|\mathcal{S}_f|} \sum_{s \in \mathcal{S}_f} \sum_{p \in \mathcal{G}_s} \omega_{sp}^{\text{obs}} \log \left[\max \left\{ \hat{f}_s(p; \theta), 10^{-12} \right\} \right], \quad (44)$$

where $\mathcal{S}_f$ is the set of flow-profile comparisons and $\mathcal{G}_s$ is the observed flow grid for segment s . The 10^{-12} floor is a numerical safeguard applied after smoothing and normalization.

The fitted *in vitro* objective is the negative weighted mean log-likelihood

$$\mathcal{L}_{\text{vitro}}(\theta) = - \left[w_{\text{growth}} \bar{\ell}_{\text{growth}}(\theta) + w_{\text{ploidy}} \bar{\ell}_{\text{ploidy}}(\theta) + w_{\text{flow}} \bar{\ell}_{\text{flow}}(\theta) \right]. \quad (45)$$

The three terms compare the simulated passage lineages with growth-rate, chromosome-number, and flow-density data, respectively. The reported fits used equal component weights, $w_{\text{growth}} = w_{\text{ploidy}} = w_{\text{flow}} = 1$ (Supplementary Table 15).

Joint fitting of *in vivo* and *in vitro* data. The joint fit uses a single optimizer vector while evaluating separate transformed parameter vectors for the *in vivo* and *in vitro* likelihood components. Soft-coupled parameters are represented by a center-delta parameterization: for each soft-coupled parameter, the optimizer carries one transformed center variable, c_Q , and one additional offset variable, δ_Q . The context-specific transformed values are reconstructed from these two optimizer variables during objective evaluation. Parameters that appear only in one observation model are retained as context-specific observation-model parameters. In the analysis reported here, all 14 overlapping active biological parameters— $O_{2,c}$, μ_{hp} , p_{misseg} , $k_{o,\text{mis}}$, s_{max} , β_{buf} , n_{exp} , n_O , α_{O_2} , γ_{growth} , λ_{max} , $p_{\text{mis,base}}$, p_{wgd} , and γ_{μ} —were soft-coupled across contexts; no overlapping active biological parameter was hard-shared (Supplementary Table 14). Thus, the strength and chromosome-number dependence of resource-stress growth suppression were allowed to differ between the *in vivo* and *in vitro* components, while regularization discouraged unsupported divergence.

The minimized joint objective is

$$\mathcal{L}_{\text{joint}}(\theta) = w_{\text{vivo}} \mathcal{L}_{\text{vivo}}(\theta_{\text{vivo}}) + w_{\text{vitro}} \mathcal{L}_{\text{vitro}}(\theta_{\text{vitro}}) + \mathcal{L}_{\text{soft}}(\theta), \quad (46)$$

where θ_{vivo} and θ_{vitro} are the context-specific parameter vectors reconstructed from the joint optimizer vector and passed to the two likelihood components. The reported analysis used $w_{\text{vivo}} = w_{\text{vitro}} = 1$ (Supplementary Table 15). For each soft-coupled natural-scale parameter Q , let

$$z_Q = T_Q(Q)$$

denote its transformed optimizer-scale value, where T_Q is the transformation used for that parameter during optimization. The separate *in vivo* and *in vitro* fits define context-specific transformed bounds $[l_Q^{\text{vivo}}, u_Q^{\text{vivo}}]$ and $[l_Q^{\text{vitro}}, u_Q^{\text{vitro}}]$. In the joint fit, these component-specific bounds are used

only to define a single active joint admissible interval:

$$l_Q^{\text{joint}} = \min(l_Q^{\text{vivo}}, l_Q^{\text{vitro}}), \quad u_Q^{\text{joint}} = \max(u_Q^{\text{vivo}}, u_Q^{\text{vitro}}). \quad (47)$$

Thus, the original component-specific bounds record the provenance of the separate fits, whereas the joint optimizer uses the union interval $[l_Q^{\text{joint}}, u_Q^{\text{joint}}]$ as the active transformed-scale admissible range. During objective evaluation, the transformed values passed to the two components are reconstructed as

$$z_Q^{\text{vivo}} = c_Q + \frac{\delta_Q}{2}, \quad z_Q^{\text{vitro}} = c_Q - \frac{\delta_Q}{2}. \quad (48)$$

The reconstructed values must satisfy

$$z_Q^{\text{vivo}} \in [l_Q^{\text{joint}}, u_Q^{\text{joint}}], \quad z_Q^{\text{vitro}} \in [l_Q^{\text{joint}}, u_Q^{\text{joint}}]. \quad (49)$$

Parameter vectors that do not yield admissible reconstructed transformed values are outside the joint feasible domain and are not evaluated as valid likelihood points. The soft-coupling penalty is applied to the offset variable,

$$\mathcal{L}_{\text{soft}}(\theta) = \sum_{Q \in \mathcal{S}} \frac{c^2}{2} \left[1 - \exp \left\{ - \left(\frac{\delta_Q}{c\sigma_Q} \right)^2 \right\} \right], \quad (50)$$

where $\mathcal{S}$ is the set of soft-coupled parameters, σ_Q is the transformed-scale soft-coupling scale, and c controls the transition from near-quadratic behavior to saturation. For the reported fits, $\sigma_Q = 0.65$ for all 14 parameters and $c = 0.4$ (Supplementary Table 15). For $|\delta_Q| \ll c\sigma_Q$, each term approaches $\frac{1}{2}(\delta_Q/\sigma_Q)^2$; for large separations it saturates at $c^2/2$, limiting the influence of a single strongly context-specific parameter. Since

$$z_Q^{\text{vivo}} - z_Q^{\text{vitro}} = \delta_Q,$$

this penalty regularizes the transformed difference between the *in vivo* and *in vitro* parameter values. For log-transformed parameters, this corresponds to regularization of fold differences rather than absolute natural-scale differences.

Differential-evolution initial-population and optimizer-endpoint analysis. For Figure 5D, “initial distribution” denotes the population passed to differential evolution at the start of each joint optimization. It is not a distribution induced from the configured fitting intervals and is not obtained by weighting candidate values with the Welsch objective penalty. One endpoint ensemble was defined for each of C01, C02, and C03 using separate-*in vivo* seeds 366, 25, and 311, respectively. Each retained *in vivo* warm start was paired with the common separate-*in vitro* seed-10 anchor (Supplementary Tables 7 and 8).

The initial populations were deterministically reconstructed with the fitting code and saved run inputs. For each C family, joint seeds 1–500 were reset individually and the same `joint_deoptim_initial_population` routine used by the fit was called. The optimizer contained 40 coordinates, so the enforced population size was

$$N_P = \max(80, 10 \times 40) = 400.$$

Warm-start initialization was enabled with $\sigma_N = 0.1216$. For an ordinary transformed coordinate with warm-start value z_0 , the routine sampled a normal distribution centered at z_0 , with standard deviation $\sigma_N|z_0|$, truncated to the active optimizer bounds. When $|z_0| \leq 10^{-12}$, the active bound width replaced $|z_0|$ as the scale reference. For each soft-coupled parameter, the center was sampled first; the feasible bounds for the transformed context difference δ_Q were then recomputed from that sampled center so that both reconstructed context values remained inside the joint union interval. The delta coordinate was sampled from a truncated normal centered on the warm-start delta after clipping that mean to its center-dependent feasible interval, using the same scale rule. The first member of every 400-member population was replaced by the exact warm-start vector. Thus, each family-parameter initial distribution contained 200,000 values (500 seeds times 400 population members; Supplementary Table 8).

For every soft-coupled coordinate, the transformed context values were reconstructed as

$$z_Q^{\text{vivo}} = c_Q + \frac{\delta_Q}{2}, \quad z_Q^{\text{vitro}} = c_Q - \frac{\delta_Q}{2}. \quad (51)$$

For $\log_{10}$ -transformed parameters, natural values were obtained as

$$Q_{\text{vivo}} = 10^{z_Q^{\text{vivo}}}, \quad Q_{\text{vitro}} = 10^{z_Q^{\text{vitro}}}; \quad (52)$$

identity-scale coordinates already equaled the natural context values. Figure 5D displays these two natural-scale marginals separately rather than plotting their ratio. For every C family and coupled parameter, all 500 endpoints were required to have finite positive context values, to satisfy the joint feasible-domain check, and to require no post-optimization projection (Supplementary Table 8).

For each context-specific DE initial population and its corresponding endpoint ensemble, the natural-scale median and type-8 5th and 95th percentiles were calculated. The percentiles were retained for the numerical concentration audit and its source tables, but Figure 5D displays only optimizer-endpoint medians, as asterisks on the shared coordinate axis; initial-population median points and separate percentile spans are not drawn. Kernel densities were evaluated on a 401-point parameter-specific grid shared by the two contexts, both numerical roles, and all three families. Density estimation used the configured optimizer scale— $\log_{10}$ for log-transformed parameters and identity otherwise—while logarithmic axes were labeled with natural parameter values. Each density was divided by its own maximum; therefore only horizontal location, width, and shape, not vertical peak height, were compared (Supplementary Table 8).

The row order in Figure 5D used the single best retained joint winner from each C family. For parameter Q and family $f \in \{\text{C01}, \text{C02}, \text{C03}\}$, the natural-scale ratio was $R_{Qf} = Q_f^{\text{vivo}}/Q_f^{\text{vitro}}$. Parameters were ordered by increasing $\left| \frac{1}{3} \sum_f R_{Qf} - 1 \right|$, with parameter name as the deterministic tie breaker. This ordering is a display rule for proximity to cross-context parity, not an inferential test.

Boundary proximity in Figure 5D was defined from the complete joint-union fitting interval, not from a distribution quantile. For lower and upper bounds L_Q and U_Q on the optimizer coordinate, the two red background regions were

$$[L_Q, L_Q + 0.05(U_Q - L_Q)] \quad \text{and} \quad [L_Q + 0.95(U_Q - L_Q), U_Q]. \quad (53)$$

Thus each region occupies exactly 5% of the plotted horizontal interval. For $\log_{10}$ -optimized param-

eters these limits were calculated in $\log_{10}$ optimizer space and the tick labels were then converted back to natural units; identity-scale parameters required no conversion. The three retained families used the same joint-union bounds for each parameter.

As an auxiliary directional audit, the paired values were also transformed to $\log_2(Q_{\text{vivo}}/Q_{\text{vitro}})$. A parameter-family contrast was called higher *in vivo* when its endpoint 5th percentile exceeded zero, lower *in vivo* when its 95th percentile was below zero, and overlapping equality otherwise. Cross-family direction agreement required the same non-overlap classification in all three families. Paired-ratio fitted-solution concentration relative to initialization was summarized as

$$R_{W,Qf} = \frac{q_{0.95,Qf}^{\text{endpoint}} - q_{0.05,Qf}^{\text{endpoint}}}{q_{0.95,Qf}^{\text{DE init}} - q_{0.05,Qf}^{\text{DE init}}}. \quad (54)$$

We additionally reported, separately for *in vivo* and *in vitro*, the fraction of the 500 endpoints that lay at the corresponding active parameter bound (Supplementary Table 8).

The fixed red edge bands show whether a fitted marginal approaches the extremes of the declared joint fitting domain, while the source-table 5th–95th percentile summaries and $R_{W,Qf}$ provide the separate numerical-concentration audit. Both are conditional on the tested warm starts, bounds, objective, and search settings. Neither the initial populations nor the endpoints are posterior or confidence distributions; they do not add biological replication or establish structural identifiability. In particular, a zero endpoint span may result from many numerical starts converging to the same endpoint, whereas high active-bound occupancy indicates optimizer pressure against the declared fitting domain rather than a resolved biological optimum.

Warm-start initialization for the joint fit. The main joint analysis used three landscape-informed separate *in vivo* fits as warm-start representatives. C01, C02, and C03 denote the three regions of the pooled 14-parameter warm-start embedding. These regions were obtained by k-means clustering of the 500 final *in vivo* positions in the pooled t-SNE embedding described below, not from the *in vivo*-only 18-parameter landscape in Figure 4B. Candidate partitions with $k = 2, \dots, 8$ were fitted with 100 starts and evaluated by average silhouette.²⁸ Although $k = 4$ gave the numerical maximum (0.816651), it improved on $k = 3$ by only 0.000957 and created a singleton region. We therefore retained the smaller non-singleton $k = 3$ partition (average silhouette 0.815694; region sizes 99, 385, and 16) as a parsimonious warm-start coverage design. This choice was audited with 100 one-start reruns on the frozen coordinates, 100 80%-endpoint subsamples drawn without replacement with re-evaluation of $k = 2, \dots, 8$, and comparison with a $k = 3$ partition fitted directly in standardized 14-parameter endpoint space (Supplementary Figure 6; Supplementary Table 7). The archived reproducibility bundle retains all 228,000 frozen t-SNE coordinates but not the corresponding 228,000 standardized 14-parameter input vectors; t-SNE seed and perplexity sensitivity therefore could not be re-estimated from the archived bundle, and the coordinate-space audits condition on the saved embedding. These checks assess reproducibility of an optimizer-search partition; they do not validate biological subtypes or make t-SNE coordinates mechanistic variables.

Within each primary region, the separate-*in vivo* endpoint with the lowest standalone objective was retained. This selected seeds 366, 25, and 311 for C01, C02, and C03, respectively. Each was paired with the same lowest-objective separate *in vitro* fit (seed 10) on the transformed optimizer scale. Thus, the main design explored three *in vivo* primary regions against one fixed *in vitro*

anchor rather than a balanced cross-product of both landscapes (Supplementary Table 7).

For each soft-coupled parameter Q , let $z_Q^{\text{vivo},0}$ and $z_Q^{\text{vitro},0}$ denote the transformed values from one such C-family configuration. The initial center and offset are set to

$$c_Q^0 = \frac{z_Q^{\text{vivo},0} + z_Q^{\text{vitro},0}}{2}, \quad \delta_Q^0 = z_Q^{\text{vivo},0} - z_Q^{\text{vitro},0}. \quad (55)$$

This initialization exactly reconstructs the two separate-fit transformed values:

$$z_Q^{\text{vivo},0} = c_Q^0 + \frac{\delta_Q^0}{2}, \quad z_Q^{\text{vitro},0} = c_Q^0 - \frac{\delta_Q^0}{2}.$$

Parameters that appear only in one component, such as context-specific observation-model parameters, are initialized from the corresponding separate-fit solution.

The warm start must be representable under the active joint union bounds in Eq. (47); otherwise, initialization fails explicitly. This ensures that the joint fit begins from the selected separate-fit solutions only when those solutions are valid under the declared joint feasible domain.

Optimization and data inclusion. Optimization attempted a parallel differential-evolution global stage followed by serial L-BFGS-B refinement; a local result was retained only when it passed the implementation checks and improved the objective. Separate *in vivo* and *in vitro* searches each used 500 numerical seeds. In the main joint analysis, each of the three retained C-family configurations used 500 numerical seeds, for 1,500 optimizer runs. All 1,500 selected runs recorded an attempted local stage with convergence code 52 and `optimizer_local_accepted=FALSE`; the reported joint solutions are therefore the retained differential-evolution endpoints. Every selected joint differential-evolution run ended by the configured relative-tolerance/step-tolerance early-stopping rule within the 500-iteration target. The separate *in vitro* convergence summary labeled 0/500 differential-evolution runs as converged; the best retained seed-10 local step was accepted but returned convergence code 1. In the separate *in vivo* search, the convergence flag was recorded for all 500 starts, 29 objectives were within 1% and 241 were within 5% of seed 25, and the selected seed-25 local step was accepted with code 52 while two of 20 active parameters were exactly at bounds. The corresponding diagnostics are summarized in Figure 3G, Supplementary Figures 4 and 2, and Supplementary Tables 6, 4, and 7. In all cases, “best” denotes the lowest retained objective among the supplied searches, not a proven global optimum. Seed-to-seed spread was used to assess numerical search robustness and was not interpreted as biological replication or inferential uncertainty.

For the Figure 6A multi-seed analysis, eligibility was defined independently within C01, C02, and C03 because the families had different objective baselines. A retained endpoint first had to have a finite recorded full joint MAP objective, a valid unique seed identifier, exactly one finite value for each of the 14 response parameters in both contexts, feasibility both before projection and at the reported solution, no applied parameter projection, and an available local model configuration. Eligible endpoints were then ordered by the complete retained joint MAP objective, including the fitted-data terms and soft-coupling penalty. The primary ensemble was the lowest objective decile within each family (50 of 500 endpoints); nested lowest-5% (25 endpoints) and lowest-20% (100 endpoints) sets were used as threshold-sensitivity analyses. Within an eligibility set, endpoints received equal weight; objective values determined inclusion but were not converted into posterior-

like weights. The decile rule is an operational numerical-fit filter, not a likelihood-ratio cutoff, posterior credible set, or confidence region. Every retained operator evaluation additionally had to return finite dominant-mode summaries with status `ok`, and the realized forced missegregation probability had to agree with the requested value within 10^{-8} . To prevent repeated convergence to an identical numerical solution from being mistaken for independent parameter evidence, context-specific exact signatures were formed from the ordered natural-scale values of all 14 response parameters within each family. The lowest-objective representative of each signature was evaluated once and its original optimizer-seed multiplicity was restored for seed-weighted summaries; an equal-weight unique-endpoint summary was retained as a sensitivity analysis. Seed-weighted and unique-endpoint-weighted modal claims were compared at all three objective thresholds. Exact inclusion, exclusion, endpoint multiplicity, threshold, and deduplication diagnostics are reported in Supplementary Figure 6 and Supplementary Tables 7 and 10.

Optimizer-ensemble sensitivity analyses. For coupled parameter Q , warm-start family $f \in \{C01, C02, C03\}$, and numerical start $s \in \{1, \dots, 500\}$, the context contrast was

$$D_{Qfs} = \log_2 \left(\frac{Q_{fs}^{\text{vivo}}}{Q_{fs}^{\text{vitro}}} \right). \quad (56)$$

Within each C family, the 500 endpoints contributed equal weight to the endpoint-class fractions and the empirical distribution shown in Supplementary Figure 3. Natural-scale ratios were classified as higher *in vitro* for $0 < Q_{\text{vivo}}/Q_{\text{vitro}} \leq 0.8$, approximately equal for $0.8 < Q_{\text{vivo}}/Q_{\text{vitro}} < 1.2$, and higher *in vivo* for ratios at least 1.2. The figure displays the three class fractions and the corresponding D_{Qfs} distributions separately for C01, C02, and C03 rather than reducing each family to a dominant label or pooling the 1,500 runs. Horizontal violins were drawn only for family-parameter groups with more than one distinct endpoint ratio; a group with 500 identical ratios was represented by its coincident median symbol rather than an artificial density width. For visualization, the log-ratio axis was limited on the left to -5.5 , below the observed minimum, and the data-empty interval $6.2 < D_{Qfs} < 12.2$ was removed with an explicit axis break. Values at or above 12.2 were retained in a compressed continuation, including all 500 coincident C03 $p_{\text{mis,base}}$ ratios at 12.43. This display transformation changes neither classification nor summary calculations. Because all three families share the same *in vitro* anchor, cross-family agreement describes numerical-search robustness rather than independent evidence across experimental replicates. These analyses characterize optimization and warm-start sensitivity; they are not posterior uncertainty estimates or structural-identifiability tests. The parameter-specific endpoint distributions and search design are reported in Supplementary Tables 7 and 8.

Retained *in vivo* observation times satisfy $0 < t_{i,1} < \dots < t_{i,n_i} \leq T_i^{(H)}$, so day-0 burden points are excluded and longitudinal follow-up is administratively truncated at harvest. No censoring likelihood was used. Only tumors with both burden and terminal chromosome-number data are retained. Terminal endpoint discrepancies are evaluated directly from the single-cell chromosome-number observations under the Gaussian-mixture likelihood in Eq. (35), and tumor-level endpoint losses are aggregated equally within each initial-ploidy cohort before applying cohort balancing as in Eq. (37).

Representative *in vivo* trajectory and terminal-distribution visualization. The lowest-objective retained separate *in vivo* fit (seed 25) was propagated at one-day reporting intervals for the eight paired tumors. The four 2N-initiated and four 4N-initiated trajectories were summarized separately (Supplementary Tables 1 and 2). At each day, the viable model state, the dead model state assigned to loss associated with resource stress, and the CIN-associated dead model state were converted to tumor-volume components and averaged over the tumor trajectories available on that day before being plotted as stacked areas. The fitted effective oxygen state was plotted directly in percent. Viable-cell mean ploidy was calculated within each tumor as

$$\bar{P}_i(t) = \frac{\sum_{N \in \mathcal{N}} N x_{i,N}^L(t)}{N_{\text{unit}} \sum_{N \in \mathcal{N}} x_{i,N}^L(t)}, \quad N_{\text{unit}} = 22. \quad (57)$$

The tumor-specific effective- O_2 and $\bar{P}_i(t)$ trajectories were then averaged over the same available trajectories at each day. Both cohort means were displayed without standardization on the numerical 0–5 left axis, while burden used the right axis. Sharing a coordinate axis is a visual alignment only and does not imply common physical units. Both longitudinal panels use a 0–100-day x-axis; model curves stop at the last available modeled time rather than being extrapolated to day 100.

Observed tumor burden was summarized only at recorded measurement days. For cohort c , day t , and n_{ct} observed tumors, the black curve is the arithmetic mean $\bar{B}_c^{\text{obs}}(t)$, and the gray band is $\bar{B}_c^{\text{obs}}(t) \pm s_c(t)$, where $s_c(t)$ is the between-tumor sample standard deviation, truncated at zero for the lower display boundary. All available observations at each recorded day contributed to these descriptive summaries. The band is not measurement error, a confidence interval, or model uncertainty. The 120 selected burden measurements were cross-checked by tumor and day between the independently prepared Figure 1 and Figure 4 analysis tables, with exact agreement (Supplementary Table 1).

For the terminal comparison, the observed single-cell chromosome-number values and predicted terminal chromosome-state weights from the seed-25 fit were converted to ploidy by division by $N_{\text{unit}} = 22$. Within each cohort and source, each tumor was normalized to unit total weight and then assigned one-quarter of the cohort distribution. This equal-tumor weighting prevents tumors with more profiled cells from dominating the observed cohort distribution. Observed and predicted distributions were displayed on the common ploidy range 1–7 with weighted box summaries (Supplementary Tables 1 and 2).

Fixed-oxygen attractor analysis. For each of the 500 separate *in vivo* parameter sets, the chromosome-number model was evaluated at fixed oxygen without parameter refitting. At each oxygen level, the chromosome-transition generator was combined with the state-specific effective death rate to form

$$M(O_2) = G_{\text{div}}(O_2) - \text{diag}\{\mu_{\text{eff}}(N, O_2)\}. \quad (58)$$

The eigenvalue of M with the largest real part, λ_1 , was calculated numerically, and the real part of its associated eigenvector was retained. The eigenvector sign was oriented to have a non-negative sum, negative numerical components were truncated to zero, and the vector was normalized to unit

mass. With normalized dominant eigenvector v_N , dominant mean ploidy was

$$\bar{P}_{\text{dom}}(O_2) = \frac{1}{N_{\text{unit}}} \sum_{N \in \mathcal{N}} N v_N. \quad (59)$$

The spectral gap was defined as the difference between the largest and second-largest eigenvalue real parts, $\text{Re}(\lambda_1) - \text{Re}(\lambda_2)$. The operator was evaluated at 201 fixed oxygen concentrations from 0% through 5% O_2 in increments of 0.025 percentage points. The continuous dominant mean ploidy from Eq. (59), without thresholding at ploidy 2, was retained for the Figure 4B association analysis (Supplementary Tables 3 and 4). These are analytical evaluations of the fitted linear operator, not stochastic simulations or oxygen-intervention experiments.

Post-fit continuous single-parameter ploidy-association analysis. At each of the 201 oxygen concentrations, the analysis units were the 500 parameter sets obtained from distinct numerical optimizer seeds, not biological replicates or posterior draws. Each of 18 prespecified fitted *in vivo* parameters was analyzed separately; the observation/volume parameters ρ_{2N} and σ_{burden} were excluded (Supplementary Tables 3 and 4). Let $R_{\theta,ij}$ be the average rank of fitted parameter θ across the 500 endpoints and let $R_{P,ij}$ be the average rank of continuous dominant mean ploidy at oxygen value $O_{2,j}$. Spearman association was calculated as

$$\rho_{\theta}(O_{2,j}) = \frac{\text{cov}(R_{\theta,ij}, R_{P,ij})}{\text{sd}(R_{\theta,ij}) \text{sd}(R_{P,ij})}. \quad (60)$$

Ties received average ranks. Natural-scale parameter values were used for this calculation; applying the configured $\log_{10}$ transformation to strictly positive parameters gives the same ranks and therefore the same ρ .

For each parameter, the displayed peak association magnitude was

$$M_{\theta} = \max_{j=1, \dots, 201} |\rho_{\theta}(O_{2,j})|. \quad (61)$$

Let j_{θ}^* be the lowest oxygen-grid index attaining M_{θ} . Figure 4B rows were first divided by the sign of $\rho_{\theta}(O_{2,j_{\theta}^*})$, with the positive group displayed before the negative group, and were then ordered by decreasing M_{θ} within each group; configured parameter order was used only to break an exact tie. The adjacent dot position reports M_{θ} , while its fill reports the signed value $\rho_{\theta}(O_{2,j_{\theta}^*})$. Selecting a maximum over 201 oxygen values emphasizes the strongest oxygen-specific association and does not measure persistence across the grid. The full signed heatmap is therefore retained to show whether that peak direction persists or reverses with oxygen. Neither M_{θ} nor the signed peak correlation is a causal effect, hypothesis-test statistic, multivariable importance estimate, or out-of-sample prediction measure.

Separate *in vivo* parameter-landscape embedding and exploratory clustering. For Figure 4C, the differential-evolution initial population was deterministically reconstructed for each of the 500 separate *in vivo* fits from its optimizer seed, ordered 20-variable parameter table, transformed bounds, and effective initialization rule. Although the saved configuration recorded `de_init_mode=hybrid`, the single-stage backend explicitly used uniform initialization; all 500 fit

logs recorded `init_mode=uniform`, zero local candidates, and 255 uniform candidates after the parameter-table initializer. The full 20-variable population was reconstructed before feature selection to preserve the optimizer random-number sequence. Candidate 1, which reproduced the parameter-table initializer, was then removed from every 256-member population. The observation/calibration variables ρ_{2N} and σ_{burden} were excluded, leaving 18 mechanistic parameters and 127,500 retained initial-population rows. These were combined with the 500 best endpoints, giving 128,000 *in vivo*-only parameter vectors. Each parameter was transformed on its configured natural or $\log_{10}$ optimizer scale and z-scored across the combined data. The vectors were embedded in two dimensions by t-SNE using random seed 123, perplexity 30, Barnes–Hut approximation parameter $\theta = 0.5$, and 1,000 iterations (Supplementary Table 4).

The 500 best endpoints were retained and k-means clustering was applied to their two-dimensional t-SNE coordinates for $k = 2, \dots, 8$. The value maximizing the average silhouette coefficient was selected; this procedure selected $k = 3$ with average silhouette 0.655 and cluster sizes of 43, 412, and 45 solutions (Supplementary Table 4). Figure 4B displays each original parameter value on a shared $\log_{10}$ coordinate, independently of whether its optimizer transformation was logarithmic or identity. Strictly positive values were mapped to $\log_{10}(x)$. Because one configured fitting interval had a zero lower bound, a dedicated display coordinate one decade below the smallest positive bound, initial value, or fitted endpoint was labeled “0”; in this analysis the display floor was 10^{-7} . The upper pink half violin is a kernel-density estimate of the 500 best retained endpoints on the displayed $\log_{10}$ coordinate, and its green point marks seed 25, verified as rank 1 under the canonical 500-seed fit objective. The lower gray glyph spans the configured lower and upper fitting bounds and its black point marks the configured initial value. These configured ranges and optimizer endpoints are not measurement bounds, posterior draws, confidence intervals, or biological replicates.

For Supplementary Figure 1, each natural-scale endpoint was instead transformed on its configured optimizer scale and mapped to the unit interval defined by its configured transformed lower and upper bounds. The plotted prior-referenced position was $z_{\text{prior}} = (u - 0.5)/\sqrt{1/12}$, where u is the bound-scaled value; zero therefore denotes the midpoint of the configured interval and one unit is the standard deviation of a uniform distribution over that interval. The endpoint values were stratified by the three exploratory t-SNE groups and shown in the same positive-peak then negative-peak order as Figure 4B, with decreasing M_θ within each group. These summaries remain distributions of optimizer-derived fitted endpoints, not a Bayesian posterior; their percentiles are not credible intervals, measurement bounds, or biological confidence intervals. Solid t-SNE boundaries are convex hulls of all assigned endpoints and therefore contain every plotted member, but they remain descriptive membership outlines. Because the clusters and displayed parameters were derived from the same fitted ensemble, cluster-associated parameter differences are descriptive fitted-landscape structure rather than independent validation, biological subtype evidence, or causal parameter effects (Supplementary Table 4).

For the exploratory Figure 4D stratification, the three cluster distributions were compared separately for each of the 18 parameters with a Kruskal–Wallis rank-sum test on z_{prior} (Supplementary Table 4). This transformation is monotone within each parameter, so it preserves the ranks and test statistic obtained from the corresponding configured transformed parameter values. The 18 omnibus P values were adjusted together by the Benjamini–Hochberg procedure. For parameter

θ , the displayed effect size was

$$\epsilon_{\theta}^2 = \max\left\{0, \frac{H_{\theta} - k + 1}{n - k}\right\}, \quad (62)$$

where H_{θ} is the Kruskal–Wallis statistic, $k = 3$ groups, and $n = 500$ fitted endpoints. Parameters with adjusted $q < 0.05$ were ranked by decreasing ϵ_{θ}^2 ; adjusted q and then the Figure 4B display order were prespecified tie breakers, and the first six were shown. This test uses numerical optimizer endpoints rather than biological replicates. Moreover, because the tested parameters were also inputs to the t-SNE and cluster assignment, the adjusted values and effect sizes describe internal cluster separation and cannot serve as independent validation or biological-subtype evidence.

Pooled parameter-landscape embedding and joint-fit warm-start selection. A separate pooled landscape was constructed from optimizer population samples labeled as initial points and the final fitted solutions from the separate *in vivo* and *in vitro* calibrations. This landscape was used to design the joint-fitting warm starts shown in Figure 5C; it is a joint-fitting initialization result rather than an embedding of optimized joint-fit endpoints, and it is distinct from the 18-parameter *in vivo*-only landscape above. Fourteen parameters shared by the two contexts were used: $\lambda_{\max}$, $p_{\text{mis,base}}$, p_{misseg} , $k_{o,\text{mis}}$, $s_{\max}$, β_{buf} , n_{exp} , p_{wgd} , α_{O_2} , γ_{growth} , μ_{hp} , γ_{μ} , $O_{2,c}$, and n_O . Parameters were transformed on their configured natural or $\log_{10}$ scales and standardized jointly across the pooled data. By configuration, the first stored population row for each seed was removed in both contexts. In the *in vivo* export that row represents an optimizer initializer, whereas the *in vitro* export does not distinguish it from an ordinary sampled population row; this asymmetric provenance is a limitation of the warm-start landscape, not of the component likelihoods. The resulting 228,000 parameter vectors—127,500 *in vivo* population points, 99,500 *in vitro* population points, and 500 final solutions from each context—were embedded in two dimensions by t-SNE²⁹ using random seed 123, perplexity 30, Barnes–Hut approximation parameter $\theta = 0.5$, and 1,000 iterations. The retained t-SNE axes are unitless visualization coordinates that preserve local neighborhood structure only approximately; neither axis is a fitted parameter, a latent biological variable, or a mechanistic “distance.” The primary-region selection procedure, stability audits, and three retained warm-start representatives are specified above, in Supplementary Figure 6, and in Supplementary Table 7.

Weak-gap dominant-ploidy regime robustness. For Supplementary Figure 7, we reused the same 201 oxygen values, 60 log-spaced fixed effective per-chromosome missegregation probabilities, three C families, and family-specific lowest-objective 50-joint-endpoint ensembles used in Figure 6A. Exact duplicate 14-parameter endpoints were identified separately for the *in vivo* and *in vitro* vectors, evaluated once, and restored to their original optimizer-seed multiplicities before pointwise aggregation; both contexts yielded 50, 15, and 50 unique signatures for C01, C02, and C03, respectively (Supplementary Tables 10 and 13). The rank-1 eigenvector was oriented to have positive total mass and normalized with the same nonnegative rule used for Figure 6A. Its chromosome-number-weighted mean divided by N_{unit} defined dominant mean ploidy $P_{\text{esc}}(O_2, p)$ for endpoint e , family s , context c , oxygen concentration O_2 , and forced effective missegregation probability p . Pointwise rank-1 medians and 10th and 90th percentiles were required to agree with the corresponding context-specific Figure 6A source surface to numerical tolerance. The fraction with $P_{\text{esc}} \geq 2$ was retained only as an additional source-surface crosscheck and was not used to define the high class.

A grid cell was classified as weak-gap when at least half of its 50 endpoints had $\text{Re}(\lambda_1) - \text{Re}(\lambda_2) <$

0.005, matching the Figure 6A overlay. For each endpoint, dominant mean ploidy was classified as low when $P_{es} \leq 2$, intermediate when $2 < P_{es} < 4$, and high when $P_{es} \geq 4$. At each grid cell, q_{low} , $q_{\text{intermediate}}$, and q_{high} were the corresponding fractions among the 50 accepted optimizer-seed endpoints, and $\max(q_{\text{low}}, q_{\text{intermediate}}, q_{\text{high}})$ was regime consensus. A cell was assigned a stable class when the corresponding endpoint fraction was at least 0.9 and was called mixed otherwise. Exact across-fit dispersion was reported as the 90th minus 10th percentile of P_{es} (Supplementary Table 13).

Local grid sensitivity was calculated independently for every context-specific endpoint vector before aggregation. The neighborhood of a grid cell contained its available immediately adjacent oxygen values and immediately adjacent fixed-misseggregation values, for at most four neighbors. The endpoint-level regime-switch indicator equaled one when any neighbor changed among the low, intermediate, and high classes defined above. The corresponding local-change amplitude was the largest absolute difference in P_{esc} among those neighbors. Supplementary Figure 7B reports the endpoint fraction with a local three-class switch, panel C reports the across-endpoint percentile spread at the focal cell, and panel D reports the empirical cumulative distribution of the endpoint-median local-change amplitude across weak-gap cells, separately for *in vivo* and *in vitro*. Boundary cells used only their available neighbors. These grid-local quantities measure sensitivity to one numerical evaluation step; they are not derivatives, finite-time transition rates, probabilities that cells switch state, or biological confidence levels. Higher eigenmodes were retained in the reproducibility audit but were not used as alternative population distributions because non-dominant eigenvectors may be signed or complex.

Fixed-oxygen response-curve classification and reliability. For each final separate-fit parameter set in each context, dominant mean ploidy and spectral gap were evaluated from 0% to 5% O₂ in increments of 0.025 percentage points, giving 201 oxygen values per solution and 100,500 seed-oxygen evaluations per context. The same context-appropriate operator implementation and the same response classifier were used for both sets. Each response curve was smoothed with a degree-2 Gaussian-family LOESS fit with span 0.20 under the versioned rule `loess_persistent_v1`. A smoothing spline with `spar=0.65` was used if LOESS failed or produced values outside the implemented plausibility envelope; if smoothing still failed, the raw finite values were retained. Finite differences of the smoothed curve were classified as positive, negative, or effectively zero using a tolerance equal to the larger of 10^{-6} and 10^{-4} times the fitted ploidy range. Persistent segments were required to contain at least three grid intervals, span at least 2% of the oxygen range, and have amplitude at least the larger of 0.01 ploidy units and 3% of the fitted range. A terminal plateau required at least 10% of the oxygen range after the last persistent segment and a trailing amplitude no greater than the segment-amplitude threshold. Curves with a fitted range no greater than 0.05 were classified as approximately flat. Remaining curves were classified from the ordered persistent slope signs as monotone increasing, monotone decreasing, a single increase or decrease followed by a plateau, U-shaped, inverted U-shaped, or complex nonmonotone. For a curve with multiple persistent signs to be reduced to a monotone class, the total reverse-direction amplitude could not exceed 5% of the larger of the forward amplitude and fitted range (Supplementary Table 9).

For seed-level reliability summaries, a response curve was flagged as unreliable when at least 25% of its oxygen-grid values had a spectral gap below 0.005. Curves were flagged for caution when any grid value had a gap below 0.005 or when at least 10% of grid values had a gap below 0.01; all

remaining curves were labeled reliable (Supplementary Table 9). These diagnostics indicate weak separation between the first two eigendirections and are not additional fitted-data hypothesis tests.

For the family-specific response surfaces in Figure 6A, every endpoint in the family-specific lowest-objective decile supplied both its context-specific *in vivo* and *in vitro* parameter vectors, giving 50 vectors per context and family. Each vector was evaluated under its own model context on the same 201-point oxygen grid and on 60 log-spaced effective per-chromosome missegregation probabilities from 0.005 through 0.5, giving 12,060 operator evaluations per context-specific endpoint vector. To isolate this probability as a standardized input axis, $p_{\text{mis},\text{base}}$ was set to the grid value and the stress-linked increment p_{misseg} was set to zero; all other parameters retained that endpoint’s fitted context-specific values. This construction makes the per-chromosome missegregation probability constant across chromosome states at each grid point. The realized population-average probability was recalculated from the dominant eigenvector as $\sum_N v_N p_{\text{mis}}(N, O_2)$ and was required to agree with the requested input to within 10^{-8} . The main panel displays C01, C02, and C03 as columns, with *in vivo* and *in vitro* as aligned rows (Supplementary Table 10).

At each oxygen–missegregation grid point and within each context, the displayed surface is the median dominant mean ploidy across the 50 eligible joint endpoints. Purple diagonal hatching marks positions where at least half of the context-specific endpoint vectors had spectral gap below 0.005, and a purple dotted contour delimits each such region; white crosses mark sampled positions where fewer than 80% agree on whether dominant mean ploidy is at least 2. The unmodified fixed- O_2 trajectory was also calculated separately for all 50 endpoint vectors in each context: the black curve is the pointwise median of the model-implied population-average missegregation probability and the gray band is its 10th–90th percentile interval. The surface therefore summarizes a standardized post-fit response ensemble, whereas the trajectory indicates the portion of that plane traversed by the corresponding unmodified fitted functions (Supplementary Table 10).

Figure 6B applies the same standardized fixed-input operator construction at higher missegregation resolution in both contexts. The fixed-probability grid was $p_i = 0.005, 0.006, \dots, 0.500$, giving 496 values at an interval of 0.001. For each context c and each p_i , one forward curve connects the 201 pointwise medians $\{\tilde{P}_c(O_{2,j}, p_i) : j = 1, \dots, 201\}$, where $\tilde{P}_c$ is the median dominant mean ploidy across the 50 eligible context-specific endpoint vectors in that C family. Values at all 496×201 conditions were calculated directly; no interpolation was used. Exact signatures were identified separately within each context, one representative of each signature was evaluated, and its original optimizer-seed multiplicity was restored before calculating the 50-endpoint pointwise median. In both contexts, the C01, C02, and C03 families contained 50, 15, and 50 exact unique endpoint vectors, respectively, requiring 11,465,040 distinct operator evaluations per context and 22,930,080 across the paired display. Four forward trajectories at $p_{\text{miss},\text{eff}} = 0.01, 0.10, 0.20$, and 0.30 were overlaid on the inverse map in black using solid, dash-dot-dot, dot-dash, and dotted lines, respectively. For each context, family, and oxygen value, the red summary curve was calculated as $\bar{P}_c(O_{2,j}) = 496^{-1} \sum_{i=1}^{496} \tilde{P}_c(O_{2,j}, p_i)$. This arithmetic mean assigns equal weight to every evaluated fixed probability and is a visualization summary of the forced-input grid rather than a fitted or probability-weighted model trajectory. These five overlays repeat or summarize existing evaluated forward curves and do not constitute additional model calculations. All context–family subpanels share the oxygen and ploidy scales (Supplementary Table 10).

The heatmaps in Figure 6B numerically inverted the context-specific endpoint-level dense curves rather than the pointwise median curve. The target dominant-mean-ploidy grid was 1.000–7.000

in increments of 0.025, giving 241 targets at each of the same 201 oxygen concentrations. For every unique context-specific endpoint vector and oxygen value, the 496 ordered $(p_i, P_{esc}(O_2, p_i))$ points were partitioned into monotonic branches and represented by piecewise-linear segments. All intersections with each target ploidy were retained; coincident intersections at a turning point were collapsed within a probability tolerance of 10^{-10} , and a target coincident with a flat segment was classified as having multiple solutions. Exact endpoint multiplicity was then restored to recover the original 50-seed weighting. A grid cell was assigned a stable unique inverse when at least 80% of the seed weight supplied exactly one solution and fewer than 20% supplied multiple solutions. Cells in which at least 20% supplied multiple solutions were classified as multiple-solution cells; all remaining cells were classified as having no stable unique inverse, including targets outside the endpoint-supported range. For stable unique cells, Figure 6B shows the seed-weighted empirical median inverse probability; empirical 10th and 90th percentiles were retained in the source table. Substitution into the same context-specific piecewise-linear forward curves was used as a numerical reconstruction check. This inversion introduces interpolation between already evaluated p_i values but no new model fit or operator evaluation. Target ploidy is an inverse-query coordinate, not a clamped model state, and gray or hatched regions do not support one biological “required” missegregation probability (Supplementary Table 10).

The nested lowest-5% ensemble used the same complete surfaces. To test sensitivity to widening the objective filter to 20% without generating unused full-resolution maps for ranks 51–100, those additional endpoints were evaluated on the complete 201-point unmodified trajectory and on a prespecified diagnostic union: all 60 missegregation values at 0%, 1%, and 5% O_2 , together with missegregation slices of 0.005, 0.0481 (the grid point nearest 0.05), and 0.5 across all 201 oxygen values. This 774-point union identifies the presence of both ploidy regimes at the three reference oxygen levels, the direction of oxygen response at three missegregation levels, and the evaluated-domain spectral-gap reliability fraction. Dominant mean ploidy at least 2 defined the high-ploidy regime. An O_2 -response direction was labeled increasing or decreasing when the 5%-minus-0% change exceeded +0.05 or was below −0.05 ploidy units, respectively, and approximately flat otherwise. Qualitative claim support was tabulated separately for the 5%, 10%, and 20% objective sets, both over optimizer seeds and after exact parameter-endpoint deduplication. These empirical support fractions quantify agreement among numerical optimizer endpoints and are not biological confidence levels or posterior probabilities (Supplementary Table 10).

Embedding of oxygen-response classes and fit objectives. For the response-class diagnostic, the 500 final endpoints from each context were taken from the same pooled 14-parameter joint-warm-start t-SNE embedding described immediately above; no second t-SNE embedding was estimated. Supplementary Figure 5C therefore uses two-dimensional pooled-embedding coordinates and does not use the *in vivo*-only 18-parameter Figure 4B coordinates. The coordinate labels denote only the first and second axes returned by t-SNE: they are unitless, may change with embedding settings, and have no direct mechanistic or biological interpretation. Endpoints were joined by context and optimizer seed to their smoothed fixed- O_2 response classes and complete fitted MAP objective values. Circles and triangles distinguish *in vivo* and *in vitro*; within every nonempty context–class combination, a black outline marks its lowest-objective endpoint. Supplementary Figure 5D subtracts the minimum objective within each context before comparing class distributions, so its vertical scale is relative fit quality and does not equate the two context-specific

objectives. The class and objective overlays were used descriptively to assess landscape overlap, not as independent tests of class separation (Supplementary Tables 9 and 4).

Visualization and reporting. For visualization and reporting in cell-number units, observed tumor volume (Eq. (24)) is converted to an approximate cell-count range using the specified diploid packing-density bounds:

$$N_{ij}^{\text{obs,low}} = V_{ij}^{\text{obs}} \rho_{2N,\text{min}}, \quad N_{ij}^{\text{obs,mid}} = V_{ij}^{\text{obs}} \sqrt{\rho_{2N,\text{min}} \rho_{2N,\text{max}}}, \quad N_{ij}^{\text{obs,high}} = V_{ij}^{\text{obs}} \rho_{2N,\text{max}}. \quad (63)$$

This conversion is used for plotting/interpretation only and is not part of the fitted likelihood, which is evaluated directly in the volume domain via Eq. (27).

1.4 Parameter Tables

The active fitted parameters and their calibration bounds are summarized in Table 14; fixed numerical quantities are summarized in Table 15. Values and bounds are reported on the natural scale. The bounds encode the configuration of the reported analyses and should not be interpreted as likelihood-based confidence intervals. Some ranges were informed by published biological scales, whereas parameters without reliable direct measurements were assigned engineering bounds for numerical calibration.

For the reported joint fits, the *in vivo*/*in vitro* bounds for each overlapping active biological parameter are merged into the joint union interval in Eq. (47), and all 14 such parameters are represented by the center-delta construction in Eq. (48). Parameters used only by one context retain their component-specific bounds.

Supplementary Table 14: Active fitted parameters and natural-scale bounds used by the separate and joint analyses reported here. A dash means that the parameter is not active in that component. For the 14 overlapping biological parameters, the joint admissible interval is the union of the separate *in vivo* and *in vitro* intervals. These bounds are calibration constraints, not confidence intervals.

Parameter	<i>In vivo</i> bounds	<i>In vitro</i> bounds	Joint bounds	Role in the reported model
λ_{max}	0.005–0.300	0.005–1.500	0.005–1.500	Maximum division rate (day^{-1}); soft-coupled.
$p_{\text{mis,base}}$	10^{-6} –0.100	10^{-6} –0.005	10^{-6} –0.100	Baseline per-chromosome missegregation probability; soft-coupled.
p_{misseg}	0.005–0.500	0.005–0.400	0.005–0.500	Maximal stress-induced missegregation increment; soft-coupled.
$k_{o,\text{mis}}$	0.001–0.020	10^{-5} –0.020	10^{-5} –0.020	Death-hazard half-saturation scale for stress-induced missegregation (day^{-1}); soft-coupled.
s_{max}	0–1	0–1	0–1	Maximum per-copy post-missegregation survival factor; soft-coupled.
β_{buf}	0.01–10	0.01–10	0.01–10	Ploidy-dependent post-missegregation viability-loss strength; soft-coupled.
n_{exp}	0.1–10	0.1–10	0.1–10	Exponent controlling ploidy dependence of post-missegregation survival; soft-coupled.
p_{wgd}	10^{-5} – 10^{-3}	10^{-5} – 10^{-3}	10^{-5} – 10^{-3}	Constant per-division WGD probability; soft-coupled.
α_{O_2}	0.5–4.0	0.5–4.0	0.5–4.0	Strength of resource-stress-dependent growth damping; soft-coupled.

Parameter	<i>In vivo</i> bounds	<i>In vitro</i> bounds	Joint bounds	Role in the reported model
γ_{growth}	1.5–4.5	1.5–4.5	1.5–4.5	Chromosome-number dependence of growth damping; soft-coupled.
μ_{hp}	0.005–0.300	0.005–0.300	0.005–0.300	Stress-associated death scale (day^{-1}); soft-coupled.
γ_{μ}	1.2–3.5	1.2–3.5	1.2–3.5	Chromosome-number penalty on stress-associated death; soft-coupled.
$O_{2,c}$	0.1–2.5	10^{-4} –2.5	10^{-4} –2.5	Critical level for the oxygen-linked resource-stress function (% O_2); soft-coupled.
n_O	0.5–5.0	10^{-4} –5.0	10^{-4} –5.0	Hill exponent for the oxygen-linked resource-stress function; soft-coupled.
S_0	2.0–5.0	–	<i>In vivo</i> only	Low-burden effective oxygen supply level (% O_2).
κ_O	0.1–1.2	–	<i>In vivo</i> only	Oxygen-drop amplitude in the supply-demand target.
η	1.0–2.0	–	<i>In vivo</i> only	Chromosome-number-weighted oxygen-demand exponent.
ρ_{2N}	5.0×10^4 – 5.0×10^5	–	<i>In vivo</i> only	Diploid-cell density (cells mm^{-3}).
k_{clear}	0.005–0.300	–	<i>In vivo</i> only	Dead-biomass clearance rate (day^{-1}).
σ_{burden}	0.100–0.150	–	<i>In vivo</i> only	Log-scale burden observation standard deviation.
σ_{growth}	–	0.001–1.000	<i>In vitro</i> only	Growth-rate observation standard deviation.
σ_{kary}	–	0.01–10.0	<i>In vitro</i> only	Chromosome-number observation smoothing/error scale.
$\mu_{2N,0}$	–	45.725–49.725	<i>In vitro</i> only	Initial near-diploid Gaussian chromosome-count mean.
$\sigma_{2N,0}$	–	0.5–3.0	<i>In vitro</i> only	Initial near-diploid Gaussian chromosome-count SD.
$\mu_{4N,0}$	–	83.667–91.667	<i>In vitro</i> only	Initial near-tetraploid Gaussian chromosome-count mean.
$\sigma_{4N,0}$	–	1.5–6.0	<i>In vitro</i> only	Initial near-tetraploid Gaussian chromosome-count SD.

Supplementary Table 15: Fixed numerical settings used by the reported resource-stress supply-demand model and calibration workflow. Values are reported on the natural scale; settings for inactive optional modules are identified explicitly.

Parameter	Value	Unit	Description	Reference
τ_{O_2}	0.1	day	Oxygen relaxation timescale.	Fixed numerical relaxation setting; empirical workflow choice rather than a literature-derived parameter.
$O_{2,\text{cap}}$	5.0	% O ₂	Maximum oxygen level used to cap the supply-demand oxygen trajectory, inherited from the S_0 upper bound.	Bertout et al. (2008); ³⁰
$O_{2,\text{min}}$	0	% O ₂	Lower oxygen floor used in the logarithmic oxygen target relation.	McKeown (2014). ³¹
N_{ref}	10^6	cells	Reference viable burden scale used in the oxygen supply-demand relation.	Physical/numerical anoxia floor used to bound oxygen values; model constraint rather than literature-estimated threshold.
N_{unit}	22	autosomal chromosomes	Modeled autosomal chromosome count corresponding to ploidy 1.	Oxygen-feedback normalization constant; empirical model scaling value rather than a measured biological quantity.
N_{min}	22	autosomal chromosomes	Minimum autosomal chromosome-count state retained in the discrete state space.	Model convention defining one haploid autosome set as ploidy 1; definition rather than literature-estimated parameter.
N_{max}	154	autosomal chromosomes	Maximum autosomal chromosome-count state retained in the discrete state space.	Discrete state-space lower bound anchored to the model ploidy convention; grid-design choice.
Δt	0.05	day	Simulation time step.	Bielski et al. (2018); ¹ Zack et al. (2013). ³²
K	10^{12}	cells	Carrying-capacity setting for the optional logistic crowding term; crowding was disabled in the reported fits.	Simulation time step selected for numerical resolution and stability; computational setting rather than biological parameter.
$N(0)$	10^6	cells	Initial total tumor population size.	Waclaw et al. (2015). ³³
σ_{ploidy}	0.12	chromosomes	Fixed endpoint autosomal chromosome-number observation-noise scale in the harvest likelihood.	Initial-condition scale for the simulation/calibration workflow; empirical setup choice.
λ_{nec}	1	unitless	Relative weight of the terminal necrosis loss in the <i>in vivo</i> objective (<code>lambda_necrosis</code> in the run configuration).	Fixed likelihood observation-noise term; empirical measurement-error setting in the calibration workflow.
σ_{nec}	0.75	logit-fraction units	Scale used to standardize terminal necrosis discrepancies (<code>sigma_necrosis_logit</code> in the run configuration).	Reported calibration setting.
ϵ_{nec}	10^{-4}	fraction	Clipping constant for observed and predicted necrosis fractions before the logit transform (<code>necrosis_fraction_eps</code> in the run configuration).	Reported calibration setting.
λ_{prior}	0.03	unitless	Relative weight of the soft-prior contribution to the <i>in vivo</i> objective.	Numerical safeguard in the reported calibration.

Parameter	Value	Unit	Description	Reference
$\varepsilon_{\log B}$	10^{-12}	mm^3	Volume-domain floor used inside $\log\{\max(V, \varepsilon_{\log B})\}$ for burden comparisons.	Numerical floor preventing $\log(0)$; computational safeguard.
$N_{\min, \text{pop}}$	10^{-12}	cells	Minimum population floor used for numerical stability.	Population floor used to avoid numerical underflow or zero states; computational safeguard.

Supplementary Table 16: Function level contrasts reconstructed from the three selected joint fit winners displayed in Figure 5. Ranges span the C01, C02, and C03 winner values. For survival summaries, the third and fourth columns report the *in vivo* and *in vitro* values. For complete effective missegregation probability, the third column reports the *in vivo*/*in vitro* ratio range and the fourth column is not applicable. The three optimizer selected fits are neither biological replicates nor confidence interval samples.

Derived quantity	Reference state	<i>In vivo</i> value or ratio range	<i>In vitro</i> reference value	Direction across winners
Per-copy post-missegregation survival	$N = 44$	0.728–0.819	0.204	higher <i>in vivo</i> , 3/3
Per-copy post-missegregation survival	$N = 88$	0.904–0.914	0.837	higher <i>in vivo</i> , 3/3
Survival gradient, $s_{88} - s_{44}$	$44 \rightarrow 88$	0.0849–0.186	0.633	larger <i>in vitro</i> , 3/3
Complete effective missegregation-probability ratio	0% O ₂ , $N = 44$	11.133–20.715	not applicable	> 1 , 3/3
Complete effective missegregation-probability ratio	0% O ₂ , $N = 88$	11.178–20.611	not applicable	> 1 , 3/3
Complete effective missegregation-probability ratio	1% O ₂ , $N = 44$	7.421–17.037	not applicable	> 1 , 3/3
Complete effective missegregation-probability ratio	1% O ₂ , $N = 88$	10.354–19.624	not applicable	> 1 , 3/3
Complete effective missegregation-probability ratio	5% O ₂ , $N = 44$	0.738–5.448	not applicable	> 1 , 2/3
Complete effective missegregation-probability ratio	5% O ₂ , $N = 88$	3.210–5.697	not applicable	> 1 , 3/3

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